

Honeywell

**TPS Builder
Configuration Tool
User's Guide**

BL20-420

TPS R420

03/2013

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Honeywell Process Solutions
1860 W. Rose Garden Lane
Phoenix, AZ 85027 USA
1-800 822-7673

About This Document

This document provides basic instruction on how to use the Honeywell TPS Builder configuration tool. This guide explains how to perform all of the tasks associated with TPS Builder, including configuring **TotalPlant** Network (TPN) node and data points, and working with engineering drawings.

No attempt is made in the guide to explain the functionality of TPS Builder or the many different options available for configuration.



ATTENTION

With TPN R684, Enhanced Universal Control Network (EUCN) is introduced. In EUCN, the traditional coaxial cable system is no longer utilized and is replaced with a Fault Tolerant Ethernet (FTE) based communication network.

Throughout the document, unless specifically mentioned, the terms HPM, NIM, and UCN are interchangeably used with the terms EHPM, ENIM, and EUCN respectively.

Release Information

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References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
<i>TPS Builder Installation, Implementation, and Administration Guide</i>	BL08
<i>TPS Builder CL Editor User's Guide</i>	BL21

Support and Other Contacts

United States and Canada

Contact: Honeywell Solution Support Center
Phone: 1-800-822-7673
Calls are answered by dispatcher between 6:00 am and 4:00 pm Mountain Standard Time. Emergency calls outside normal working hours are received by an answering service and returned within one hour.
Facsimile: 1-973-455-5000
Mail: Honeywell TAC, MS L17
1860 W. Garden Lane
Phoenix, AZ, 85027 USA

Europe, Middle East, and Africa (EMEA)

Contact: Honeywell TAC-EMEA
Phone: +32-2-728-2345
Facsimile: +32-2-728-2696
Mail: TAC-BE02
Hermes Plaza
Hermeslaan, 1H
B-1831 Diegem, Belgium
Email: tac-be02@honeywell.com

Pacific

Contact: Honeywell Global TAC – Pacific
Phone: 1300-364-822 (toll free within Australia)
+61-8-9362-9559 (outside Australia)
Facsimile: +61-8-9362-9564
Mail: Honeywell Limited Australia
5 Kitchener Way
Burswood 6100, Western Australia
Email: GTAC@honeywell.com

India

Contact: Honeywell Global TAC – India
Phone: +91-20- 6603-9400
Facsimile: +91-20- 6603-9800
Mail: Honeywell Automation India Ltd
56 and 57, Hadapsar Industrial Estate
Hadapsar, Pune –411 013, India
Email: Global-TAC-India@honeywell.com

Korea

Contact: Honeywell Global TAC – Korea
Phone: +82-80-782-2255 (toll free within Korea)
Facsimile: +82-2-792-9015
Mail: Honeywell Co., Ltd
4F, Sangam IT Tower, , B4-4Block
1590, DMC Sangam-dong, Mapo-gu
Seoul, 121-835, Korea
Email: Global-TAC-Korea@honeywell.com

People's Republic of China

Contact: Honeywell Global TAC – China
Phone: +86- 21-2219-6888
800-820-0237
400-820-0386
Mail: Honeywell (China) Co., Ltd
33/F, Tower A, City Center, 100 Zunyi Rd.
Shanghai 200051, People's Republic of China
Email: Global-TAC-China@honeywell.com

Singapore

Contact: Honeywell Global TAC – South East Asia
Phone: +65-6823-2215
Facsimile: +65-6445-3033
Mail: Honeywell Private Limited
Honeywell Building
17, Changi Business Park Central 1
Singapore 486073
Email: GTAC-SEA@honeywell.com

Japan

Contact: Honeywell Global TAC – Japan
Phone: +81-3-6730-7160
Facsimile: +81-3-6730-7228
Mail: Honeywell Japan Inc.
New Pier Takeshiba, South Tower Building,
20th Floor, 1-16-1 Kaigan, Minato-ku,
Tokyo 105-0022, Japan
Email: Global-TAC-JapanJA25@honeywell.com

Elsewhere

Call your nearest Honeywell office.

World Wide Web

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Contents

1. Introduction

1.1 Overview

Document scope

This document provides basic instruction on how to use the Honeywell TPS Builder configuration tool. This guide explains how to perform all of the tasks associated with TPS Builder, including configuring **TotalPlant** Network (TPN) node and data points, and working with engineering drawings.

No attempt is made in the guide to explain the functionality of neither TPS Builder nor the many different options available for configuration.

Intended audience

You may be using TPS Builder on one of a variety of platforms, such as the Honeywell Global User Station (GUS), a stand-alone personal computer, or a network.

This guide is intended for individuals who are familiar with the platform on which TPS Builder will be used. This includes familiarity with the computer, its operating system, and networks when applicable.

1. Introduction

1.1. Overview

What do you want to do?

The following table provides references in this document for the most common TPS Builder activities.

If you want to...	Then go to...
Learn the basic terminology and workings of TPS Builder	TPS Builder Fundamentals, Overview
Begin working in TPS Builder	- Getting Started, Start TPS Builder - Using the TPS Builder Workspace, Overview
Work with libraries and templates	- Using Libraries, Overview - Using Templates, Overview
Create projects and strategies	- Working with Projects, Overview - Working with Strategies, Overview
Create engineering drawings	Working with Drawings, Overview
Access recorded TPS Builder and implementation data	- Working with Logs, Overview - Working with Reports, Overview

1.2 Getting Started

Start TPS Builder

To start TPS Builder from your desktop, refer to the *TPS Builder Installation, Implementation, and Administration Guide*, beginning with the Introduction for instructions on how to install TPS Builder.

Click **Start**, then point to **Programs > TPS Builder > TPS Builder**.

Working in the TPS Builder workspace

The TPS Builder workspace follows standard Microsoft Windows' conventions in terms of activities such as positioning, sizing or minimizing the windows within the workspace.

For tips on working in this environment, refer to "Using the TPS Builder Workspace," beginning with the "Overview."

For details on configuring various workspace and usage aspects of TPS Builder, refer to the *TPS Builder Installation, Implementation, and Administration Guide*, Software Configuration.

Quit TPS Builder

Use one of the following methods to quit TPS Builder:

- Click the **X** button in the upper right corner,

OR

- From the **File** menu, choose **Exit**.

If any unsaved drawings are open, you will be asked whether or not to save them. Click the desired answer in the message box.

When TPS Builder quits, your workspace is saved.

1. Introduction
1.2. Getting Started

2. TPS Builder Fundamentals

2.1 Overview

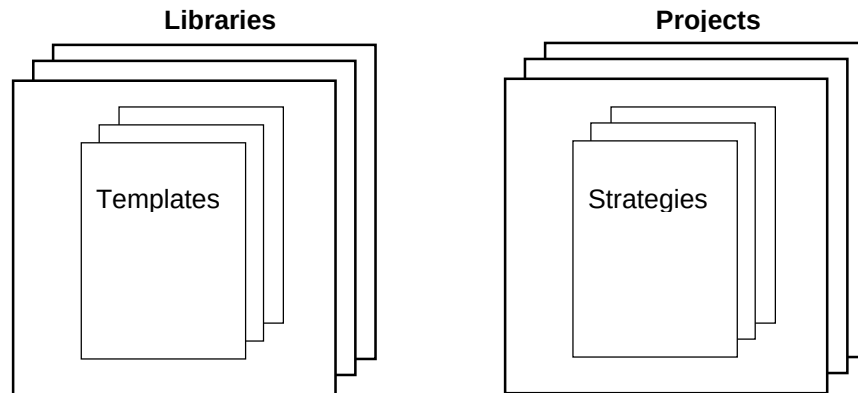
Learning the terminology

The following terms are all closely related parts of TPS Builder:

- Libraries
- Templates
- Projects
- Strategies
- Blocks
- Modules
- Composition Views
- Drawings

Understanding basic TPS Builder terminology will help you use it efficiently. The remainder of this section explains each of these terms as they apply to TPS Builder.

In the simplest terms, the TPS Builder repository contains libraries of templates and projects composed of strategies:

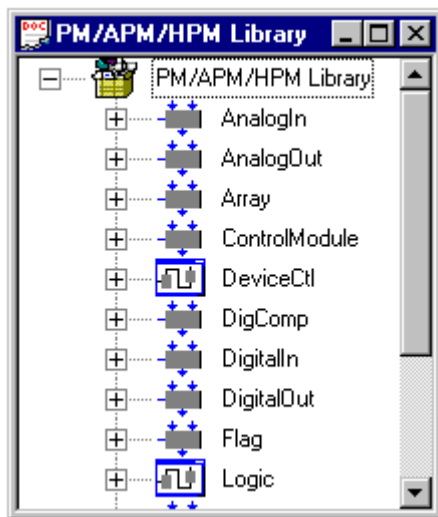


2. TPS Builder Fundamentals

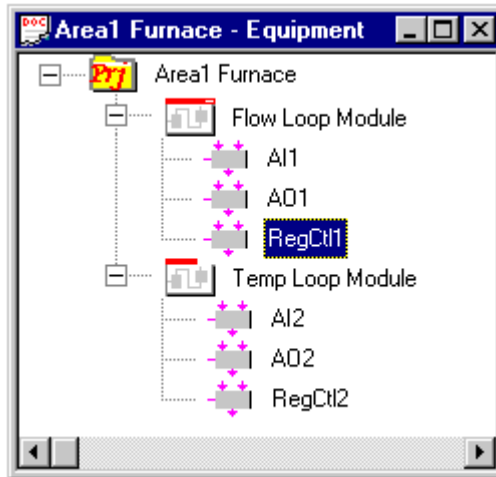
2.1. Overview

The following figures show how these parts are represented in TPS Builder.

**Typical Honeywell-Supplied
Library and Templates**



**Typical User-Created
Project and Strategies**



Planning

Planning is a key success factor for efficient use of TPS Builder. Read this fundamentals section, and begin to think about and plan for such factors as

- Types and specialization of templates and libraries
- Project organization
- Types of views needed
- Drawings – size and color of drawing symbols, connectors, and so on

Creating projects and specializing templates on a test basis will help you understand and define your needs.

2.2 Strategies

What is a strategy?

A TPS Builder strategy is the configuration information for some element in the TPS system (or another system), intended for loading to the system. A strategy contains loadable data, such as point names, parameter values, and slot assignments.

A strategy can be either a simple block that represents a point, or a module that represents a composite, such as a control loop.

How a strategy is created

A strategy is created from a template. Templates for creating strategies are included in TPS Builder libraries. As you configure a system, you create your own strategies, giving them unique names, such as AI100 for a particular analog input point.

Strategy blocks

Individual strategies can be thought of as “blocks.” In TPS Builder, blocks equate to individual TPS Network points or logic blocks and gates.

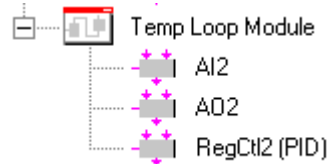
In addition to the engineering aspects of a TPS Process Network point (such as parameters, behavior, or programs if applicable), a block also carries with it TPS Builder-specific characteristics, such as how TPS Builder blocks are represented in a drawing.

Block appearance

TPS Builder has two basic ways of showing blocks:

- as icons in outline form within a project, as shown in the following figure, and
- as a drawing symbol in a control drawing (refer to Control Drawings)

Strategy modules

Strategy blocks can be combined in a strategy “module,” to represent a control scheme. This example shows a module, as well as Analog Input, Analog Output, and Regulatory Control blocks (points). Modules can contain other modules as well as strategy blocks.	 The diagram illustrates a 'Temp Loop Module' represented by a red-outlined box. Inside the box, three strategy blocks are shown: 'AI2' (Analog Input), 'AO2' (Analog Output), and 'RegCtl2 (PID)' (Regulatory Control). These blocks are connected by dashed lines, indicating their functional relationship within the module.
---	--

2. TPS Builder Fundamentals

2.2. Strategies

Honeywell-provided modules

In TPS Builder, the following items are considered modules:

- TPS Network nodes
- Device Control points
- Logic points



ATTENTION

Each type of module can contain only elements that are pertinent to its configuration.

TPS Builder also includes “empty” modules, which are not loaded to the TPS system. These modules allow you to group strategies in a number of ways.

Device control and logic points

Device control and logic points are treated as modules within TPS Builder because these points can include a mix of logic capability, such as logic blocks, flags, numerics, and input and output connections. These capabilities are represented as strategy blocks within the device control or logic modules.

Module appearance

TPS Builder represents a module as:

- an icon in an outline form within a project (as in the previous figure)
- a drawing symbol in a control drawing
- a drawing

Reference: Control Drawings.

2.3 Templates

What is a template?

A TPS Builder template can be thought of as a reusable strategy that presents all configuration requirements for that strategy. Honeywell provides a set of system templates (for example, AnalogIn).

Data that will be common to a class of strategies can be included in a template, leaving only unique parameter values to be configured.

Templates can be grouped into modules. This enables you to build templates for loops, or larger control schemes.

How templates are used

You use Honeywell-provided templates to create your own strategies, entering specific configuration data to meet your needs (for example, for an analog input point, or a flow loop strategy).

You can create specialized templates as well (for example, FlowInput). You can also create template modules, that is, groupings of templates that you want to use again and again (such as a common flow loop configuration).

Honeywell-provided templates

Honeywell provides libraries that include system templates, which include (as applicable) parameter values as defined in the TPS system parameter reference dictionaries, except in cases where this information differs from the operation of the PED (Parameter Entry Display).

2. TPS Builder Fundamentals

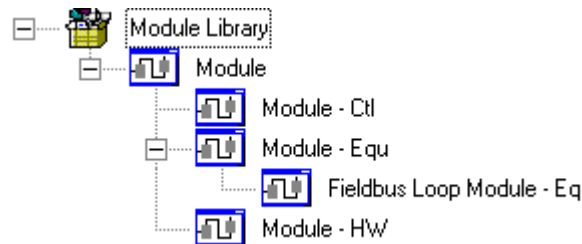
2.3. Templates

Depending on which TPS Builder libraries you purchase and install, templates are provided for the following implementation tasks:

- Creating TPS Network points—one for each data point type, such as Analog Input points
- Configuring logic inside Device Control and Logic points—one for each internal logic element of the points, such as gates, inputs, and outputs
- Configuring Hiway Gateway (HG) points—one for each data point type, such as Analog Input points
- Configuring TPS Network (UCN) nodes and Hiway Gateway (HG) boxes—one for each node and box type, such as the High-Performance Process Manager (HPM) and Multifunction Controller (MC)
- Configuring Fieldbus devices
- Creating control schemes—"Empty" template modules

"Generic" template modules

One empty module exists for each basic composition view (refer to Composition Views):

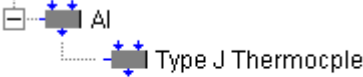


Templates or strategies can be combined in modules to represent a control scheme. For example, a temperature loop template can be created from analog input, regulatory control (PID), and analog output point templates.

In TPS Builder this scheme would be represented as a module, and it would be created by first using one of the empty modules.

Template customization and specialization

You can customize or specialize the templates provided by Honeywell to represent configurations that will be used repeatedly. Common configuration data is included in a specialized template, so that only unique aspects of a strategy need to be configured to complete its definition.

Template Action	Examples
Customize: Changing the configuration of a template	You may want all temperature-related strategies to utilize a temperature scale of degrees Kelvin. You can define all related parameters on templates accordingly.
Specialize: Creating a new template	You may want to specialize an AI template to represent a Type J thermocouple:  Another example of specialization is that it can be particularly useful for setting up logic blocks for different logic mixes that occur on a regular basis.



ATTENTION

The hierarchy for templates in a library is different than that for strategies within a project—specialization as opposed to components.

Specialization relationship

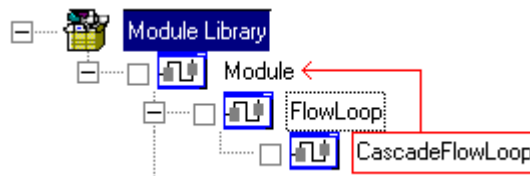
Specialized templates maintain a relationship with their “parent.” If you make changes to the template-defining attributes of the parent, the changes will also be applied to its specialized templates (and strategies created from it). The specialization relationship also affects how templates move from one library to another.

To break the relationship between a specialized template and its parent, drag the specialized template onto a new (applicable) parent somewhere higher in the hierarchy.

2. TPS Builder Fundamentals

2.3. Templates

Consider the following example:



Here, if you drag CascadeFlowLoop onto Module you will break its relationship with FlowLoop. You can only perform this change by moving a template up in a hierarchy. In the previous example, you could not move FlowLoop under CascadeFlowLoop, for instance.

Template-defining attribute

A template includes various elements, including parameter groups, parameters, parameter values, connections, and nested templates. Any of these can be designated as "template defining" on a particular template, which means that it will be inherited by other templates specialized from it, or strategies created from it. It can be changed only on the template on which it was originally set as template defining.

Template-defining rules

Some template elements are always template defining and others can have the attribute set by users. The following table summarizes the rules regarding setting of the template-defining attribute.

Element	Rule that Applies
• Parameter Groups • Parameters • In Template Modules: – Contained templates – Connections	Element is always template defining.
Parameter values	User can specify as template defining.
The value for the NAME parameter	Element should never be set as template defining.

Template-defining example

You can choose parameter values to be template-defining. For example:

Template X

Temperature parameter = °K is set as Template DEFINING

You can modify this parameter only in Template X.

Template-defined elements

A relationship is maintained between a template and the strategies created from or templates specialized from it. If template-defining data is changed, the change is reflected in all strategies and templates based upon the original template.

From the perspective of the strategy and specialized template, the data is “template-defined,” that is, dictated by the template upon which they are based.

Continuing the previous example:

Strategy A, based on Template X

Temperature parameter = °K = Template DEFINED

- You cannot modify this parameter for Strategy A.
- If this parameter is changed in Template X, it will also change here.

Template Y, based on Template X

Temperature parameter = °K = Template DEFINED

- You cannot modify this parameter in Template Y.
- If this parameter is changed in Template X, it will also change here.

2. TPS Builder Fundamentals

2.3. Templates

Updating derived controls with template changes

To improve the consistency of template support for strategies, use the update derived controls to update changes made in template drawings. This updating means that changes in a template are propagated to the template's specialized templates and existing strategies.

Template change propagation makes templates more powerful than in previous TPS Builder releases. It saves the time of getting changes into the drawings of derived controls so that drawing rework is significantly reduced. Not all rework can be eliminated because template changes may affect derived drawings that need to be revised for clarity. For example, when control symbols have been added both to a template and to one of its strategies, then after propagation, the strategy may have overlapping symbols. Therefore, the drawings of all controls affected by propagating template updates should be opened and checked before the drawings are used.

Effects of template change propagation

Because the effects of template change propagation can be extensive, template changes should be planned. In particular, template changes should not be done without considering how many derived controls will be affected. (Although selection is available when performing updates, the Relationship Browser and the Dependencies button in the Identification dialog are two additional tools for this consideration.)

Due to the complexity of control drawings, a set of rules controls the propagation. In general, a property is not propagated when it has been changed on the derived control. The following chart explains what is propagated during an update of template changes.

Rules for propagation during update

Drawing Elements	Propagation for Updating Template Changes
Build symbol positions, sizes, fonts, and colors	Changes are propagated unless a change in the same block position or size has been made on the derived control. Thus, changes specific to a derived control drawing are preserved.
Build symbol icons	Changes are always propagated.
Displayed parameters	Changes are always propagated. The order does not matter because parameters are listed alphabetically.

Drawing Elements	Propagation for Updating Template Changes
Connectors	Changes are propagated. New connectors are added after connectors added specifically on the derived control.
Connection lines and named connections	Connection line changes are not propagated because autorouting handles connection lines. Name connections are propagated.
Connection tables	Connection tables and changes to them are propagated.
Embedded objects	Embedded objects cannot be opened and compared, so each embedded object is considered to be a single drawing element. Unless the position and size of an embedded object match, the object is always propagated. This may result in having multiple copies of an object in a drawing. Also, if an embedded object is updated in both a template and one of its derived controls, the update on the derived control will be overwritten if left in the same position; but if it is moved, then the updated from the template will be added to the drawing.
Drawing map	The drawing map is propagated unless there has been a change to the derived map's dimensions.
Title block	Not applicable because title block data is project-specific and is not stored in drawing files.

Updating can be done in three different ways.

1	Changes made to a specific template can be propagated.
2	Changes necessary to update a single derived control can be propagated.
3	All outstanding changes in a repository can be propagated in a mass update.

2. TPS Builder Fundamentals

2.3. Templates

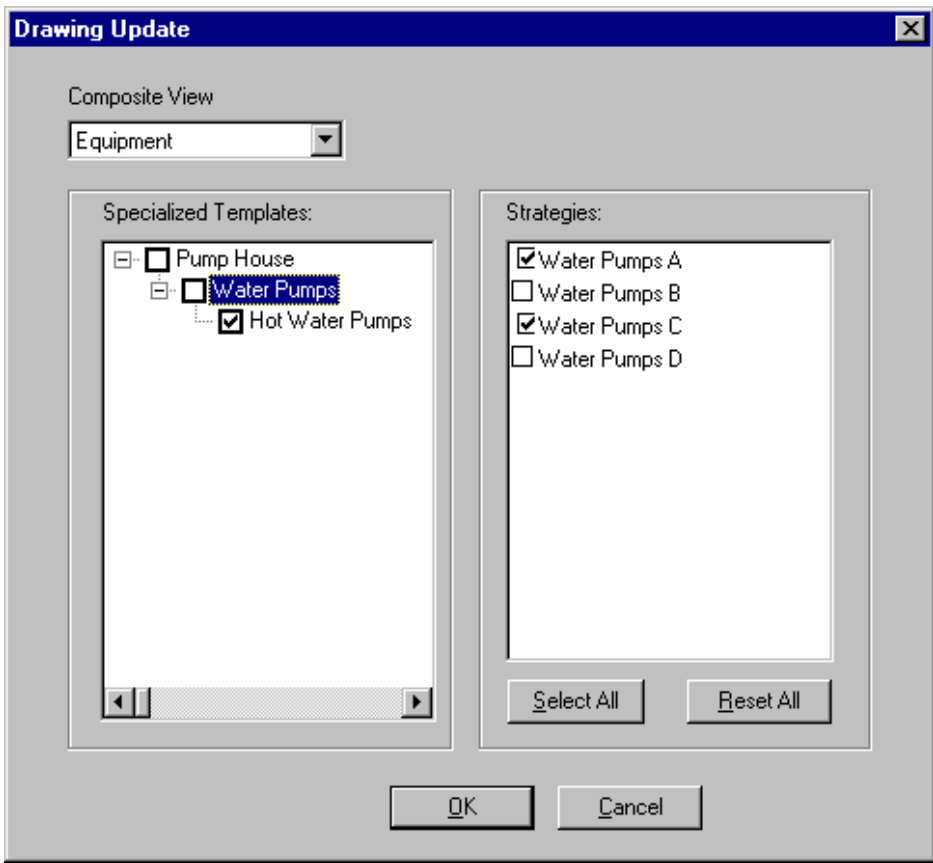
Using the checkbox to update

Use the checkboxes to select templates and strategies for update. The selections are maintained as the view is changed or as the template focus is changed. This allows any combination of templates and strategies in different views to be selected. It also allows any combination of strategies for different templates to be selected.

Changes are propagated from the starting module to all selected modules. If a module on a branch is not selected, but one further down the branch is selected, both will be updated because the changes must be propagated through any intermediate modules to the selected modules.

- **Changes to a particular derived control.** When a drawing is opened, TPS Builder traces its lineage to see whether one of its “ancestor” templates has changes that need to be propagated. If so, the changes are propagated only along the one derivation line before the drawing is opened. This limited propagation is also done before a strategy is templated.
- **After changes have been made and saved in a module,** the changes can be propagated. To propagate changes from a single template to all derived controls follow this procedure.

Step	Action	Result/Comment
1	In the library tree, select the module and use the mouse right button to open the context menu. Select Update Derived Drawings . Select a view.	The specialization tree on the left lists the selected template and its specialized templates.
2	Select template(s) using checkbox(es).	The focus causes its strategies to be listed on the right.

Step	Action	Result/Comment
 The image shows a 'Drawing Update' dialog box. At the top, there's a 'Composite View' dropdown menu set to 'Equipment'. Below this, there are two main sections: 'Specialized Templates' and 'Strategies'. In 'Specialized Templates', there's a tree view with 'Pump House' (unchecked), 'Water Pumps' (checked), and 'Hot Water Pumps' (checked). In 'Strategies', there are four checkboxes: 'Water Pumps A' (checked), 'Water Pumps B' (unchecked), 'Water Pumps C' (checked), and 'Water Pumps D' (unchecked). At the bottom of the dialog are 'OK' and 'Cancel' buttons. Below the 'Strategies' section are 'Select All' and 'Reset All' buttons.		
3	Check the box(es) of the strategy(ies) you want to update. Click OK to begin the update.	The update begins.
4	Multiple templates may be selected for update using the check boxes.	The update begins according to the boxes checked.

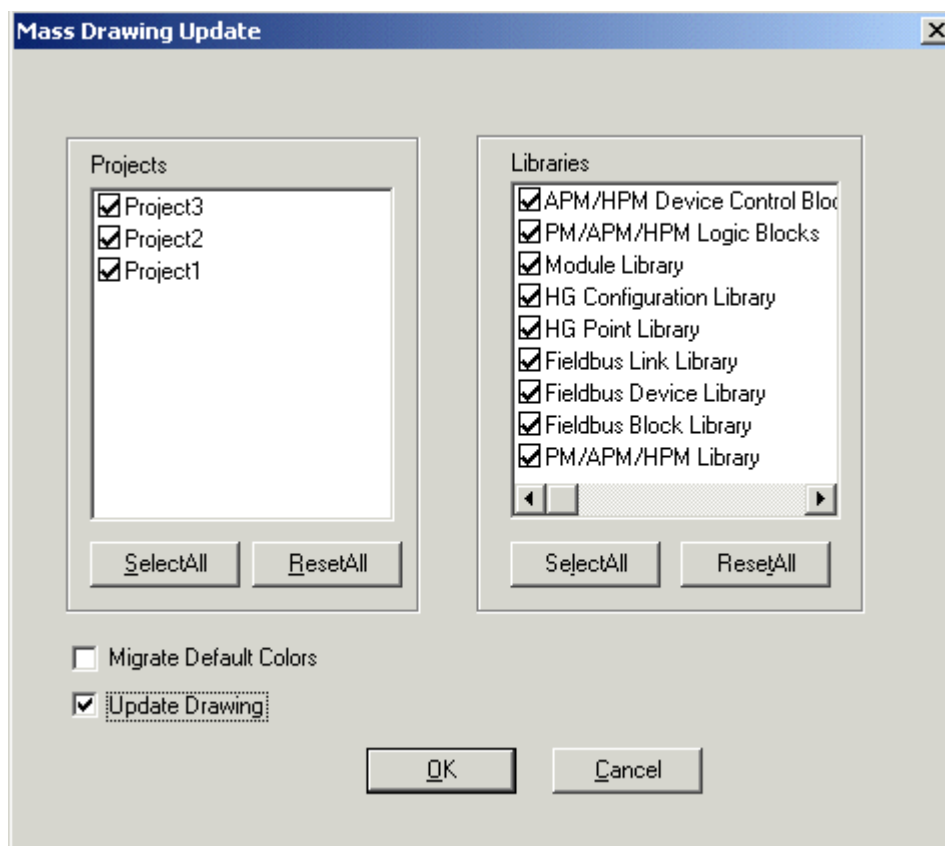
Selections are maintained as the composite view changes or the template focus changes. Any combination can be used for each.

Mass update

Update Drawing

Use the following procedure for a Mass Update of selected projects and library.

Step	Action	Result/Comment
1	Select Tools>Update Drawing .	The Mass Drawing Update dialog box appears, listing all the projects and libraries.



Step	Action	Result/Comment
2	Select the required combination of projects and libraries. To select all the projects and libraries click SelectAll in the Projects section and also click SelectAll in Libraries section.	For the selected <i>project(s)</i> , all unpropagated template changes will be propagated to strategies in all views. For selected <i>libraries</i> , all unpropagated template changes will be propagated to specialized templates in all views.
3	Select Update Drawing and click OK .	The updates for selected projects and library begin.

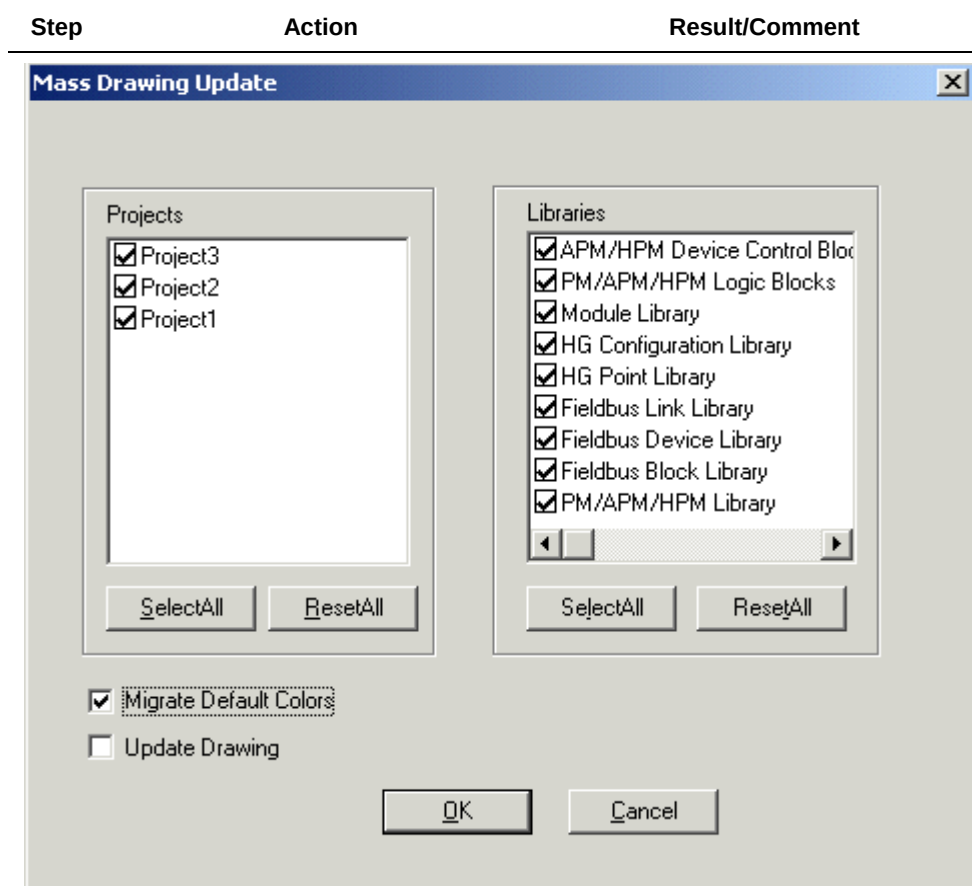
Migrate Default Colors

Use the following procedure for a Mass Update of block, module, and arrow colors.

Step	Action	Result/Comment
1	Select Tools>Update Drawings	The Mass Drawing Update dialog box appears, listing all the projects and libraries.

2. TPS Builder Fundamentals

2.3. Templates



- 2 Select the required combination of projects and libraries.

To select all the projects and libraries click **SelectAll** in the **Projects** section and also click **SelectAll** in **Libraries** section.

Step	Action	Result/Comment
3	Select Migrate Default Colors and Click OK .	For each of the selected project(s) and library(ies), its corresponding drawing files will be converted to reflect the new default settings. The new default settings are: • All the block colors are changed to white • Module color are changed to cyan • Arrow color are changed to black
CAUTION	It is strongly recommended that you take a backup of the repository before attempting to migrate so that you can revert back to the old look if you are not satisfied with the new look after migration. It will be a laborious process to change the colors and settings back to its original should you not like the new settings as there is no 'Mass Update' functionality available to revert back all the drawings to its previous settings. It has to be done manually and individually.	

3D Blocks in drawing and migration

To view the drawings in 3D, choose **Drawing Settings > Three Dimensional Blocks**. A check mark appears corresponding to the option **Three Dimensional Blocks** indicating that the 3D option is selected. When this option is unchecked the drawings are displayed in 2D.

On migrating the database, by default the drawings are displayed in 3D.

Keeping track of changes

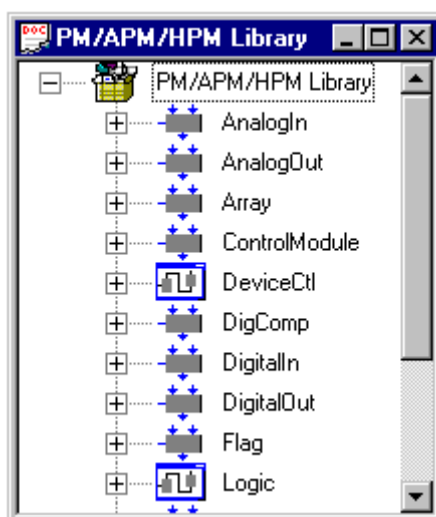
TPS Builder keeps track of derived controls that need to be updated. If a strategy needing update is selected in a project window, the message "Drawing Update needed" is displayed on the left side of the status bar.

Updates to drawings are recorded in the audit trail. If an update fails, a message is recorded in the error log.

2.4 Libraries

What is a library?

TPS Builder Libraries are convenient template managers that enable you to view, access, and organize templates. Honeywell provides libraries, and you can also create your own. The following shows a Honeywell-provided library. Note the library icon, and icons for templates:



Honeywell-provided libraries

Libraries provided by Honeywell are called “system” libraries. When you purchase TPS Builder, you select the base TPS Builder application plus one or more subsystems containing related groups of libraries.

Available libraries include:

• LCN/UCN libraries	• Module library
• Fieldbus libraries	• Data Hiway libraries
• CLM Platform Library	

LCN/UCN libraries

The Local Control Network/Universal Control Network libraries contain the following templates:

Library	Contains Templates for
AM	Each Application Module data point (such as Custom, Switch, Timer)
PM/APM/HPM	Each Process Manager-family data point (such as Analog Input, Logic, Regulatory Control)
PM/APM/HPM Logic	Each internal logic element of PM-family Logic points (such as gates, inputs, outputs)
APM/HPM Device Control Blocks	Each internal logic element of APM or HPM Device Control points (such as gates, inputs, outputs)
LM/SM	Each Logic Manager/Safety Manager data point
Module	“Empty” modules for creating template or strategy control schemes
UCN Configuration	Each supported TPS Network (Universal Control Network) node type (such as PM, APM, HPM, and NIM)

Fieldbus libraries

Fieldbus-specific libraries include:

Library	Contains Templates for
Fieldbus Block	- Standard Fieldbus function blocks and transducer blocks - Manufacturer-specific function and transducer blocks
Fieldbus Device	Fieldbus devices supported in the R61X.XX release of Fieldbus
Fieldbus Link	Fieldbus Links and Link Schedules

2. TPS Builder Fundamentals

2.4. Libraries

Data Hiway libraries

The Data Hiway libraries include:

Library	Contains Templates for
CG Point	ACCRDP and ACIDP points
HG Configuration	Each supported TPS Network (Hiway Gateway) box type (such as CB, DHP, MC)
HG Point	Each Hiway Gateway data point (such as Analog Input, Control Counter, and Timer)

Module library

The Module library contains templates for creating template or strategy control schemes. One empty module is provided for each standard composition view:

- Control
- Equipment
- Hardware

In addition, a Fieldbus Loop module is provided.

CLM Platform Library

The CLM Platform library would house the following CLM template types:

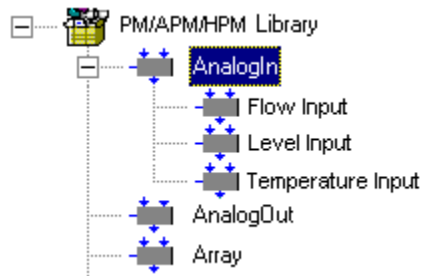
- CLMI
- PIT
- PITG

Other CLM templates that are provided in TPS Builder are distributed into different libraries based on the support to different kinds of networks.

Library organization

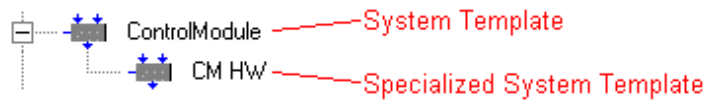
Libraries list their contents (template blocks and/or modules) alphabetically in an outline fashion that shows the specialization hierarchy of all templates.

The following figure shows an example:



Library structure

TPS point and logic libraries are structured as follows. Each point or logic element in the library is structured as shown in this figure and discussed in the following paragraphs.



- A system template is provided. It contains default parameter values as defined in the TPS system parameter reference dictionaries, except in cases where this information differs from the operation of the PED (Parameter Entry Display).



ATTENTION

Do not modify this template.

- Specialized templates can be created from this template, but strategies cannot. A strategy can be created only from a template with a view(s) assigned, and system templates have no composition views assigned.
- A specialized version of the system template, with specific views assigned, is provided. This template contains all default values of the system template (and nothing more). Specialized templates and strategies can be created from this template.

2. TPS Builder Fundamentals

2.4. Libraries

A system template can be specialized, for example, to add a Control-view version. Other specializations can be made from the system template, or any other template, as needed. These specializations will initially appear in the library in which they are created. They can be moved to user libraries if desired.



ATTENTION

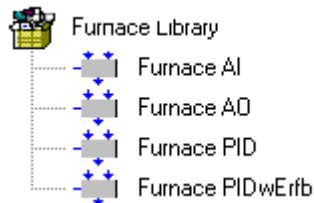
Some libraries contain only system templates with a view(s) assigned.

User libraries

A user library contains specialized templates of your choosing. The intent is to provide a convenient, single location for templates that will be used frequently. It can eliminate the need to open several libraries, and eliminate scrolling within libraries that contain many infrequently used templates.

You can place any specialized templates in user libraries.

Consider a plant that utilizes many furnaces. Furnace controls include analog input, regulatory control (PID or PIDwErfb), and analog output points. The following figure shows a user library based on these needs:



Certain characteristics for these points will be the same for all furnaces. In TPS Builder, you can create a user library of template modules and blocks so that all common furnace-related characteristics are defined in the templates.

Planning for libraries

Planning for user libraries will help you make the most of TPS Builder. Take time to evaluate your plant's needs, existing implementation strategies, and the engineering environment to determine the best application of user libraries. For large organizations in which several engineers are involved, consider the value of user libraries for optimum consistency in the creation of TPS Builder projects.

Beginning with TPS Builder R130, you can take one of these approaches regarding the distribution of libraries:

- A central engineering group can create a database with system templates plus specialized templates that they want to distribute. The database typically will not have any project data. This database can then be shipped to users in the organization when they begin to use TPS Builder. This approach may be best when consistency is of utmost importance.
- Specialized templates, and user libraries if desired, can be created and distributed on an as-needed basis, by using TPS Builder's import/export functionality. This approach allows a faster rollout of TPS Builder.

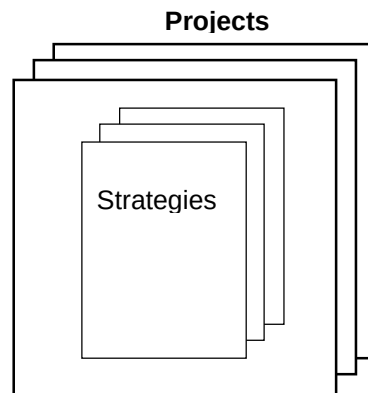
In addition to the libraries and templates themselves, consider adding your own support information to TPS Builder, in the form of online Help. With TPS Builder R130 and later, you can attach specific Help topics to libraries and templates.

2.5 Projects

What is a project?

TPS Builder projects are easy-to-use strategy managers. They are what you create to organize your implementation data.

In addition to serving as an organizational tool, projects define the scope for checking. This means that when making data flow connections, only names in the current project are recognized as valid names.



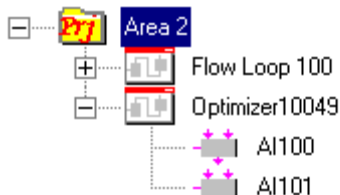
How many projects?

You can create any number of projects in TPS Builder.

A single project, in most cases, would be very difficult to manage (as discussed in “Project size”). Thus, in nearly all cases, you will create multiple projects that organize your system implementation into logical groupings.

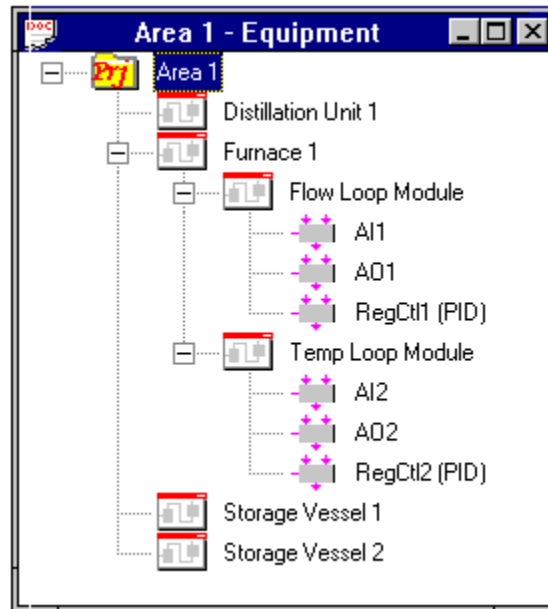
Project organization

Projects list their contents (strategy blocks and/or modules) alphabetically in an outline fashion that shows the hierarchy of all strategies, as shown in the following example.



Project example

The following figure shows a project based on a plant area. Similar projects would be developed for all areas of the plant. Note the project is in Equipment view.



Project size

Projects can be any size. Creating large projects can lead to several problems, however:

- Additions and changes are more difficult and time consuming. For example, adding an extra unit to a large existing system could require a significant amount of overhead.
- Overall performance will be affected.
- The checking function will take much longer to run — or you may need to restrict checking.
- TPS Builder database backups will take more time and require much more hard disk space.

You must plan and evaluate your implementation needs to create projects of appropriate sizes.

R130 limitations

TPS Builder R130 does not allow copying or moving strategies between projects.

2. TPS Builder Fundamentals

2.6. Composition Views

Establishing separate databases

If you do not want projects to have the same database, you can set up multiple databases. If you do so, you can then switch from one database to another.

Refer to the procedures for creating and accessing different databases in the *TPS Builder Installation, Implementation, and Administration Guide*, Database Administration.

2.6 Composition Views

Definition

TPS Builder enables you to look at projects and drawings from different perspectives called “composition views.” These views organize strategies into different hierarchies. One example would be a hierarchy based on equipment association.

Default views

Default views include:

View	Description
Equipment	Allows you to use an equipment association as a basis for organizing strategies. Typically, the strategies in this view will represent transmitters, valves, pumps, and so forth. In this view you might also include lower level control functions, for example, the PID block in a loop is kept together with the input and output.
Hardware	Allows you to use hardware addressing as a basis for organizing strategies (such as networks, nodes, and points assigned to them).
Control	Allows you to organize higher level control functionality, for example, optimization or a mass balance implemented in an Application Module.
Fieldbus Link	This view is used only if you are configuring Fieldbus entities. It shows configuration of the project in physical Fieldbus terms. Fieldbus Link View shows each Fieldbus Link (segment) and the devices assigned to the link. Each device includes associated Virtual Field Devices (VFDs), as well as function and transducer blocks appearing in each VFD.

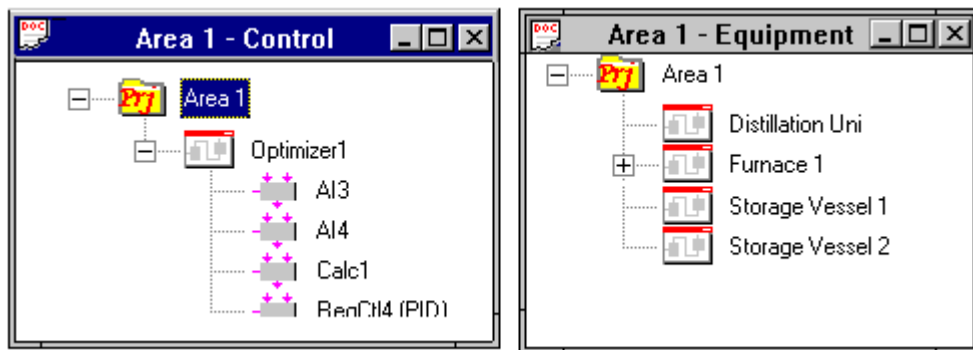
You can also create your own composition views.

Using views

How the TPS Builder views are actually used is your decision. The intent is to provide a mechanism that allows you to view projects and drawings in different ways. Once you have defined view usage, you can open a project in each view, one window per view, simultaneously if desired.

View example

Consider the Area 1 project. It includes an optimizer module for Control view only, and a distillation unit, furnace, and two storage vessels for Equipment view, as shown in the following figure.



2. TPS Builder Fundamentals

2.6. Composition Views

View membership

A view membership is the view in which a strategy will appear. View membership of a template determines the view membership of strategies created from it (and templates specialized from it). For instance, all strategies created from an unmodified AI_EqHW will be an analog input visible in equipment and hardware views of its project (but not control view).

Honeywell-supplied templates include default view memberships as follows:

- System TPS point/logic element templates have no view membership.
- For each TPS point/logic element, Honeywell supplies a specialized template with equipment and hardware views.
- TPS Network node and Data Hiway box templates are supplied with hardware view.
- Several "generic" template modules are provided, one for each view (control, equipment, hardware, and fieldbus link).

Rules for changing view membership

Due to the complex relationships between templates and strategies or templates specialized from them, certain restrictions apply for changing view memberships:

Item	Rule
Strategies	You cannot change the view membership of a strategy.
System Templates	You cannot modify the view membership status of original system templates.
Specialized Templates	You can change the view membership of a specialized template, only if it has not been used to create a strategy or to specialize a template.

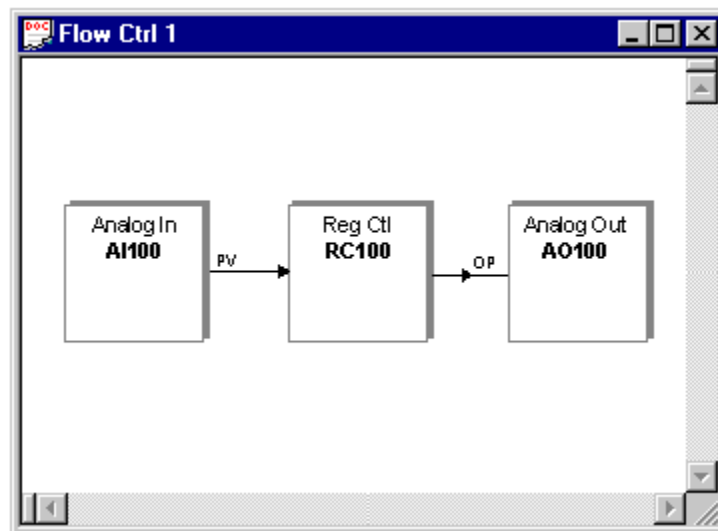
View membership for a template must be set before it is used.

2.7 Control Drawings

What is a control drawing?

A TPS Builder control drawing is a graphical representation of a strategy or template module, its components, and their connections.

A strategy or template module may have one control drawing per composition view. The following simple example shows the equipment view drawing for the Flow Control 1 module. This module has three connected strategy blocks.



Drawing activities

Once a template or strategy module is created, all configuration activity can take place within a drawing, including:

- Adding blocks or modules to (or deleting them from) the module.
- Making connections between blocks or modules.
- Adding or deleting any parameter as a connector, including selecting project-up parameters as connectors. Refer to online help for information on Project Up, which is a control attribute.

2. TPS Builder Fundamentals

2.7. Control Drawings

- Accessing parameter forms for configuration.
- Setting properties (colors, fonts, etc.) of the drawing symbols.

Printing

Drawings can be printed for documentation purposes.

Drawing options

You can customize the drawing in several ways, so that it represents the information to suit your needs. Consider these examples:

- Define size and colors for each drawing symbol, including labels, connectors.
- Specify the information that will appear on each strategy's drawing symbol.
- Add picture(s) to a drawing symbol.
- Use line or named connections.
- Position drawing symbols as desired.
- Annotate a control strategy by adding objects, such as a bitmap drawing or a text file.
- Link or embed any OLE file, such as a Microsoft Word document or Excel spreadsheet.

The Control Drawing-to-GUS Export Tool is an option that enables you to convert TPS Builder control drawings to GUS Display Builder (.pct) drawing files.

2.8 Data Transfer in TPS Builder

Description

Data transfer is the process of readying TPS Builder data for the control system on which it will reside, or making control system data accessible to TPS Builder. The following TPS Builder functions support data transfer.

Downloading

Configuration data for the following entities can be downloaded from TPS Builder directly to the TotalPlant Network (TPN) or Fieldbus network:

- PM family (PM, APM, HPM, and EHPM) point and node configuration data
- Network Interface Module (NIM/ENIM)
- Fieldbus Link

In this document, refer to "Downloading Entities" in the "Creating an EB or EL File" section.

Backbuilding

This function is the importing of TPS system entity data, from Exception Build (EB) JS files, or other files, into the TPS Builder database. These entities appear in the form of TPS Builder strategy blocks (and in the case of Logic and Device Control points, strategy modules), within a selected project. Data that can be backbuilt into TPS Builder include point and node data, and custom data segments.

File Type	Description
EB	Exception Build (EB) files are text files that define TPS Network entities by listing their parameter names and values.
JS	JS files are source text files that correspond to logic blocks configured for each Data Hiway Multifunction or Advanced Multifunction controller.
CL	Control Language (CL) files contain custom data segments (CDS) and parameter lists that can be backbuilt.



REFERENCE - INTERNAL

In this document, refer to the "Importing TPS Entity Data from TPS Files (Backbuilding)" section. Begin with the "Overview."

Forward Building

This function is the creation of a text file, named with an extension of .EB, of TPS system entity (node and point) parameter data in a TPS Builder project. The EB file follows the syntactic and format specifications required by the TPS system platform.

This function also allows creation of a text file, named with an extension of .JS, for Logic Blocks of Multifunction controllers. The JS file follows the syntactic and format specifications required by the Multifunction Controller.



REFERENCE - INTERNAL

EB files: In this document, refer to the "Importing Creating an EB or EL File" section. Begin with the "Overview."

JS files: Refer to the [TPS Builder Hiway Gateway online Help](#).

EL file building

This function is the creation of an Entity List (EL) file for use with functions such as forward building. An EL file is a text file, named with an extension of .EL, containing names of TPS system entities in a TPS Builder project.



REFERENCE - INTERNAL

In this document, refer to the "Importing Creating an EB or EL File" section. Begin with the "Overview"

LCN Workbook and Point Builder data

LCN Workbook and Point Builder projects can be converted into TPS Builder projects. With R130 and later, this conversion must be performed by backbuilding EB files. Refer to your LCN Workbook or Point Builder documentation for instructions on crating EB files from within those applications.

Importing data from spreadsheet or database

The TPS Builder Application Programming Interface (API) allows you to import data from an existing spreadsheet or database into TPS Builder. An example script shows how to import data from a Microsoft Excel spreadsheet into TPS Builder.



REFERENCE - INTERNAL

Refer to the [TPS Builder Installation, Implementation, and Administration Guide](#).

Transferring files

Moving files to and from the TPS system can be accomplished using Honeywell File Transfer, which is available for the Global User Station (GUS) or network-based Windows Professional or NT or XP personal computers.

2.9 Fieldbus in TPS Builder

Description

TPS Builder functions for TPS Fieldbus allow you to fully configure Fieldbus networks offline, and to download the configuration to the Fieldbus network. Offline configuration includes building of the link schedule and function block schedules.

TPS Builder also allows you to configure the TPN strategies required to support a Fieldbus network, and to download the strategies to the TPS Network.

Fieldbus checking

Fieldbus configuration can be checked, including Fieldbus-specific checks.

Value files

TPS Builder supports the Fieldbus Foundation Common File Format; it is able to create Value files for a device from TPS Builder data. Value files created elsewhere can be used to backbuild values into already existing devices. Device configuration values can be uploaded from the device on the Fieldbus network.

Metadata files

TPS Builder creates Detail display Metadata files from Fieldbus templates for use in TPS Network Detail displays.

Related documents

The following documents include details about fieldbus and how to configure fieldbus devices.

- *TPS Fieldbus Overview*
- *Fieldbus Implementation Guide*

2.10 Control Language in TPS Builder

Description

The Honeywell Control Language (CL) Editor allows you to efficiently write CL programs using Borland Codewright (version 7.5 or later), to which CL functionality has been added. In addition to all of StarBase's features, CL Editor provides

- Complete CL editing
- CL syntax checking, and checking against the TPS Builder database
- Online CL error help
- A TPS Strategy Browser for locating TPS Builder strategy parameters

Syntax checking

Syntax checking is provided for CL for the Process Manager family, Application Module (AM) and A^XM, Multifunction Controller (MC) and Advanced MC.

Accessing CL Editor

If Borland Codewright and CL Editor have been installed, you can access CL Editor from within TPS Builder by associating a CL program with an applicable strategy block, or from **Start > TPS Builder > CL Editor**.

If Codewright and/or CL Editor have not been installed, you can install them as explained in the Codewright documentation and the *TPS Builder Installation, Implementation, and Administration Guide*, "Getting Started with CL Editor", "Installation"

Using CL Editor

To use CL Editor, refer to:

- the Codewright documentation and its online Help. Codewright is supplied with TPS Builder.
- *CL Editor User's Guide*

Note that for TPS Builder R130, if you access CL Editor from within TPS Builder, you must quit CL Editor in order to return to TPS Builder.

3. Using the TPS Builder Workspace

3.1 Overview

Setting up the workspace

TPS Builder provides a workspace in which you can open any number of project, library, and drawing windows.



REFERENCE - INTERNAL

Basic details regarding the TPS Builder workspace are contained in the following document:

TPS Builder Installation, Implementation, and Administration Guide, "Set Up the Workspace"

Basic workspace operations

The TPS Builder workspace is a standard Windows environment in which you can perform any of the following actions to facilitate your work:

- Size and position library, project, and drawing windows.
- Tile or cascade windows.
- Minimize windows.
- Arrange and restore minimized windows.
- Close windows.

Setting up composition views

In the workspace, projects are opened in particular composition views. Refer to the following section and document for details about composition views:

- This document, Fundamentals section: "Composition Views"
- *TPS Builder Installation, Implementation, and Administration Guide: "Set Up Composition Views"*

3.2 Workspace Tips

Saving the workspace

Any time you have an arrangement that you like, do not close the project, library, and drawing windows when you quit TPS Builder. Leave them open for your next session. When you restart TPS Builder, the workspace will be as it was at the time it was closed.

Maximizing drawing size

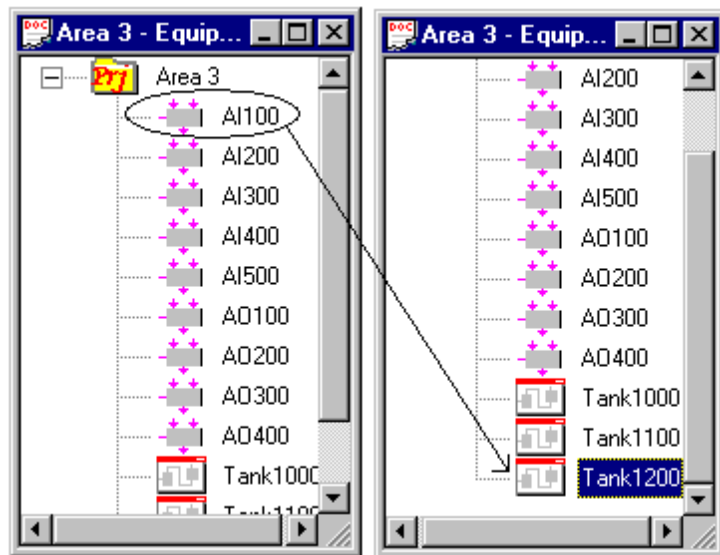
In many cases, you will want drawings to be as large as possible, in which case, you can maximize them. For additional space, you can hide the toolbar and/or status bar temporarily.

Working in a large project

In a large project, you may not be able to see two (or more) needed items at the same time. Consider strategy block AI100 that you want to drag into module Tank1200. The project containing these items is so large that you cannot see them at the same time, even if the project window is maximized.

To solve this problem, open the project in the same view, twice, as shown in this figure.

You will then have two windows for the same project, and you can scroll to make one strategy (AI100) visible in one window and another (Tank1200) visible in the other window. Drag AI100 from its window to module Tank1200 in the other window.



To use this approach, make sure both windows are in the same view.

4. Using Libraries

4.1 Overview

Fundamentals



REFERENCE - INTERNAL

Basic definitions and details regarding TPS Builder libraries are contained in this document's Fundamentals section. Refer to Libraries.

4.2 Basic Library Procedures

Open a library

Follow these steps to open an existing library:

Step	Action
1	From the File menu, choose Open .
2	In the Open dialog box, click the Library option button. A list of available libraries appears.
3	Choose the required library and click OK.

Create a user library



ATTENTION

System templates cannot be moved to a user library, but all specialized templates (including those provided by Honeywell) can be moved.

To create a user library, follow these steps:

Step	Action
1	From the File menu, choose New > Library .
2	In the New Library dialog box, enter a name of up to 50 characters and click OK .
3	Open the libraries containing the templates you want to place in the new library.

4. Using Libraries

4.2. Basic Library Procedures

Step	Action
4	Drag a template to be placed in the new library to the library so that its name is highlighted
5	Once the name is highlighted, release the mouse button.
6	Repeat for other templates as needed.

Enter library identification details

Every TPS Builder user library includes unique identification information, which can be useful for tracking and library management. You enter some of the information and TPS Builder records some of the information presented in this dialog box.

To enter user library identification details:

Step	Action
1	Create or open the user library for which details are to be added.
2	Inside the library window, right-click the library name. On the resulting menu, choose Identification .
3	In the Identification dialog box, complete one or more of the following actions: • Add a description. • Modify the library name or author. Note: Modifying the name here will modify it in the library window. • Enter a version number using your numbering scheme. • Select the TPS Network release to be applied to the library (such as R520 or R600).
4	When the identification details are completed, click OK to add these details to the library.

For system libraries, you can select the TPS Network release to be applied to the library (such as R520 or R600).

Add user-specific Help to a library

A WinHelp topic can be associated with any TPS Builder library. In this way, you can add important, site-specific information and instructions, or personal usage notes, regarding the use of the library.

Step	Action
1	Using any Help authoring program (for example, Microsoft Help Workshop or RoboHELP by eHelp Corp.), create one or more WinHelp files containing topics specific to the TPS Builder library(ies) of interest. Maintain the Help source files in a location of your choice.
2	Place the compiled Help file(s) in the HWIAC/databases/TPS Builder/data directory (or equivalent location, if you did not use the default installation). Include the .hlp file and related .cnt (contents) file if one exists. Important! Place the files in the specified directory so that they will be included in any migration or export/import activities.
3	Identify the Help file name, plus numerical topic ID for each library Help topic to be used.
4	Open the library to which Help is to be associated.
5	Inside the library window, right-click the library name. On the resulting menu, choose Identification .
6	In the Identification dialog box, enter the Help file name and topic id to be associated with the library. This entry is not case-sensitive, and it is not necessary to include the ".hlp" extension.
7	Click OK .

View user-specific library Help

Once you have added Help to a library, you can view it by using these steps:

Step	Action
1	Open the library of interest.
2	Inside the library window, right-click the library name. On the resulting menu, choose Library Help . The Help topic will open.

4. Using Libraries

4.2. Basic Library Procedures

Save a library

No special action is needed to "save" a library. When a library is created, it is saved. As soon as templates are added to the library, they are part of it.

Delete a user library

Follow these steps to delete a user library:

Step	Action
1	Move or delete all templates from the library before. Only user libraries can be deleted (a library must be empty before you can delete it).
2	From the File menu, choose Delete .
3	In the Delete dialog box, select the Library option.
4	In the User Libraries list, select the library/libraries to be deleted.
5	Click OK .
6	In response to the resulting confirmation message, click Yes to delete the library.

Close a library

To close a library, click the **X** button in the upper right corner of the library window.

4.3 Working with Templates in Libraries

Add templates to a library

Templates are added to libraries by one of two methods:

- Creating new templates within a library. Refer to the Using Templates section, “Specialize a template,” in this manual.
- Dragging templates from one library to another. Refer to Using Templates section, “Moving Templates,” in this manual.

Return templates to system library

If you have moved templates to user libraries and want to return them to the system library they were originally a part of, then drag the template from the user library to the correct system library. (It must be the correct library. You cannot place a PM-family template in the AM library, for example.)

TPS Builder will place the template in its correct location in the library hierarchy.

Delete templates from a library

Under certain circumstances, templates can be deleted from libraries. For templates that can be deleted, one template at a time can be chosen for deletion. Refer to Using Templates section, Deleting Template.

4. Using Libraries

4.3. Working with Templates in Libraries

5. Using Templates

5.1 Overview

Fundamentals



REFERENCE - INTERNAL

Basic definitions and details regarding TPS Builder templates are contained in this document's Fundamentals section. Refer to: "Templates" in this manual.

Template usage

Templates are used to make strategies, as explained in "Working with Strategies," Creating Strategies - Basic Procedures.

To determine template usage, follow these steps:

Step	Action
1	Right-click the template and choose Identification on the resulting menu.
2	Click the Dependencies button.
3	When finished, click Close to close the Template Dependencies dialog box.
4	Click OK or Cancel to close the Identification dialog box.

5.2 Accessing and Creating Templates

Access templates

All templates are contained in libraries. To access a template, open the library that contains it (**File > Open > Library**).

If the library contents are collapsed, expand them by clicking the + (plus) sign to the left of the library name. Continue to expand library contents until you locate the needed template.

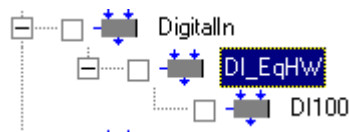
To access the elements (template blocks and/or modules) of user-created template modules, double-click the template module to open its drawing. You cannot view/access template elements contained in a template module in a library.

Specialize a template

Follow these steps to specialize a template (block or module):

Step	Action
1	Open the library containing the template you want to specialize. Note: If any template drawings are open that contain specializations of the template you are going to specialize, close them.
2	In the library, right-click the template and choose Specialize from the resulting menu.
3	In the Specialize Template dialog box, enter a name for the new template and click OK.

In the following example, DI100 is a specialized template. Note the hierarchy established in the library.



Note: If you are specializing a template module, open the new module drawing and rename all of the templates contained in the module, according to your naming conventions.

Create a template module

Template modules are kept in libraries, for easy access during project creation. You will probably keep template modules in a user library, or in the Module library.

Follow these steps to create a template module to be used as a reusable, generic control engineering scheme:

Step	Action
1	Open the libraries containing the templates you want to use. If you will keep the new template module in a user library, open that library as well.
2	Size and position the library windows in a manner that is convenient for dragging templates from one library to the other.
3	Specialize a template module. If it is to reside in a library other than the one in which it is created, move it to that library.
4	Double-click the new template module to open its drawing.
5	Add template blocks and/or modules to the drawing by dragging them from the open libraries. If desired, rename all items added to template.
6	When the template module is populated as desired, identify any connections that you want to make in the template module. Connections created in a template module are template defining, and will not be modifiable in any strategies created from the template (nor in any templates specialized from it). Note: The connection relationship between template blocks is established, without actually configuring the specific connection-related parameters.
7	Choose Save from the File menu.



ATTENTION

Template blocks and/or modules added to the drawing are named automatically by TPS Builder. You will probably want to rename these items using your own naming conventions.

5. Using Templates

5.2. Accessing and Creating Templates

Make connections in template drawings

You can make connections in template drawings. Note:

- Actual connection parameters are not configured.
- When you create a strategy from the template module, the connections parameters are configured.

Connections are discussed in the “Drawing Connection Overview” section of this document.

View template module contents

The contents of a template module can be viewed only in a drawing (not in a library). To view a template drawing, follow these steps:

Step	Action
1	Open the library containing the template module of interest.
2	Double-click the module icon.
3	If the Open Drawing dialog box appears, double-click the view in which you want to open the drawing.

5.3 Modifying Templates

Basic template identification

Templates include unique identification information, which can be useful for tracking and record-keeping purposes. Identification details are as follows:

Type of Template	Identification Details
System	You can view, but not enter, system template identification details (Name, Description, Author, or Version). If you modify parameter values on a system template, the Last Modified By and Last Modified Date will change in accordance with system information regarding the person and date/time of the change. You can designate a template to be used for backbuilding. Only one template per point type can be chosen for backbuilding purposes. Refer to the Backbuilding section of this document, How to choose a template for backbuilding.
Specialized	You can include unique identification information for specialized templates: • Name • Author • Description • Version

How to enter template identification details

To include unique identification information for specialized templates, follow these steps:

Step	Action
1	Open the library containing the specialized template for which identification details are to be added.
2	Right-click the template name and on the resulting menu, choose Identification .

5. Using Templates

5.3. Modifying Templates

Step	Action
3	In the Identification dialog box, complete one or more of the following actions: • Add a description. • Modify the template name or author. Note: Modifying the name here will modify it in the library window. • Enter a version number using your numbering scheme.
4	When the identification details are completed, click OK to add these details to the template.

How to view template default parameter values

Every system template includes default parameter values. These are values that will be included unless you change them. To view default values:

Step	Action
1	Open a library.
2	Right-click a system template that has not been modified, and on the resulting menu, choose Parameters .
3	Scroll on each tab of the configuration form, and click different tabs to find a desired parameter.



TIP

To quickly find a specific parameter, set **Options > Parameter Configuration > Alphabetical Flat List** for an alphabetized list of parameters.

Default values source

For TPS system entities, the defaults match those assigned by the system, as defined in the TPS system parameter reference dictionaries.

How to change template values

You can change a value on a template if the value is not template-defined. The changed value will be used on all strategies created from, or templates specialized from, the template.



CAUTION

It is strongly recommended that you do not modify the Honeywell-provided system templates (although nothing will prevent you from doing so). The recommendation is made because you might inadvertently add too many constraints, such as membership definition or defining parameters as template defining.

It will be much easier to leave the Honeywell-provided template unmodified, than to try to "undo" the constraints that have been put in place. If you leave the system template unmodified, a "clean" template will always be available.

To change a template value, follow these steps.

Step	Action
1	Double-click the template in which you want to change values.
2	In the parameter configuration form, locate the parameter for which you want to change a value.
3	Enter or choose an appropriate value.
4	Repeat as needed for other parameters.
5	When finished, click OK .

5. Using Templates

5.3. Modifying Templates

-
- 6 If you want a particular parameter's values to be template-defined (unchangeable on the strategies and specialized templates created from the template), then you need to set the template-defining attribute for the parameter(s).

Refer to the following discussions in this document for details on the template-defining attribute:

- TPS Builder Fundamentals section, Templates
 - Improving Configuration Consistency and Efficiency section:
 - Template-Defining Fundamentals
 - Configuring the Template-Defining Attribute
-

How to recall a default value

If you change a default value in a template, you can recall the default value by placing your cursor in the parameter field and pressing Ctrl+D.

If no default exists, the pressing Ctrl+D will not change the entry. For example, no default exists for PTDESC. The field is initially blank. If you add something to this field, pressing Ctrl+D will not remove it.

How to change composition view membership



ATTENTION

You can change the view membership only for a node or a specialized template that has *not* been used to create strategies or other specialized templates.

To change the view membership of a node or specialized template that has not yet been used, follow these steps:

Step	Action
1	Right-click the template name and on the resulting menu, choose View Membership .
2	Make changes as needed in the View Membership dialog box.
3	Click OK .

5.4 Moving Templates

Which templates can be moved

All specialized templates (including those provided by Honeywell) can be moved from one library to another as follows.

A template **can** be moved:

- From system library to user library
- From user library back to its own system library
- From user library to user library, with certain constraints as discussed here

A template **cannot** be moved:

- From system library to system library

Hierarchy considerations

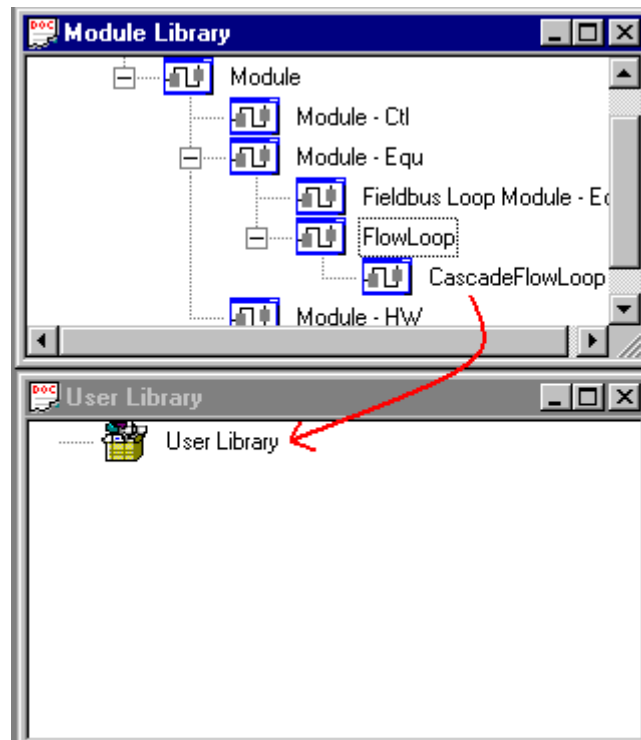
If you move a specialized template, it will move, but its "parent" will not move (if you do not select it). If you move a parent, templates specialized from it will move as well. This is true even if you have moved the specialized templates to other libraries. (All children will be reunited with their parent in its new location.)

5. Using Templates

5.4. Moving Templates

In the following example,

- CascadeFlowLoop can be moved by itself, or
- You can move FlowLoop, in which case it and CascadeFlowLoop will both move.



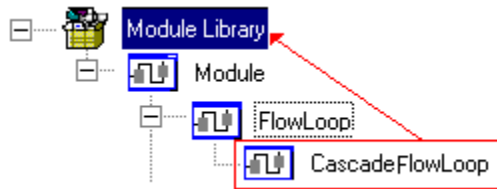
Specialization relationship remains

Moving templates does not eliminate specialization relationships. Therefore, if you make changes to a template, the changes will occur in all templates specialized from it, no matter where they reside.

In addition, moving related templates to a common library will cause them to be displayed in their hierarchical relationship (as the two flow loop templates appear in the previous example).

Breaking a specialization relationship

To break the relationship between a specialized template and its parent, drag the specialized template onto a new parent. For example, in the following example, dragging CascadeFlowLoop onto Module will break the relationship with FlowLoop:



If you do this, you can move both flow loop templates independently of one another.



ATTENTION

If you break a specialization relationship in this way, you cannot re-associate the template with its original parent.

How to move a template

To move a template, follow these steps:

Step	Action
1	Open the libraries that contain/will contain the template to be moved (the source and destination libraries).
2	Size and position the library windows in a manner that is convenient for dragging the templates from one library to the other. Note: If moving template modules, ensure that all template drawings related to the templates you are going to move are closed. This includes any template(s) specialized from the one(s) you want to move.
3	Select a template in the source library.
4	Drag the selected template to the destination library.



ATTENTION

It is important to remember this difference in dragging templates:

- Dragging to a project makes a copy of the template (creates a strategy from the template).
- Dragging to a library moves the template.

5.5 Deleting Templates

What can be deleted?

Templates can be deleted if they are specialized, but have no template subordinates.

"Template subordinates" means templates specialized from the template you want to delete. These subordinates may be in any library -- TPS Builder maintains the specialization relationship across libraries.

Note that although specialized templates provided by Honeywell can be deleted if they have no subordinates, in most cases should not be.

Only one template can be deleted at a time.

What cannot be deleted?

None of the following types of templates can be deleted:

Template Type	Comments
System	Honeywell-supplied, non-view-assigned templates.
Specialized with template subordinates	See What can be deleted?
Certain templates, if strategies have been created from them	These templates can be deleted only if the strategies can also be deleted. See the following discussion of template-strategy relationships.

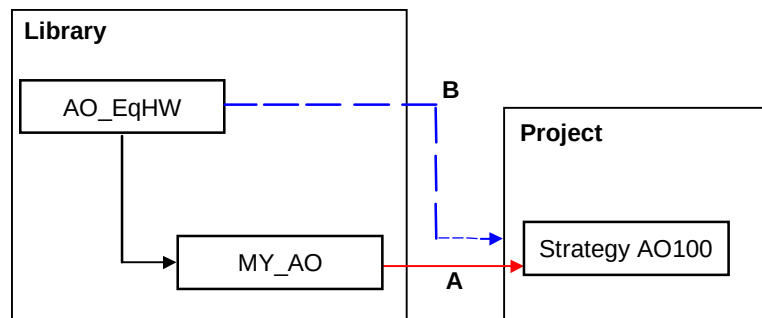
What happens to related strategies?

If a specialized template has no subordinates, but has been used to create strategies, you can delete it in most cases. You must decide whether

- to delete the strategies along with the template, or
- to reassign another template association to the strategies. This option works as long as the composition view is the same.

Example

In the following example, you want to delete the MY_AO template. If you choose to keep the strategies created from it, then AO100 will become associated with AO_EqHW after you delete MY_AO. Otherwise, AO100 will be deleted when MY_AO is deleted.



A = Association prior to deletion.

B = New association if you do not delete items created from the template you are deleting.

This example assumes that all three items share the same composition view.

Deleting template modules

Deleting a template module will delete all of its contents (unlike strategy modules, where you can delete the module but save its contents.)

How to delete a template



CAUTION

No "undo" function exists for deleting templates. Pay careful attention to your actions so that you delete only what you want to delete.

To delete a template, follow these steps:

Step	Action
1	Read the considerations in the previous paragraphs and decide what to do about any associations to the template to be deleted. Also, if a template module is to be deleted, ensure that you want to delete all templates contained in the module.
2	In a library, right-click, the template to be deleted.
3	In the resulting menu, choose Delete .
4	In the resulting dialog box, choose whether or not to delete related strategies. A confirmation message appears.
5	If needed, click Show Dependencies to view the template's dependencies. Click Close to return to the confirmation message.
6	Click Yes to delete the template.

6. Working with Strategies

6.1 Overview

Fundamentals



REFERENCE - INTERNAL

Basic definitions and details regarding TPS Builder strategies are contained in this document's Fundamentals section. Refer to Strategies.

Basic strategy identification

Every strategy block and module can include unique identification information, which can be useful for tracking and project management.

6.2 Creating Strategies - Basic Procedures

Create a strategy block

To create a strategy block follow these steps:

Step	Action
1	Open the project or drawing in which you want the strategy block to reside.
2	Open the library containing the template you want to use.
3	Size and position the windows in a manner that is convenient for dragging the needed template from the library to the project or drawing window. Note: • If creating the strategy block in a module in the project, make sure the module is visible. • If creating the strategy block in a drawing, make sure the portion of the drawing where you want to place the block is visible.
4	Drag the template to the project or the drawing. If placing the strategy block in a module in a project, make sure the module name highlights before releasing the mouse button.

Create a strategy module

You can create a strategy module in four basic ways:

Method	Description	Reference
In a project	Manually drag template or strategy blocks (or modules) to a module in a project. This is a quick way to establish the basic contents of the module without defining the layout of the drawing.	Create a module in a project
In a drawing	Manually drag template or strategy blocks (or modules) into a module drawing. This allows you to create your drawing at the same time you create the module.	Creating and Modifying Drawings

Method	Description	Reference
By Module Generation	If you are creating a project that includes a number of strategies that have already been configured with connection relationships, you can establish an "empty" module and TPS Builder will populate it nearly automatically, based on the connection relationships.	Module Generation
Use a Module Template	If a particular strategy grouping could be used several times, you can create a template module, then use that template module to create strategy modules.	Create a Module Template

Create a module in a project

This procedure allows you to create a new strategy module and add strategy blocks and/or modules to it quickly. When taking this approach, the strategies are not positioned correctly within the module's drawing (you will need to do that as a separate step). In most cases, therefore, you will use the drawing drag-and-drop approach so that you can create the strategy module and drawing at the same time.

Follow these steps to create a strategy module within a project:

Step	Action
1	Plan the module, identifying the composition view and strategies to be used. The strategies may already be in the project, or you may create them during this procedure.
2	Open the Module library and library(ies) containing the templates you want to use.
3	Size and position the project and library windows in a manner that is convenient for dragging templates from the library(ies) to the project.
4	In the Module library, locate the template module to be used. Remember to use one that has view membership matching the project/module.
5	Drag the template module from the library to the project. Position the cursor over the project name so that it is highlighted; or if you are adding a module to an existing module, position the cursor over the existing module name so that it is highlighted. Make sure the project or module name is highlighted. If not, TPS Builder may place the module incorrectly.

6. Working with Strategies

6.2. Creating Strategies - Basic Procedures

Step	Action
6	Once the name is highlighted, release the mouse button.
7	When the New Strategy dialog box appears, enter a name for the new strategy module and click OK . The name must be unique within the project.
8	Now, add strategies to the new module:
<u>If you want to...</u>	<u>Then...</u>
Move strategies that are in the project	Choose one or more strategies (using click, Ctrl+click, or Shift+click) and drag them into the new module so that the module name highlights.
Add new strategies	Drag one template at a time into the new module so that the module name highlights.

View module contents

To view the strategies in a module in a project, expand the module by clicking the  to the left of its icon. Strategies are listed alphabetically within the module.

Enter strategy identification details

Follow these steps to enter strategy identification details:

Step	Action
1	Open the project containing the strategy (block or module) for which identification details are to be added.
2	Right-click the strategy name and on the resulting menu, choose Identification .
3	In the Identification dialog box, complete one or more of the following actions: • Add a description. • Modify the project name or author. Note: Modifying the name here will modify it in the project window. • Enter a version number using your numbering scheme.
4	When the identification details are completed, click OK to add these details to the strategy.

Duplicate a strategy

To make a copy of a strategy block or module, follow these steps:

Step	Action
1	Right-click the strategy to be duplicated.
2	On the resulting menu, choose Duplicate. TPS Builder will make a copy of the strategy, assigning a new name (and new names for any components, if it is a strategy module).
3	Rename the new strategy according to your naming conventions.

6.3 Generating a Module Automatically

Description

If you are creating a project that includes a number of strategies for which connection relationships have been configured, you can populate an existing module ("empty" or containing other strategies) nearly automatically. The generation is based on the connection relationships.

This method is particularly useful when many TPS Network points have been or will be backbuilt as strategy blocks into TPS Builder.

Overall process

As described in the following paragraphs, the main stages of module generation include:

- Preparation
- Configuring the drawing and populating the module
- Evaluation

6. Working with Strategies

6.3. Generating a Module Automatically

Prepare for module generation

Follow these steps to prepare a project for Module Generation:

Step	Action
1	Choose the composition view to be used and open the project in that view.
2	Ensure that all strategy blocks and modules to be included are in the project, at the highest level in the project (they cannot be in other modules). These are the strategy blocks/modules with connection relationships already configured.
3	If you are not familiar with the data connection relationships between these strategies, generate a connectivity map so that you can determine the module structure that will best represent these relationships. Refer to PM-Family Connectivity Maps for details.
4	Determine a "starting point" for the module that is the main vantage point from which you want to view inputs and outputs. (For example, one of the starting input points related to a particular valve control.) Note: The starting point must be an individual strategy block, not a module (therefore, this excludes logic and device control points).
5	Create a module, or identify an existing module to be used. Refer to Create a module in a project in this section for details.

Configure the drawing and populate the module

After completing the preparation, follow these steps to configure and populate the module.

Step	Action
1	In your project, locate the starting-point strategy block.
2	Drag that strategy block to the new module; before releasing the mouse button, press and hold the Ctrl key.
3	Release the mouse button, then the Ctrl key.
4	Complete the Generate Module dialog box.

Step	Action
5	Click the Generate button. TPS Builder will evaluate all connection relationships and the module options set, then generate a drawing based on those factors. Note: If you click the Cancel button in the progress indicator while the module generation is in progress, TPS Builder will stop the generation and display an incomplete drawing. To remove it, close it without saving.



TIP

If you do not like the results of the module generation, you can close the drawing without saving it and repeat the process using another starting point.

Evaluate and finalize the module

After you use module generation to create a module, follow these steps to finalize it:

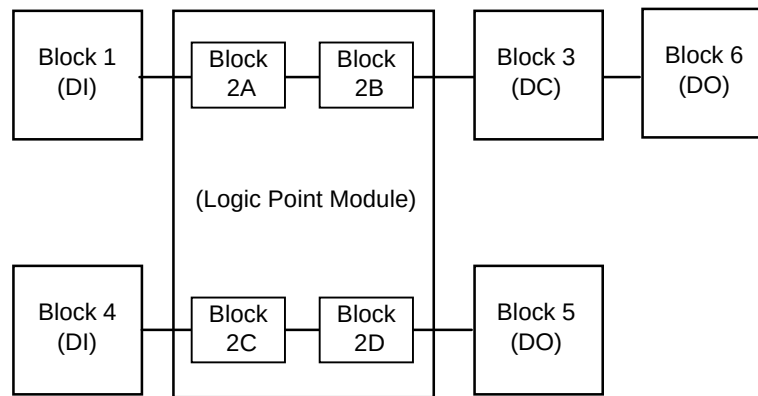
Step	Action								
1	Evaluate the drawing to determine if it meets your needs.								
2	<table><tr><th><u>If the drawing is...</u></th><th><u>Then...</u></th></tr><tr><td>OK</td><td>Go to step 3.</td></tr><tr><td>Not OK</td><td>Close the drawing and answer "No" when prompted. Try different generation options, such as a different starting point, more drawing columns or fewer connections, and repeat the module generation.</td></tr><tr><td>OK, except that some strategies should not be part of it</td><td>Select one of the strategies, then right-click and choose Remove from the resulting menu. It will be removed from the drawing, but not the project. Repeat as needed.</td></tr></table>	<u>If the drawing is...</u>	<u>Then...</u>	OK	Go to step 3.	Not OK	Close the drawing and answer "No" when prompted. Try different generation options, such as a different starting point, more drawing columns or fewer connections, and repeat the module generation.	OK, except that some strategies should not be part of it	Select one of the strategies, then right-click and choose Remove from the resulting menu. It will be removed from the drawing, but not the project. Repeat as needed.
<u>If the drawing is...</u>	<u>Then...</u>								
OK	Go to step 3.								
Not OK	Close the drawing and answer "No" when prompted. Try different generation options, such as a different starting point, more drawing columns or fewer connections, and repeat the module generation.								
OK, except that some strategies should not be part of it	Select one of the strategies, then right-click and choose Remove from the resulting menu. It will be removed from the drawing, but not the project. Repeat as needed.								
3	When the drawing is acceptable, save it (and add the module to your project) by choosing File > Save and answering " Yes " when prompted.								

6. Working with Strategies

6.3. Generating a Module Automatically

Module generation examples

The following TPS Network example shows a typical connection relationship between digital input, logic, digital composite, and digital output points, in terms of TPS Builder strategies and modules.

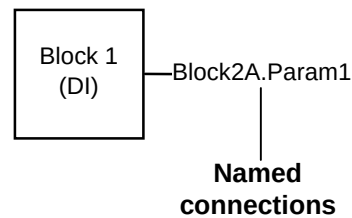


The effects of different module generation options are presented in the following paragraphs, and include:

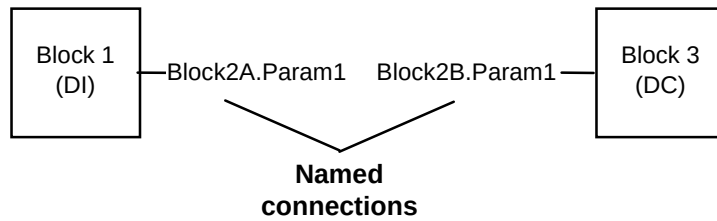
- Number of connection levels examples
- Maximum number of drawing columns examples
- Include modules examples
- Show all connections examples

Number of connection levels examples

This example shows the general result if Block 1 is chosen as the starting point for the module generation, with two connection levels, and "Include modules" not selected.



The same portion of the strategy, with three connection levels and "Include modules" not selected:



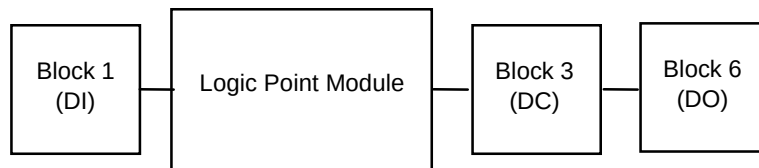
Maximum number of drawing columns examples

The following example is six columns by five rows (number of rows is determined by the number of blocks in the module). The default number of columns is ten.

1	2	3	4	5	6
2					
3					
4					
5					

Include modules examples

This example shows the general result if Block 1 is chosen as the starting point for the module generation, with four connection levels, and "Include modules" selected. Note that the number of connection levels includes those within the module, even though individual blocks in the module are not shown.

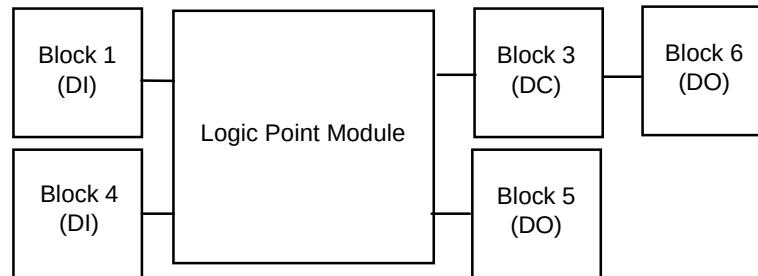


6. Working with Strategies

6.4. Working with a Template Module

Show all connections examples

This example shows the general result if Block 1 is chosen as the starting point for the module generation, with four connection levels, "Include modules" selected, and "Show all connections" selected. Note that the number of connection levels includes those within the module, even though individual blocks in the module are not shown.



6.4 Working with a Template Module

Description

If a particular grouping of strategy blocks and/or modules can be used in several different projects, create a template module and then use it to create strategy modules. This approach increases your efficiency and helps ensure consistency in your configurations.



REFERENCE - INTERNAL

In this document, refer to the [Using Templates](#) section, [Accessing and Creating Templates](#), [Create a template module](#).

How to create a strategy from a template module

Once a template module is created, follow these steps to create a specific strategy module within your project:

Step	Action
1	Open the project in the needed composition view.
2	Open the library containing the template module you want to use.
3	Size and position the project and library windows in a manner that is convenient for dragging templates from the library(ies) to the project.

Step	Action
4	Drag the template module from the library to the project. Position the cursor over the project name so that it is highlighted; or if you are adding a module to an existing module, position the cursor over the existing module name so that it is highlighted. Note: Make sure the project or module name is highlighted. If not, TPS Builder may place the module incorrectly.
5	Once the name is highlighted, release the mouse button.
6	When the New Strategy dialog box appears, enter a name for the new strategy module and click OK. The name must be unique at its level within the project.
7	Your module is populated with strategies defined by the template. You can do the following: • Configure the strategies as needed. • Add strategies. • Delete strategies that you have added. You cannot • Delete strategies defined by the template.



ATTENTION

If the template upon which your strategy module is built is changed, then your strategy module will change accordingly.

6.5 Moving Strategies

Move strategies between projects

In TPS Builder R130, you cannot move strategies from one project to another.

Move strategies between modules

Within a single project, you can move one or more strategy blocks or modules from one module to another (or to the project level). To move strategies, follow these steps:

Step	Action
1	Select, then drag a strategy to be moved to its new location. You can select multiple strategies, if desired, using Ctrl+click or Shift+click methods.
2	Drag until the targeted module (or project) name highlights.
3	Once the target name highlights, release the mouse button.



ATTENTION

Connections made before a strategy is moved remain unchanged, and are shown as named connections in drawings.

6.6 Associating/Disassociating a Program with a Strategy

Description

In TPS Builder, you can associate Honeywell Control Language (CL) programs with strategies representing:

- Application Module Custom, Regulatory, or Switch data points
- PM-family Process Module (PROCMOD) data points

The CL program may have been created in the past, or you may create or edit it using TPS Builder's CL Editor. CL Editor can be accessed from within TPS Builder, or from the **Start** menu.

If you did not initially install CL Editor, refer to the *TPS Builder Installation, Implementation & Administration* guide, Install CL Editor in the Software Installation section.

File location recommendation

It is strongly recommended that you keep TPS Builder-related programs in the *Projectx\program* directory, where *Projectx* is the name of your project. (The complete directory path depends on your TPS Builder installation.)

If the program files are not in that directory, move or copy them before associating them with strategies. (You can maintain these program files in other locations if desired.)

Create and associate a new program

Follow these steps to create and associate a program from within TPS Builder.

Step	Action
1	Open the project or drawing containing the strategy to which a program is to be associated.
2	Right-click the strategy and choose Programs on the resulting menu.
3	In the Programs dialog box, click New .
4	In the resulting dialog box, enter a name for the new file. You do not need to include the .cl extension, as TPS Builder will do that. CL Editor/Codewright starts, and a new file is created, with the name you entered. This file is placed in the <i>Projectx\program</i> directory, where <i>Projectx</i> is the name of your project. (The complete directory path depends on your TPS Builder installation.)
5	Create the CL program, check/correct it using the CL Checker, then save it. Refer to the <i>CL Editor User's Guide</i> for details.
6	Exit the application and return to TPS Builder. The file will be included in the Associated Programs listing in the Programs dialog box.
7	When finished, click OK to implement the actions taken.

**ATTENTION**

Click **Cancel** in the **Programs** dialog box to void the addition of the program (and any other actions you may have taken).

Cancel does not affect the creation of a program, only whether or not it is associated with the strategy.

6. Working with Strategies

6.6. Associating/Disassociating a Program with a Strategy

Associate an existing program

To associate an existing CL program with a strategy in TPS Builder, follow these steps:

Step	Action
1	Open the project or drawing containing the strategy with which a program is to be associated.
2	Right-click the strategy and choose Programs on the resulting menu.
3	In the Programs dialog box, click Add to access the Windows file browser.
4	Locate the file to be used. (If needed, navigate in the file browser to the directory containing the required CL file.)
5	To select a single program: Double-click the file name. The CL program is associated with the strategy and its file name appears in the Programs dialog box. To select more than one program: Click a file name, then use Ctrl+click or Shift+click methods to select others. Click Open to associate all of them at once.
6	When finished, click OK to implement the actions taken.

Edit an associated program

You can edit and save an associated program using CL Editor. Alternatively, you can edit the program from within TPS Builder by following these steps:

Step	Action
1	Open the project or drawing containing the strategy with which the program to be edited is associated.
2	Right-click the strategy and choose Programs on the resulting menu.
3	In the Programs dialog box, select (highlight) the file(s) to be edited.
4	Click the Edit button. Codewright and CL Editor will start and the file(s) will open.
5	Make changes in the file(s) as needed, then save and close them.
6	When finished, exit the CL Editor and return to TPS Builder. Note: You must exit in order to return to TPS Builder.
7	When finished, click OK to implement the actions taken.

Remove/delete a program association

To remove an association between a CL program and strategy in TPS Builder, or to remove the association and delete the program, follow these steps:

Step	Action
1	Open the project or drawing containing the strategy from which the program to be disassociated.
2	Right-click the strategy and choose Programs on the resulting menu.
3	In the Associated Programs listing, select one or more file names.
4	In the Programs dialog box, • click Remove to disassociate the CL program from the selected strategy, or • click Delete to disassociate the CL program and delete the file.
5	When finished, click OK to implement the changes.

6.7 Associating/Disassociating a Document with a Strategy

Description

In TPS Builder, you can associate any type of document with a strategy (block or module). For example, you may want to associate a design document with a strategy. You could associate a document containing maintenance information with a pump strategy.

The document may have been created in the past, or you may create or edit it from TPS Builder.

File name requirements

To create and edit documents from within TPS Builder, the needed file type-application associations must be defined for your computer.

Examples of typical file type associations include ".doc" for Microsoft Word, ".txt" for Microsoft Notepad, and ".xls" for Microsoft Excel. Refer to Microsoft NT/XP Help for more information.

6. Working with Strategies

6.7. Associating/Disassociating a Document with a Strategy

File location recommendation

It is strongly recommended that you keep documents to be associated with strategies in the *Projectx\document* directory, where *Projectx* is the name of your project. (The complete directory path depends on your TPS Builder installation.)

If the documents are not in that directory, move or copy them before associating them to strategies. (You can maintain these documents in other locations if desired.)

Create and associate a new document

To create and associate a new document from within TPS Builder by following these steps:

Step	Action
1	Open the project or drawing containing the strategy with which the document is to be associated.
2	Right-click the strategy and choose Documents on the resulting menu.
3	In the Document dialog box, click New.
4	In the resulting dialog box, enter a name for the new file. Include an extension that is associated with the application you want to use to create the document (such as ".doc" for Microsoft Word).
5	Enter information as needed into the file, then save it.
6	When finished, exit the document application and return to TPS Builder. The file will be included in the Associated Documents listing. Note: You must exit in order to return to TPS Builder.
7	When finished, click OK to implement the actions taken.



ATTENTION

Click **Cancel** in the **Documents** dialog box to void the addition of the document (and any other actions you may have taken).

Cancel does not affect the creation of a document, only whether or not it is associated with the strategy.

Associate an existing document

To associate an existing document with a strategy in TPS Builder, use the following procedure.

Step	Action
1	Open the project or drawing containing the strategy with which the document is to be associated.
2	Right-click the strategy and choose Documents on the resulting menu.
3	In the Document dialog box, click Add to access the Windows file browser. It should be in the <i>Projectx\document</i> directory, where <i>Projectx</i> is the name of your project. (The complete directory path depends on your TPS Builder installation.)
4	Locate the file to be associated. (If needed, navigate in the file browser to the directory containing the required document file.)
5	To associate a single document: Double-click the file name. The document is associated to the strategy and its file name appears in the Documents dialog box. To associate multiple documents: Use Ctrl+click or Shift+click methods to choose more than one document, then click Open to associate all of them at once.
6	When finished, click OK to implement the actions taken.

6. Working with Strategies

6.7. Associating/Disassociating a Document with a Strategy

Edit an associated document

You can edit and save an associated document using the document's application. Alternatively, you can edit an associated document from within TPS Builder by using the following procedure.



ATTENTION

This procedure starts the application used to create the document (assuming the filename includes a correctly associated extension) and opens the selected file.

If the application supports opening more than one file at a time, then you can select multiple files for editing. If the application supports opening only one file at a time (for example, Microsoft Notepad), then select only one file at a time for editing.

Step	Action
1	Open the project or drawing containing the strategy with which the document is associated.
2	Right-click the strategy and choose Documents on the resulting menu.
3	In the Document dialog box, select (highlight) the file(s) to be edited.
4	Click the Edit button. The related application will start and the file(s) will open.
5	Make changes in the file(s) as needed, then save and close the file(s).
6	When finished, exit the document application and return to TPS Builder. Note: You must <i>exit</i> in order to return to TPS Builder.
7	When finished, click OK to implement the actions taken.



ATTENTION

If you click **Cancel** in the **Documents** dialog box after editing documents, only actions taken in the **Documents** dialog box will be cancelled. The editing of the documents will not be affected.

Remove/delete an associated document

To remove an association between a document from a strategy in TPS Builder, or to remove the association and delete the document file, follow these steps:

Step	Action
1	Open the project or drawing containing the strategy from which the document is to be disassociated.
2	Right-click the strategy and choose Documents on the resulting menu.
3	In the Associated Documents listing, select one or more file names.
4	In the Document dialog box, • click Remove to disassociate the document(s) from the selected strategy, or, • click Delete to disassociate the document(s) and delete the file(s).
5	When finished, click OK to implement the changes.

6.8 Creating Multiple Strategies Using Bulk Build

Description

Bulk building is the process of using Microsoft Excel to efficiently create new strategies that are similar.



TIP

The more you know about Excel, the more efficiently you will be able to create multiple strategies using bulk build.

If you are unfamiliar with Excel, it is suggested that you obtain basic training, or do some reading, before using TPS Builder's bulk building function.

How bulk building works

Bulk building begins with a single strategy (block or module) that models the type of strategies you want to create.

Configuration data for this strategy is imported into Excel, where you can quickly multiply the model, then revise and/or add data for the new strategies to make each strategy unique. Regarding modules, data for the module and all strategies contained within it, is brought into the spreadsheet.

6. Working with Strategies

6.8. Creating Multiple Strategies Using Bulk Build

In Excel,

- Column A becomes populated with the names of parameters exposed on the configuration form at the time of the bulk build.
- Column B contains any specific configuration values present in the model's configuration form.

How to bulk build strategies



CAUTION

You can only create new strategies when using bulk build. Do *not* modify the strategy you bring into Excel as your model. TPS Builder ignores any changes.

To create multiple strategies using the bulk build method, follow these steps:

Step	Action
1	Open the project containing the strategy that will serve as your model.
2	Select the strategy, then from the Edit menu, select Bulk Build . This action starts Microsoft Excel and populates it with TPS Builder data.
3	Familiarize yourself with the layout of the TPS Builder data.
4	Select column B.
5	Position the cursor on a handle and press and hold the Ctrl key. The cursor will become a "double" plus sign when you pass it over the handle.
	
6	Drag the mouse over the number of columns equaling the number of strategies you want to create. Pressing the Ctrl key prevents Excel from automatically incrementing numerical data. Note: This action copies column B and copies it to all columns over which you dragged the mouse.

Step	Action
7	Edit configuration data as needed. For maximum efficiency, use Excel macros for all repetitive tasks (typing the same keystrokes, choosing the same commands, going through the same sequence of actions). Note: As a minimum, every strategy name must be changed so that it is unique.
8	When changes are completed, select Close from the File menu of the spreadsheet.
9	A message will ask you if you want to bulk build the strategies. Select Yes .
	ATTENTION If you select No in response to this message, you will lose your bulkbuilt data. When you select Yes , TPS Builder creates strategies based on the data in the spreadsheet.
10	If errors are found during the process, a message suggests that you view the Error log. Click OK in response to the message. Otherwise, TPS Builder will be the active window.
11	Repeat these steps if needed to bulk build more strategies.
12	When finished bulk building, exit MS Excel.

6.9 Editing Multiple Strategies Using Bulk Edit

Description

Bulk editing is the process of using Microsoft Excel to efficiently modify a selected group of strategies at one time.



TIP

The more you know about Excel, the more efficiently you will be able to edit multiple strategies using this function.

If you are unfamiliar with Excel, it is suggested that you obtain basic training, or do some reading, before using TPS Builder's bulk building function.

6. Working with Strategies

6.9. Editing Multiple Strategies Using Bulk Edit

How bulk edit works

Bulk editing begins with a group of strategies (blocks and modules) that you want to edit, and the selection of the parameters you want to edit.

Configuration data for the selected strategies is imported into Excel, where you can use Excel functionality to efficiently revise it. Regarding modules, only the data related to the module, not the strategies within it, is brought into Excel.

In Excel,

- Column A becomes populated with the names of selected parameters.
- Column B - x become populated with the parameter values for each selected strategy.



CAUTION

You can only edit existing strategies when using bulk edit. Do *not* add strategies (or parameters not selected for editing) in the spreadsheet. TPS Builder ignores such changes.

How to bulk edit strategies

To modify multiple strategies using the bulk edit method, follow these steps:

Step	Action
1	Open the project containing the strategies that are to be edited.
2	Select the strategies to be edited.
3	From the Edit menu, select Bulk Edit .
4	In the Bulk Edit dialog box, choose the parameters that you want to edit, then select OK . Note: This action starts MS Excel and populates the spreadsheet with selected strategy and parameter data.
5	Familiarize yourself with the layout of the TPS Builder data.
6	Edit configuration data as needed.



TIP

For maximum efficiency, use MS Excel macros for all repetitive tasks (typing the same keystrokes, choosing the same commands, going through the same sequence of actions).

Step	Action
7	When changes are completed, select Close from the File menu of the spreadsheet.
8	A message will ask you if you want to bulk edit the strategies. Select Yes .
	ATTENTION If you select No in response to this message, you will lose your bulkbuilt data.
9	If errors are found during the process, a message suggests that you view the Error log. Click OK in response to the message. Otherwise, TPS Builder will be the active window.
10	When you are finished bulk building and/or editing, exit MS Excel.

6.10 Templatize a Strategy

Introduction

Sometimes a set of points is backbuilt and used to produce a control strategy that could become the basis for other strategies. If the new strategies are the same as the original, then the original can be duplicated or used to start a bulk build. However, if the original requires some adaptation, then it would be convenient to produce a template from it. Then the template can be specialized and more strategies can be instantiated from the original and its specializations. The *templatize* function provides for making a template from a strategy.

Before templating a strategy, its drawing should be opened, saved, and closed. This ensures that any template changes are propagated to the strategy before it is templated.

Templatize menu item

Templatize is on the Strategy menu and is available when an eligible strategy is selected in a project window. Certain Fieldbus strategies are not eligible for templating:

- schedule
- function block
- transducer
- VFD

6. Working with Strategies

6.10. Templatize a Strategy

When **Templatize** is invoked, the User Library Selection dialog box is displayed. This dialog box enables the user to select a user library for the new template. When the **OK** button is clicked, a new template is created from the strategy.

Templatization process

The templatization process creates a template that is specialized from the strategy's original template, but it has all the values of the strategy. If the strategy is a module, then a template is created for each of its components and each new template is a specialization of the strategy's original template. This process is recursive; that is, it continues through all levels of the strategy.

The strategy and all its components are released from their original templates. If the strategy's original template is in a user library, and the new template is placed in the same library, then the new template is a specialization of the original template; otherwise, the new template is a root template and its base template is the original template.

Each component template of the new template becomes the template for its respective component in the strategy; and the original template of each strategy component becomes the base template for the respective new template. This is also done for all levels of the strategy.

All the properties of the templatized strategy are copied to the template. This includes fonts, colors, embedded build symbol images, build symbol sizes and positions, displayed parameters, connectors, connection lines, named connections, connection tables, embedded objects, drawing maps, and projected up connectors. Connections to external strategies are not translated to the new template. Template-defined parameters become template-defining. The new template also has the same view membership and attachments as the strategy.

7. Working with Projects

7.1 Overview

Fundamentals



REFERENCE - INTERNAL

Basic definitions and details regarding TPS Builder projects are contained in this document's Fundamentals section. Refer to Projects.

7.2 Creating and Viewing Projects

Choose default view for new projects

By default, TPS Builder will open a new project in Equipment view. If you would like new projects to open in another view, follow these steps:

Step	Action
1	From the Options menu, choose Set Default View .
2	In the list of existing composition views (system and custom), choose the desired view.
Result: The view you choose will remain the default view for new projects until you change it again.	



TIP

Remember that if a new project opens in a view you do not want, you can always open it in the desired view by opening the project and selecting another view.

7. Working with Projects

7.2. Creating and Viewing Projects

Create a new project

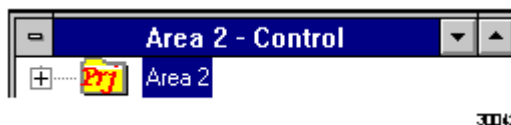
Follow these steps to create a new project in TPS Builder:

Step	Action
1	From the File menu, choose New > Project .
2	Enter a name in the New Project dialog box and click OK .
3	If desired, open the project in another view(s).
4	Add strategies to the project.

Open a project

Use these steps to open a project:

Step	Action
1	From the File menu, choose Open . By default, available projects are shown.
2	In the Open dialog box, choose the needed project(s) and the view(s) in which you want to work.
3	Click OK . The project will appear in its own window, with the project name and view in the window title bar.



Result: The project opens in the TPS Builder workspace. If you chose more than one project and/or more than one view, then the project(s) will open in multiple windows, one for each view.

Add strategies to a project

Follow these steps to add strategy blocks or modules to a project:

Step	Action
1	If the project is not open, open it in the needed view.
2	Open the library or libraries containing the templates you want to use to create strategies for the project.
3	Size and position the project and library windows in a manner convenient for dragging templates from the library(ies) to the project.
4	In a library, locate the template for the strategy block or module to be added to the project. Ensure that the view of the template matches the view of the open project.
5	Drag the template from the library to the project. Position the cursor over the project name so that it is highlighted.
6	Once the project name is highlighted, release the mouse button.
7	Enter a unique name for the strategy in the New Strategy dialog box. The name must be unique at its level within the project.
8	Repeat for all strategies to be added to the project.

Save a project

No special action is needed to "save" a project. When a project is created, it is saved. As soon as strategies are added to a project, they are part of it. (This behavior contrasts with that of a module drawing, which must be saved.)

Close a project

To close a project, click the **X** button in the upper right corner of the project window.

7.3 Deleting Projects and Project Contents

Delete a project

A project does not need to be empty to be deleted. If you delete a project containing strategies, all of the contained strategies are deleted at the same time.

Follow these steps to delete a project:

Step	Action
1	From the File menu, choose Delete .
2	In the Delete Projects dialog box, projects are listed. Select the project to be deleted. You can select more than one project by using Shift+click or Ctrl+click.
3	Click OK .
4	In response to the resulting confirmation message, click Yes to delete the project(s).

Delete strategy blocks from a project



ATTENTION

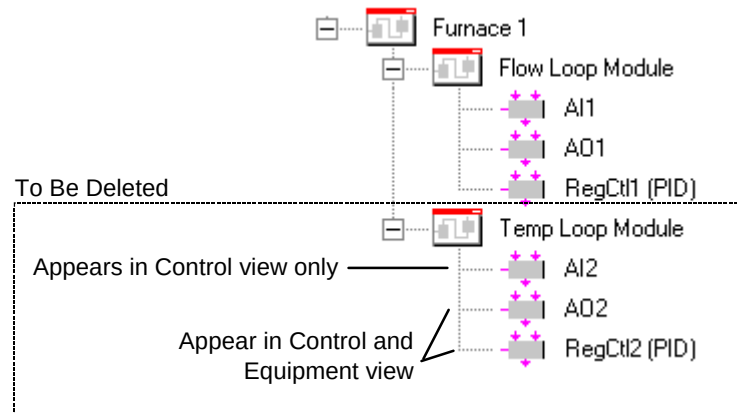
To delete strategies from a project, no drawing containing the strategy can be open. If a drawing is open, delete the strategy from the drawing. It will be deleted from the project.

To delete a strategy from a project, follow these steps:

Step	Action
1	In a project, right-click the strategy to be deleted.
2	On the resulting menu, choose Delete .
3	In the resulting dialog box, click OK .

Delete strategy modules from a project

To delete a module, you must decide whether to delete all strategies (blocks and modules) in the module, or to keep those having additional view memberships. Consider the following example.



- If you choose to delete all components of the Temp Loop, regardless of view membership, the module, AI2, AO2, and RegCtl2 will be deleted from the project
- If you choose to keep the components that appear in other views, then Temp Loop and AI2 will be deleted; AO2 and RegCtl2 will remain in the project. They will be placed directly under the project (not in any module).

To delete a strategy module, follow these steps:

Step	Action
1	In a project, right-click the module to be deleted.
2	In the resulting menu, choose Delete .
3	In the Delete Components Strategy dialog box, choose the type of deletion, as explained previously.
4	In response to the confirmation message, click Yes .

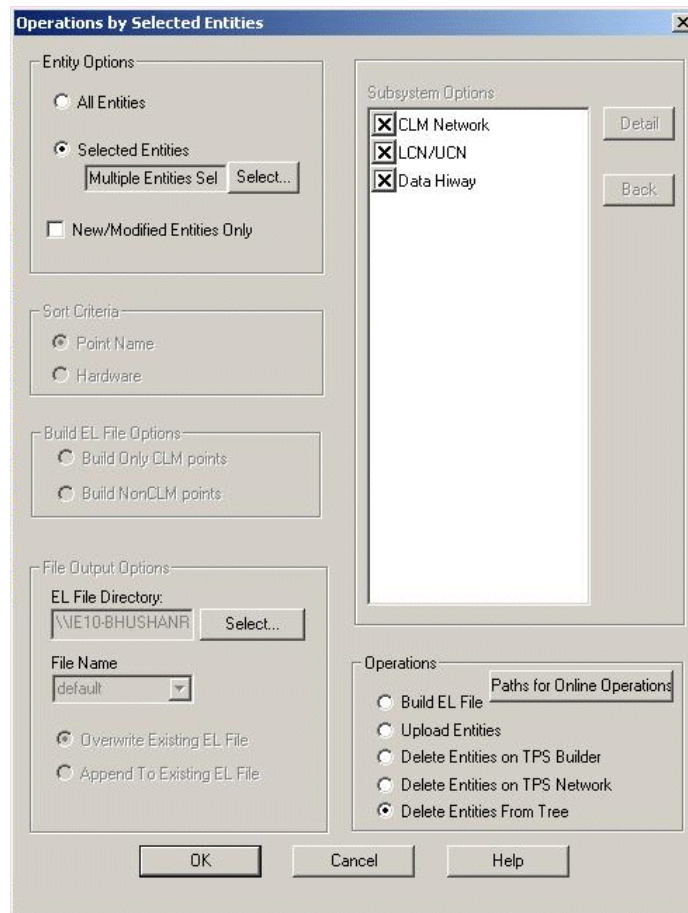
7. Working with Projects

7.3. Deleting Projects and Project Contents

Delete Multiple Strategies from Tree View

To delete multiple strategies from tree view, follow these steps:

Step	Action
1	From the Tree view, select the strategies to be deleted.
2	From the Tools menu choose Operations by Selected Entities .
3	In the Operations by Selected Entities dialog box, select the option Delete Entities From Tree .



- 4 Click **OK**.

7.4 Check a Project

Description

TPS Builder validates TPS entity data in a project according to the rules presented in the TPS system parameter reference dictionaries, except in cases where this information differs from the operation of the PED (Parameter Entry Displays).

What is checked

Specific checks that are performed include:

- Unique name checking
- Range checking against fixed limits
- Enumeration checking
- Address validation against node configuration
- Entry validation against exposure rules
- Basic connection checking (whether the parameter and tagname exist only; index checking that the connection is a valid TPS Network connection does not occur)

All entities within a project can be checked at once, or some subset of those entities can be selected for checking.

Errors and warnings are posted in the Checker log.

Validation exceptions

Note the following exceptions to validation checking:

- Tagnames are checked for uniqueness and length only. Other types of errors, such as spaces in the name, are not identified.
- If backbuilding a CDS, PKGNAME is not checked for validity.

7. Working with Projects

7.4. Check a Project

How to check a project

To check a TPS Builder project:

Step	Action
1	Plan your checking: • Determine which project is to be checked. • Decide whether you will check all entities or some subset of them. • If this project has been checked before, decide whether you need to check all entities again, or only new and modified entities.
2	Open the project to be checked.
3	If the entire project is to be checked, select the project name. If only certain entities within the project are to be checked, select them in the project window or skip to the next step and use the Check dialog box (which has several selection options).
4	From the File menu, choose Check. In the resulting dialog, the options will be set based on your choice in step 3. If a project was chosen, All Entities will be chosen. If you selected specific entities in the project window, Selected Entities will be chosen, and "Multiple Entities Selected" will appear as the selection (or a single entity name, if only one entity was chosen).
5	Complete the Check dialog box, modifying entity options if needed. (You can modify any choices you may have made in the project window, if desired.) Also, choose new checking and log overwrite options as needed.
6	When the Check dialog box is complete, click the Check button. The process will begin, and a progress indicator will display checking progress. Two checking stages will be indicated, and when the checking is complete, control will return to the Check dialog box.
7	If any errors are encountered during the check, they will be posted in the Checker log. Click the View Log button to see the log.
8	Make configuration corrections based on log information.

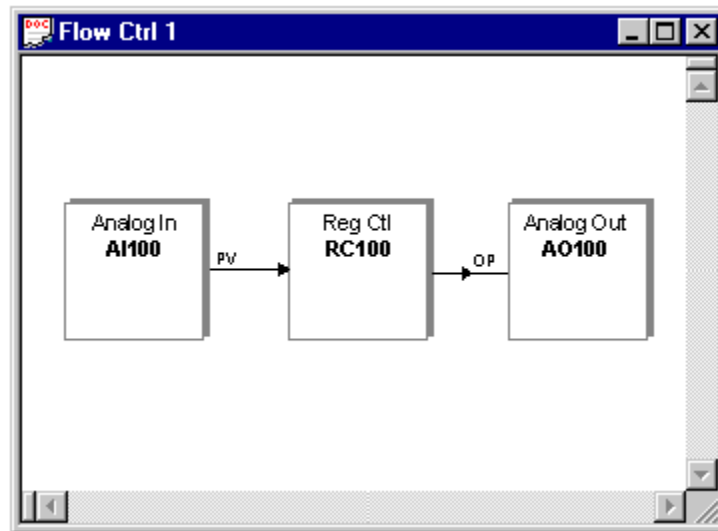
8. Working with Drawings

8.1 Overview

Definition

A drawing is intrinsically associated with Maps. See the section on “Module Drawing Maps” in this document.

A TPS Builder control drawing is a graphical representation of a strategy or template module, and its connections. The following figure is an example of a simple drawing:



Drawing views

A strategy or template module may have multiple control drawings, one per composition view (such as equipment and hardware).

If a module has multiple drawings, only one drawing view can be open at a time. That is, if a module has drawings for both Equipment and Hardware view, only one of them can be open at one time.

8. Working with Drawings

8.1. Overview

For more information on views, refer to the following sections in this guide:

- Fundamentals section, Composition Views
- Using the TPS Builder Workspace, Overview

Drawing Migration

Drawing files created in previous releases of TPS Builder, R100-R131, are automatically migrated to the R140 data structure when they are opened. They remain in the earlier format until they are opened and saved. Future releases of TPS Builder will work in the same manner, opening an existing drawing file and saving it in the format for the current release.

From R140, you can prevent propagating changes when Use Map is set to OFF on the parent template by any one of the following ways:

- 1) Double click to open the drawing. One drawing opens and if Use Map is set to OFF in the parent template, a dialog box pops up to display the error. To avoid this, the errors are logged into the log file and the user checks the log file at the end.
- 2) Right click on the template to invoke Update Derived Drawing option. As there may be more than one drawing that has to be updated and if Use Map is set to OFF on the parent template, an error message appears for each drawing. To avoid this, the errors are logged into the log file and the user checks the log file at the end.
- 3) Select Update Drawings option from the Tools menu. As there may be more than one drawing that has to be updated and if Use Map is set to OFF on the parent template, an error message appears for each drawing. To avoid this, the errors are logged into the log file and the user checks the log file at the end.

Drawing symbol size

In addition to the drawing symbol activities explained in this section, you can set a default size for drawing symbols.

For details, refer to the *TPS Builder Installation, Implementation, and Administration Guide* Software Configuration section "Define Default Size for Drawing Symbols".

3D Blocks in Drawing and Migration

To view the drawings in 3D, choose **Drawing Settings > Three Dimensional Blocks**. A check mark appears corresponding to the option **Three Dimensional Blocks** indicating that the 3D option is selected. When this option is unchecked the drawings are displayed in 2D.

On migrating the database, by default the drawings are displayed in 3D.

Conversion of TPS Builder drawings to GUS Display Builder

The Control-Drawing-to-GUS Export tool is an option that enables you to convert TPS Builder control drawings to GUS Display Builder (.pct) drawings. If you have purchased and installed this add-on, refer to specific Help in the main Help Contents.

8.2 Creating and Modifying Drawings

Introduction

Strategy module drawings are intrinsically tied to Drawing Maps. See the section on “Module Drawing Maps” in this document for more information.

Create a strategy module

The following procedure allows you to create a new strategy module, add strategy blocks and/or modules to it, and create the module’s drawing at the same time.

Step	Action
1	Plan the strategy module, identifying the project it will be a part of, the composition view to be used, and template blocks and/or modules to be included.
2	Open the Module library and library(ies) containing the templates you want to use.
3	If the project is not open, open it in the needed view.
4	Size and position the project and library windows in a manner that is convenient for dragging the template module to the project.
5	Drag the needed template module from the library to the project. In the project window, position the cursor over the project or a strategy module name so that it is highlighted.
6	Once the name is highlighted, release the mouse button.
7	In the New Strategy dialog box, enter a name for the new strategy module and click OK . The name must be unique at its level within the project.

This procedure leaves you with a new strategy module in a specific project. To work with its drawing, double-click the module.

8. Working with Drawings

8.2. Creating and Modifying Drawings

Open a drawing

To open a drawing, follow these steps:

Step	Action
1	Open the library or project containing the template or strategy module whose drawing you want to open.
2	Locate the template or strategy module and double-click it.
3	Size and position the drawing window as needed.

Add strategies to a drawing

Once a strategy module is created, you can continue to add strategies (blocks or modules) to it. To do so, follow these steps:

Step	Action
1	Open the library(ies) containing the templates you want to use.
2	If the project is not open, open it in the needed view.
3	Double-click the module to which strategies will be added. Its drawing opens.
4	Size and position the drawing and library windows in a manner that is convenient for dragging templates into the drawing.
5	Drag a needed template to the open drawing. The new strategy block/module will be added to the drawing as a drawing symbol.
6	A default name is assigned to the new strategy. This name is generally based on the template's name, with a unique numerical suffix added. If desired, rename by right clicking, choosing parameters and changing the NAME parameter.

Step	Action
7	At this point, you have several options, depending on your goals and personal preference. You may take any of the following actions on the strategy to continue: • Drag it to a more accurate location in the drawing. • Right-click the drawing symbol and choose a desired action from the mouse right-button menu (such as parameter configuration, identification, drawing symbol properties). • Double-click a strategy module to open its drawing, and complete that module. • Continue creating new strategies by dragging the needed templates, one at a time, to the drawing.
8	As soon as the drawing contains strategies that will be connected to one another, you can configure the connections. Refer to Drawing Connection Overview.
9	Periodically save the module drawing by choosing File > Save.

Remove strategies from a drawing

You can remove a strategy from a drawing without deleting it from the project. When you do this, the strategy is moved out of the module directly into the project.



ATTENTION

If a template module is used to create a strategy module, the strategy blocks based on the blocks in the template module cannot be removed from the strategy module. They are template defined and therefore "fixed" for all strategies based on that template module.

Strategies you add to this type of module can, however, be removed.

8. Working with Drawings

8.2. Creating and Modifying Drawings

Follow these steps to remove strategies from a drawing:

Step	Action
1	In an open drawing, select the strategy to be removed.
2	Right-click the selected strategy, and on the resulting menu, choose Remove .
3	Choose File > Save .

Delete strategies from a drawing



ATTENTION

If a template module is used to create a strategy module, the strategy blocks based on the blocks in the template module cannot be deleted from the strategy module. They are template defined and therefore "fixed" for all strategies based on the template module.

Strategies you add to this type of module can, however, be deleted.

Follow these steps to delete strategies from a drawing:

Step	Action						
1	In an open drawing, select the strategy to be deleted.						
2	Right-click the selected strategy, and on the resulting menu, choose Delete .						
3	<table><tr><td><u>If you are deleting...</u></td><td><u>Then...</u></td></tr><tr><td>a strategy block</td><td>click Yes in response to the confirmation message.</td></tr><tr><td>a strategy module</td><td>complete the Delete Component Strategies dialog box.</td></tr></table>	<u>If you are deleting...</u>	<u>Then...</u>	a strategy block	click Yes in response to the confirmation message.	a strategy module	complete the Delete Component Strategies dialog box.
<u>If you are deleting...</u>	<u>Then...</u>						
a strategy block	click Yes in response to the confirmation message.						
a strategy module	complete the Delete Component Strategies dialog box.						
4	Choose File > Save .						

Note: If connections were made to or from the strategy you deleted, the line(s) is(are) removed and parameter configuration forms of the remaining strategies are cleared of entries related to the connection.

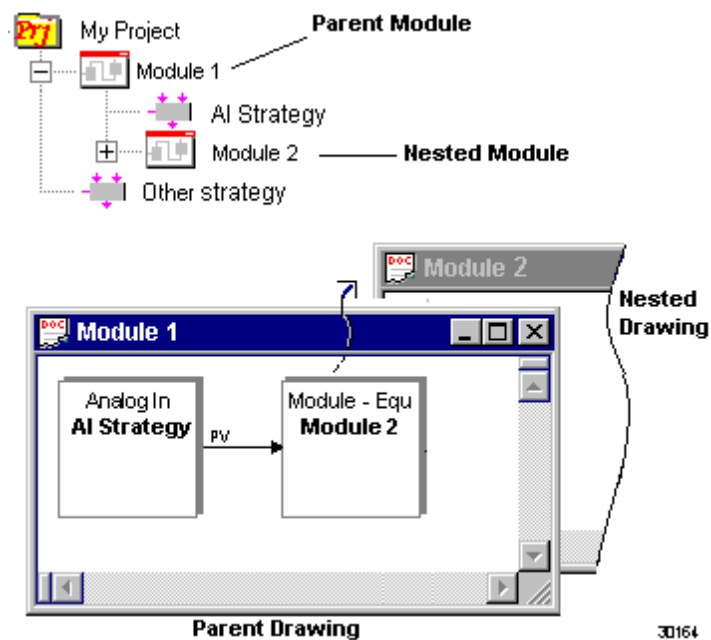
Save drawing

Plan to save periodically as you create or modify a drawing. To save a drawing:

Step	Action
1	Ensure the drawing is the active window.
2	From the File menu, choose Save .
3	If the drawing is within another drawing (is “nested”), and both drawings are open, see the following paragraph.

Save a nested drawing

A drawing may be within another drawing (be “nested”), as shown in this figure:



8. Working with Drawings

8.2. Creating and Modifying Drawings

Nested drawings cannot be saved “by themselves” if the parent drawing is also open. To save a nested drawing:

Step	Action
1	Make the parent drawing the active window.
2	From the File menu, choose Save .



TIP

If you are going to create or make changes to both a parent drawing and a drawing nested within it, take one of two approaches:

- Open only one of the drawings at a time, work on, and save it, or
 - If both drawings are to be open at the same time, work on one drawing at a time and save it before moving to the other drawing.
-

8.3 Inserting Objects into a Drawing

Types of associations

Information and documents can be associated with TPS Builder drawings and strategies as follows:

Type of Association	Usage and Notes
Embedded documents in a drawing	Mainly used for annotation purposes. An embedded document is stored in the control drawing itself and therefore is self-contained in TPS Builder. This method increases the size of the drawing.
Linked documents in a drawing	Used to provide additional information about the control scheme represented by the drawing. A linked document is kept in its original place, only a link is stored in the drawing. This method is typically used if there are multiple users for the document, for example, a maintenance document for a pump. If you have ten of those pumps, you may want to link to one document.

Insert by pasting

To paste material you have copied from an OLE-compliant application (including a file name that you have copied in My Computer or Explorer) into a TPS Builder drawing:

Step	Action
1	Start the needed application (for example, Microsoft Word).
2	Open the file containing the material you want to paste into your TPS Builder drawing; or, for complete file, locate the file in My Computer or Explorer.
3	Select the material (or file name) and copy it. It is placed on the clipboard.
4	In TPS Builder, open the drawing you want to contain the material on the clipboard.
5	From the Edit menu, choose Paste (or press Ctrl+V). • If you paste a complete file, it becomes an embedded (not linked) object. • To paste a link to a file, choose Paste Link.

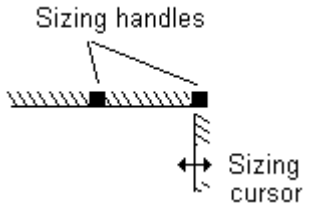
8. Working with Drawings

8.3. Inserting Objects into a Drawing

Step	Action
6	To re-position the newly inserted object, position the cursor over the object so that the cursor becomes a crosshair,  , and drag the object to a new location.

Insert using the Edit menu

To insert a new object into a TPS Builder drawing:

Step	Action
1	In TPS Builder, open the drawing in which you want to insert an object.
2	From the Edit menu, choose Insert Object.
3	In the Insert Object dialog box, choose the type of object and click OK. The application associated with your choice will start, and a window for the object will open in the TPS Builder workspace. In addition, the TPS Builder main menu bar will change to that of the associated application. Other tools may also become available (for example, a drawing toolbar in a paint application).
4	If needed, move the object. Position the cursor over the "frame" around the object so that the cursor becomes an arrow/crosshair, then drag it to the desired location. 
5	If needed, size the object. Position the cursor over one of the handles until the cursor becomes a sizing (double arrow) cursor, then drag the handle to a new size. 
6	Use the menus and tools of the associated application to create the object.
7	When finished, click outside of the object frame. You will exit the associated application and return to TPS Builder.

Insert by dragging

To drag a selection or file from an application window into TPS Builder:

Step	Action
1	Open the file containing the information you want to drag into the drawing (or My Computer or Explorer if you are going to drag a file).
2	Open the TPS Builder drawing that is to receive the selection or file.
3	Drag the selection (or file) onto a TPS Builder drawing. When you release the mouse, the selection becomes embedded.



ATTENTION

If you drag a file:

- It will become an embedded object (not linked).
 - It will be represented as an icon. To view the file, double-click it. The file will open in its native application window.
 - Its icon name will be a generic identification of the file type. If desired, add a file-specific title by inserting an object with the file name. Position the name object and icon together on the TPS Builder drawing.
-

8. Working with Drawings

8.3. Inserting Objects into a Drawing

Edit an inserted object

To edit an object that has been inserted into TPS Builder, follow these steps:

Step	Action
1	Left-click the object to select it.
2	Right-click the object to access its menu.
3	On the resulting menu, choose the desired action, Edit or Open. • Edit starts the associated application and opens the object window within the TPS Builder drawing. • Open starts the associated application and opens the object window separately from TPS Builder.
4	If you choose Edit, make changes as needed. When finished, click in the TPS Builder workspace to close the associated application and return to TPS Builder.
5	If you choose Open , make changes as needed. When finished, return to TPS Builder by choosing File > Exit & Return to <i>ModuleName</i> , where <i>ModuleName</i> is the name of your drawing.

Delete an inserted object

The following steps allow you to delete an object that has been inserted in a drawing.



CAUTION

If deleting an object, be certain you want to remove it. No confirmation message appears, and no undo function exists.

If you do delete an object by mistake, close the TPS Builder drawing without saving it; the object will be restored.

Step	Action
1	Right-click the object to be removed.
2	On the resulting menu, choose Delete .

8.4 Drawing Symbol Properties Overview

Description

TPS Builder allows you to customize the appearance of the symbols in your drawing. Properties that can be set are as follows:

- Colors
- Pictures
- Connectors and their labels
- Fonts
- Size
- Parameter names and values appearing on the drawing symbol

Strategies

For strategies, these properties apply to a drawing symbol in a particular view. That is, if you change the symbol while in Equipment view, the changes will *not* appear in Hardware view.

Once properties are set, they can be changed at any time, using the same procedures.

Templates

You can define the drawing symbol properties at the template level, so that all strategies created from the template will exhibit the properties defined. You can do this by modifying the drawing symbol properties of a system template, or by creating a specialized template with the drawing symbol properties you want.

These are not template defining properties, though, and they can be changed for any strategy after it is created. Likewise, if you change the properties for the template, these changes will not propagate to strategies already created from the template.

Size - special considerations: The ability to change the drawing symbol size for a template has some unique considerations. Refer to the discussion on setting symbol size in the following topic, Setting Drawing Symbol Properties.

8. Working with Drawings

8.4. Drawing Symbol Properties Overview

Drawing symbol considerations

Planning

Sound planning will help ensure efficient use of TPS Builder. If you want to use a particular color scheme, fonts, and/or pictures for your drawing symbols, plan the scheme first, test it, and then decide whether or not you want to modify templates to represent this scheme before you use them.

When setting drawing symbol properties, consider legibility issues regarding the display and print versions of your drawing. For example, color combinations that display well may not provide adequate contrast in a black-and-white printed copy. The type of printer and its capabilities will also affect the legibility of various color combinations. For example, if the default color for connector lines is a dark gray when your drawing is printed, this color may be too light.

Standards

Your organization may have established standards that apply to the drawing properties; if so, follow them.

Views

Remember that for strategies, these properties apply to a drawing symbol in a particular view. That is, if you change the symbol while in Equipment view, the changes will *not* appear in Hardware view.

Procedures

To change the appearance of a drawing symbol(s), refer to the following topic in this section: Setting Drawing Symbol Properties.

8.5 Setting Drawing Symbol Properties

Set drawing symbol colors

To set color properties for a drawing symbol, follow these steps:

Step	Action
1	In an active drawing, library, or project, right-click a template or strategy. On the resulting menu, choose Drawing Properties .
2	In the Properties dialog box, Color tab, choose an element, from the Property Name list, for which you want to set a color. Note that for the current release, many properties are not active. Refer to TPS Builder's online Help for specifics about the available colors. Ignore all others.
3	Click the desired color, then the Apply button. The element is now defined with that color. When choosing colors, refer to Drawing symbol considerations in the previous topic.
4	Repeat steps 2 and 3 for all elements for which you want to set specific colors.
5	Continue with other drawing properties, or click OK to close the dialog box.

8. Working with Drawings

8.5. Setting Drawing Symbol Properties

Set drawing symbol fonts

To set font properties for a drawing symbol, follow these steps:

Step	Action
1	In an active drawing, library, or project, right-click a template or strategy. On the resulting menu, choose Drawing Properties .
2	In the Properties dialog box, Fonts tab, choose the text element, from the Property Name list, for which you want to set a font. You can set fonts for the following elements of a block: • Title font: The font used for the title of the block. • Parameter font: If you define parameter names and values to be visible (using Control Attributes, then this setting defines the font in which those labels appear. Note that you can also define the color of this text using the Colors tab, ParameterLabel and ParameterValue Colors. • Connector font: If you define connector names to be visible (using Control Attributes), then this setting defines the font in which those labels appear. Note that you can also define the color of this text using the Colors tab, ConnectorLabelColor. • Font: This setting defines the font in which any named connections appear. Refer to TPS Builder's online Help for details about this property.
3	Choose a font, font size, style, and effect, then the Apply button. The text element is now defined with those properties.
4	As needed, repeat steps 2 and 3 for other text elements.
5	Continue with other drawing properties, or click OK to close the dialog box.

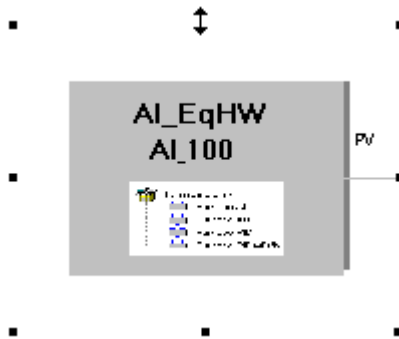
Add a picture to drawing symbols

To add a picture to a drawing symbol, follow these steps:

Step	Action
1	In an active drawing, library, or project, right-click a template or strategy. On the resulting menu, choose Drawing Properties .
2	In the Properties dialog box, Picture tab, click Browse to access the Browse Pictures dialog box. Here, locate the picture file to be used.
3	Double-click the picture file. The picture appears in the Preview section of the Picture tab.
4	Click Apply to add the picture.
5	Continue with other drawing properties, or click OK to close the dialog box.

Size a drawing symbol within a drawing

To make a drawing symbol larger or smaller in a strategy or template drawing:

Step	Action
1	Click the symbol to select it. "Handles" appear around the drawing.
2	Move the cursor over one of the handles until the cursor becomes a double-headed arrow. For example, 
3	Drag the handle as follows: • Move a centered top or bottom handle to make the object taller or shorter. • Move a centered left or right handle to make the object wider or narrower. • Move a corner handle in or out to simultaneously change the width and height of the object.
4	Position the pointer in another part of the drawing area and click the mouse button. The object is deselected and the handles disappear.

8. Working with Drawings

8.5. Setting Drawing Symbol Properties

Default sizes for drawing symbols

You can set a drawing symbol's default size. Note that:

- This size will not affect existing drawings or templates that have already had any drawing properties set.
- It will be used the first time you set any drawing properties for a template.
- It will also be used the first time you open a drawing for any of the drawing's strategies created from templates without drawing properties set.

Set a default size

You may want to perform this procedure to define sizes for templates for different types of strategies, such as control strategies, input/out strategies, and so forth.

To establish a default size:

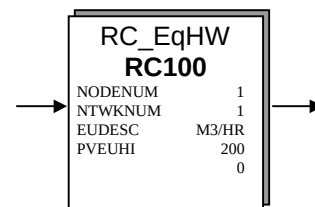
Step	Action
1	Begin by evaluating your drawing symbol needs. You will need to experiment with different sizes to determine what is appropriate. In considering drawing symbol size, you must also consider • the information that is to appear on the symbol (in addition to the strategy and template name, you may want to include parameter names and values), • the font type and size that are appropriate, and • the number of connectors for the symbol. Also consider the total number of strategies to be included in your drawings. With this information in mind, continue with the procedure.
2	From the Options menu, choose Build Symbol Default Size.
3	In the Default Size dialog box, choose a width and/or height, in pixels. Note that the initial Honeywell-provided size is 200x200 pixels.
4	Click OK.
5	Identify a template that has not had any drawing symbol properties set (for example, choose a template you have never used).
6	Create an empty strategy (or specialize a template) module to be used just for experimentation.

Step	Action
7	Open the drawing for this module.
8	Drag the template into the module drawing.
9	Evaluate the drawing symbol size on the screen and by printing a copy of your drawing. Change other properties on the strategy to match the needs you identified in step 1.
10	If the size is acceptable, you have set the default size. Go to step 12. Otherwise, go to step 11.
11	If the size is not acceptable, set a different default size and drag the template into the drawing again.
12	If you will be using this size for all templates (and subsequent strategies) of a particular category, then: In turn, select each template and set at least one other drawing symbol property (color, font, picture, parameters, or connectors. This action creates a drawing symbol properties file for the template, which includes the default size.
13	If needed, repeat the entire procedure, choosing another drawing symbol size, for your other category of templates.
14	When you are finished, delete the experimental module.

Add parameter information to a drawing symbol

You can choose parameter names and/or values to appear on the drawing symbol for a selected template or strategy block or module. This can be useful for configuring or monitoring parameters. The following example shows a strategy block with a few parameters and their values defined to appear.

In this example, various parameters and their values have been made visible. Note that the parameter name PVEULO has been omitted, to reduce clutter. Users can be instructed regarding conventions such as this.



Preparation

Before beginning, complete these steps:

8. Working with Drawings

8.5. Setting Drawing Symbol Properties

Step	Action
1	Plan which parameters you want to be visible in a drawing, and whether or not you want their values to appear.
2	Check the font to be used for the parameter information; change if needed. Refer to Set drawing symbol fonts in this topic.
3	Control attributes for strategies (blocks or modules) can be configured only if the strategy is contained in a module. Therefore, place strategies to be configured into a module if they are not already in the module.

Add parameter information

To add parameter information to a drawing symbol for a selected template or strategy block or module:

Step	Action
1	Complete the preparation described previously.
2	In any project, library, or drawing, right-click the item for which control attributes are to be configured. (If in a drawing, left-, then right-click the item.)
3	In the resulting menu, choose Control Attributes .
4	In the Control Attributes dialog box, choose the Parameters tab.
5	Define the parameter attribute: • Click a parameter name in the Parameters list. • If the parameter represents an array, enter a valid index number. • For the parameter names (labels) to be on the drawing symbol, choose Expose Parameter Label and View Labels. • To put the parameter value to be on the drawing symbol, choose View Values.
6	Click ADD →. The parameter name will be placed in the Visible Symbol Parameters list. If you chose to expose the label, the parameter will be marked as "Labeled."
7	Repeat as needed for additional parameters.
8	If needed, when finished with the Parameters tab, click another tab in the dialog box for further control attribute configuration.

Step	Action
9	Click OK when all control attributes are configured.
10	Evaluate the font used for the parameter labels. If desired, modify the font. Refer to Set drawing symbol fonts in this topic.

Modify existing parameter information

To modify the parameters after they have been added, note that you can turn **View Labels** and **View Values** on and off for parameters already labeled.

To add a parameter label, REMOVE the parameter from the **Visible Symbol Parameters** list, then add it back, ensuring that you choose the **Expose Parameter Label** option.

8.6 Drawing Connection Overview

Description

You can make connections between appropriate template or strategy blocks, or for modules for which project-up parameters have been set. The connection can be to one of these items in your current drawing, or in another drawing in the same project.

The lines depicting the connections can be automatically positioned, or you can manually position them. Once connections are made, you can re-position them as necessary.

What comprises a connector?

For a selected template or strategy block or module, parameters are set as connectors on the drawing symbol. No restrictions exist as to which parameters can be set as connectors, although some settings would be clearly illogical (for example, using the Name parameter as a connector), and would not be connectable.

You can define:

- Which parameters will appear as connectors
- The position of each connector on the drawing symbol
- Whether or not the connector is labeled
- A label other than the actual (pushed or pulled) parameter name

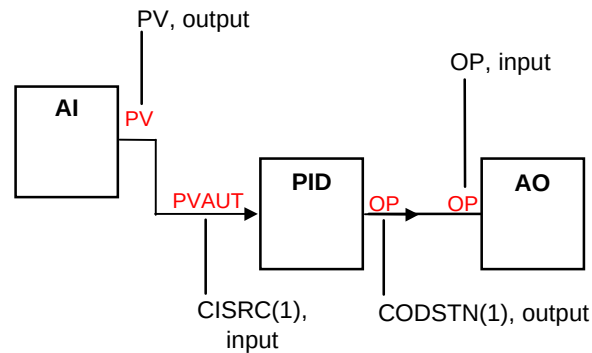
Default labels

Default labels are the names of the actual (pulled or pushed) parameters, not the parameters that are receiving/sending the values. In the following simple drawing, the

8. Working with Drawings

8.6. Drawing Connection Overview

parameters set as connectors are shown in black type, with the type of connector setting, and the actual labels are shown in red (next to the connector lines).



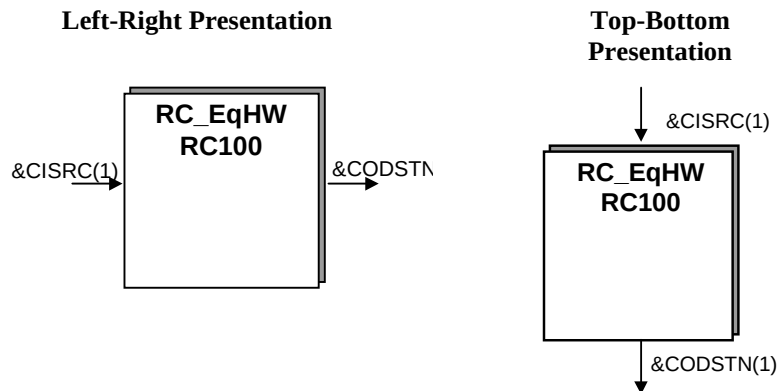
Configuration represented by connections

The behavior of connections in strategy and template drawings is slightly different, as discussed in the following table.

Drawing Type	Behavior
Strategy	For connections made in strategy drawings, TPS Builder makes entries into the appropriate parameter configuration forms.
Template	TPS Builder maintains the relationship you establish without actually configuring the connection-related parameters. When you create a strategy module from such a template, TPS Builder completes the configuration forms based on the relationships established in the template module. Connections made in a template module are considered template defining, and thus cannot be changed in any strategies created from the template (or in any templates specialized from it).

Connector location

Connectors can be placed from left-to-right on a drawing symbol, or from top-to-bottom, as shown in the following figures:



Connector labeling

Connector labels may be visible or not. You may also change a label from the actual (pushed or pulled) parameter name to something else.

8.7 Drawing Connection Preparation

Before making connections

Complete these steps before making connections in a TSP Builder drawing:

Step	Action
1	If you have not yet done so, read the Drawing Connection Overview in this section.
2	Make sure that fundamental parameter configurations are completed; for example, for a Reg Ctl point, CTLALGID must be set to other than Null.
3	Make sure that the drawing symbols are located approximately where desired in your drawing.

8. Working with Drawings

8.7. Drawing Connection Preparation

Step	Action
4	For connections within a single drawing, make sure that the drawing symbols have visible connectors for the connections you want to make. If they do not, refer to Make connectors visible. Note that templates supplied by Honeywell have, by default, the most common connectors configured as visible, with a left-right layout. (This can be changed, if desired.) For connections across drawings, the connectors do not need to be visible; TPS Builder will add the connector when the connection is defined.
5	Determine whether the connector labels need to be changed. If they do, change them as explained in: • Add or remove connector labels • Change connector labels
6	Modules (system-specific or "generic") cannot have connectors unless the applicable parameters of blocks in the module have first been designated as Project-Up. If you want to configure connectors for such a module, complete Project-Up configuration first. Refer to Project-Up Fundamentals in the Improving Consistency and Efficiency section of this document.

Make connectors visible

To make connectors visible for a drawing symbol:

Step	Action
1	In your drawing, right-click the drawing symbol to which you want to add connectors.
2	On the resulting menu, choose Control Attributes.
3	In the Connectors tab, click a parameter name in the Potential Connectors list.
4	Evaluate, and change if needed, the default settings for whether the parameter is • to be treated as passive or active, • to be output or input, and • where on the drawing symbol the connector is to appear (top-bottom or left-right arrangement).
5	If the parameter represents an array, enter a valid index number to be used for the connector.
6	If you want the connectors to be labeled with the parameter name, choose the View Labels option. This option applies to all connectors of the selected item.
7	If you want to a name other than the parameter's name to appear in your drawing, enter the desired name in the Optional Override Label box.
8	Click ADD →. The parameter name, its defined state (active or passive), and label override, (if any) are placed in the appropriate Visible Connectors group.
9	Repeat as needed for additional connectors.
10	Click OK.

8. Working with Drawings

8.7. Drawing Connection Preparation

Add or remove connector labels

Follow these steps to control the presence of connector labels for a specific drawing symbol:

Step	Action
1	In your drawing, right-click the drawing symbol of interest.
2	On the resulting menu, choose Control Attributes.
3	In the Connectors tab, • Check the View Labels option to add labels. • Clear the View Labels option to remove labels.
4	Click OK.

Remove connectors

If the drawing symbol includes unused connectors, you can remove them by following these steps:

Step	Action
1	In your drawing, right-click the drawing symbol whose unused connector(s) you want to remove.
2	On the resulting menu, choose Control Attributes.
3	In the Connectors tab, Visible Connectors groups, click on the connector you want to remove.
4	Click ← REMOVE.
5	Repeat for any other connectors to be removed.
6	Click OK.

Change connector labels

If a connector label needs to be changed, follow these steps:

Step	Action
1	In your drawing, right-click the drawing symbol whose connector labels you want to change.
2	On the resulting menu, choose Control Attributes.
3	In the Connectors tab, Visible Connectors groups, click the connector whose label you want to change.
4	In the Optional Label Override, type the new label.
5	Repeat for other connectors if needed.
6	Click OK.

8.8 Autolayout/Autorouting

Autolayout

In TPS Builder, **autolayout** uses a drawing map to place blocks so they are arranged in a matrix—inputs on the left and outputs on the right. Autolayout, to the extent possible, arranges blocks from left to right by tracing connections.

Remember that autolayout is applied only when a drawing is generated for the first time; that is, it is not applied to drawings that already exist. Although, if a drawing is empty, autolayout is applied. For example, if an empty module is added to a strategy, and points are dragged from a tree into the module, the first time it is opened TPS Builder detects the empty drawing and autolayout is applied. Autolayout and autorouting (that is, the placing of blocks and their connections) are tied to the Map function. This will be discussed later in this section.

8. Working with Drawings

8.8. Autolayout/Autorouting

Three rules apply to autolayout/autorouting for maps:

Action	Result
If an entirely new map was created in the template before the drawing is created . . .	that map is used by autolayout the first time the strategy drawing is opened, because it is empty.
If the map was edited (either the avenue dimensions were altered or the cells were merged or split), . . .	the map is not used for autolayout. The original map, as it was before the edits were made, is used.
If the drawing was produced in an earlier version of TPS Builder . . .	the map is not applied. These drawings have no map and they are not given a map. Therefore, when they are opened the function is toggled off and the other map functions are disabled.

Autolayout connectivity

Autorouting draws connection lines through avenues between map cells (see “Module Drawing Maps” which follows). As with a highway, avenues contain multiple lanes through which connection lines are routed. An avenue’s width and the number of lanes determine the spacing between the connection lines. While autolayout is applied only when a drawing is first generated, autorouting continues to be active when a drawing is edited, unless you turn it off by the toggling Connection/Use Map. When Use Map is off, Auto Line is also off.

When points are defined as inputs to another point or their output destinations are defined, as the points are dragged into a module, connections are made.

8.9 Module Drawing Maps

The types of maps

TPS Builder has one general map and two special maps:

Map Type	Description
1. The general map is for modules comprised of points and/or other modules.	The general drawing map is simply a matrix of 3 by 4 cells per letter on an A4 sized page.
2. The logic and device control drawing maps.	Columns for gates are provided between an input column on the left and an output column on the right. These two maps also provide an area to the left of the title block for placing flag and numeric blocks (only connected flags and numerics are drawn). Inputs and outputs are placed in ascending order, top to bottom. Logic outputs are not drawn unless they have at least one connection besides FL1 on the enable pin. The logic and device control drawing maps are designed to fit on one letter or A4 sized page.

Create a map

When points are placed in a module drawing, a map can be created by selecting the menu item View>Map. Cells appear as white blocks delineated by dashed lines, and avenues for connection lines appear in light gray. Connection lines are automatically routed through the avenues as long as build symbols (blocks) are within cell boundaries. When a build symbol is moved or resized such that it extends beyond the boundaries of a cell, its connection lines are converted to named connections.

In the course of rearranging blocks, blocks may not be placed entirely within a cell, causing connection lines to be automatically converted to named connections. Once blocks are rearranged, the named connections can be reconverted to connection lines by invoking **Reconnect All**. **Reconnect All** runs autorouting on the entire drawing, so any moved connection lines will be redrawn.

8. Working with Drawings

8.9. Module Drawing Maps

The map will display the function blocks such as input, output, regulatory control, composite points, and so forth. Blocks are placed in cells with “avenues” between them. Blocks can be resized by selecting them and dragging by their “handles” or by defining them in the templates. Cells can be merged or split, and avenues can be resized by editing the map. Before placing points in a module, consider the number of points, their sizes, and the number of cells on a page, as discussed in the following topic “Planning the map.”

Planning the map

The default map size for a portrait A4 or letter-sized map is 4 rows of 3 cells. For the landscape view, the cells are 3 rows of 4 cells. These cells accommodate the blocks (such as Analog Input, Regulatory Control, Analog Output, and so forth). The default size for avenues between cells is 36 pixels in either direction.

Strategy module maps can also be viewed using the Windows Print Preview function or printed using the Print function. This will give you an idea of the page layout to choose.

Before creating a map, you should do some planning, such deciding the most common number of blocks you will be placing in a strategy module. This will help you to decide the size of the cells, the width of the avenues between the cells, and the best size for your blocks. It is suggested that you use an empty module to define the number of cells, and begin at this point. Add the most common number of blocks you will have in a module, make the connections, and view the layout. You should also consider if you are going to print the drawing (see “Title Block” following) in color or in grayscale. If you are not using a color printer, consider defining the blocks as clear or gray so the block titles and parameters (if printed) will not be obscured by a dark hue. You can then use this to make a model, and “templatize” this module.

Use Map

When a drawing is open, **Connection>Use Map** toggles the use of autorouting (see preceding topic “Autolayout/Autorouting.”)

View Map menu item

When a drawing is open, the View Map toggles the displays of the drawing map. To view a map, use the following procedure.

Step	Action	Result
1	Double-click on the module to open the drawing. (It is assumed that the strategies exist within the module).	The module map is displayed. Blocks are placed in cells as described in the preceding topic, "Planning the map."
2	From the Menu, select View>Map .	

Editing a map

When it is necessary to resize a block, map cells can be edited to accommodate the larger and smaller blocks. Cells are enlarged by merging, and reduced by splitting. To resize map cells, use the following procedures:

Begin/Quit Edit

When a drawing is open, and the drawing map is displayed, it is not in edit map mode. **Begin Edit** turns on the edit map mode, displaying the cells. **Quit Edit** turns off edit map mode. These features are on the right-mouse button menu.

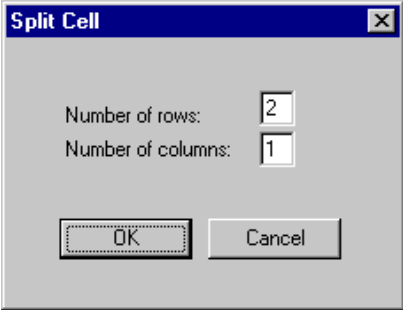
Split cell

To split a cell, use the following procedure.

Step	Action	Result
1	From the main menu, select Connections>Edit Map .	Cells are outlined in blue; dashed lines appear more vivid.
2	If you want to split a cell, select a single cell.	The cell shows that it is selected by its "handles."

8. Working with Drawings

8.9. Module Drawing Maps

Step	Action	Result
3	Right-click and the menu appears.	 The dialog box titled "Split Cell" has a close button (X) in the top right corner. It contains two input fields: "Number of rows:" with the value "2" and "Number of columns:" with the value "1". At the bottom, there are two buttons: "OK" and "Cancel".
4	Enter the number of rows/columns and click OK .	The cell will split accordingly.

Merge cells

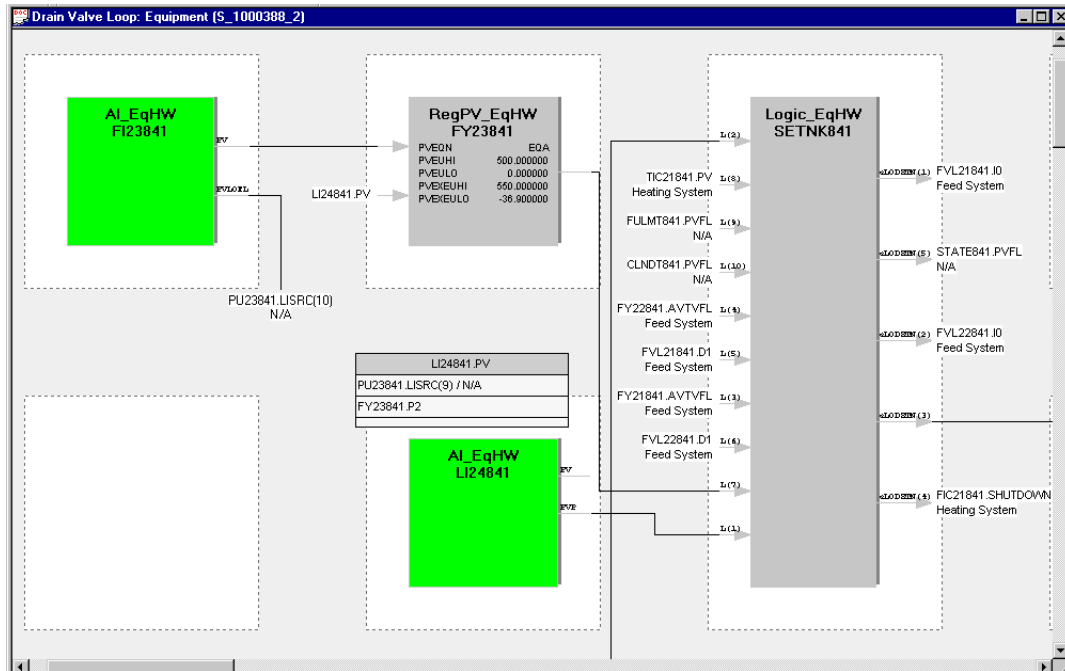
To merge cells (with the map open) use the following procedure.

Step	Action	Result
1	From the main menu, select Connections>Edit Map .	Cells are outlined in blue; dashed lines appear more vivid.
2	Hold down the Ctrl key and select the cells you want to merge. You can merge adjacent cells vertically or horizontally.	The handles for the cells appear.
3	Right-click and the menu item Merge Cells appears.	The cells are merged. See the following figure in which cells have been merged.

The following figure illustrates an example of a cell that has been merged vertically to accommodate the number of inputs in a logic block. Avenues have been widened to accommodate the view of the inputs and outputs. Parameters have been added to show values for the RegPV block. Note that some passive outputs have been converted to tables.

8. Working with Drawings

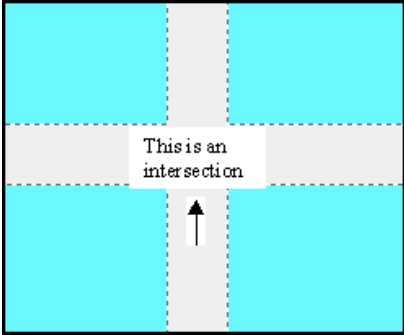
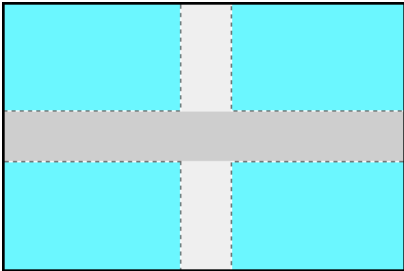
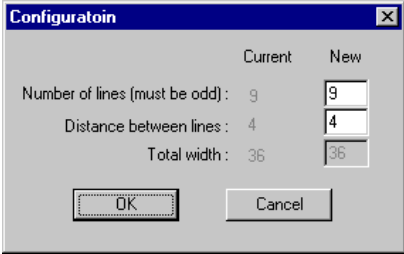
8.9. Module Drawing Maps



When a build symbol (block) is moved or resized such that it extends beyond its boundaries, its connections are converted to named connections.

Edit avenues

To edit the avenues, with the map open, use the following procedure. The distance between lines is in **pixels**.

Step	Action	Result
1	From the main menu, select Connections>Edit Map .	Cells are outlined in blue; dashed lines appear more vivid.
2	Select the avenue to be edited. Do not select an avenue at an intersection.	It turns a darker hue of gray.
		
3	Right click.	The Configuration dialog box appears.
		
4	Make the desired edits and click OK .	Cells bordering the lanes edited grow/shrink accordingly.

Drag a connection or use Switch command

Lines can be routed with more line crossings that are desired. In such a case, line segments can be moved to a different lane to reduce the number of intersections. A line segment can be moved in either of two ways:

- With the map in edit mode, a line can be dragged to an available lane.
- The switch command can be used to swap the positions of parallel line segments. They do not have to be adjacent.

Use the Switch command

Step	Action	Result
1	From the main menu, select Connections>Edit Map .	Cells are outlined in blue; dashed lines appear to delimit the avenues.
2	Hold the Ctrl key down and select both line segments.	They change color to show they are selected.
3	Right-click to display the Switch command, and select it.	The two line segments switch locations.

New map command

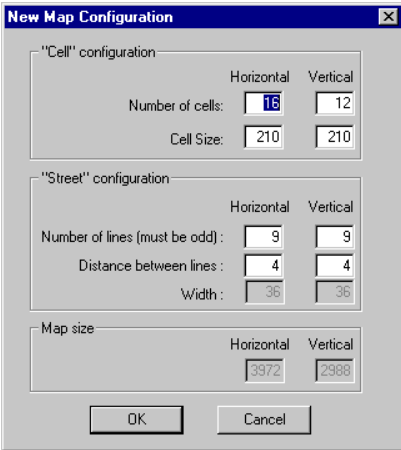
If the current map is not adequate, it can be replaced by a **New Map**. The New Map dialog box enables you to replace an existing map with a new map. You can specify the number of cells in the matrix (horizontally and vertically), the cell size, and the dimensions of the avenues. When the dialog box is closed, the new map replaces the current map. This dialog box is best used for developing a module template.

Use the following procedure to replace an existing map with a New Map.

Step	Action	Result
1	Open the module drawing, and select View>Map .	The module build symbols (blocks) are displayed in cells and connections are displayed in the avenues between cells.

8. Working with Drawings

8.9. Module Drawing Maps

Step	Action	Result
2	From the main menu, select Connections>New Map .	The New Map Configuration dialog box appears. 
3	Define the cell and line configuration as desired and click OK .	The new map is applied.

Snap to Grid

The Snap to Grid function is provided for aligning objects in a drawing. The grid is square and spacing can be set in pixels, providing for a wide range of grid spacing from very fine to very coarse. The grid can be displayed as a matrix of dots so that the nearest grid point is visible.

Snap-to-grid applies only to build symbols, embedded objects, and connection tables when they are being moved or resized. It does not apply to stationary objects or to connection lines; connection lines are governed by connector position and autorouting. When an object is placed into or moved in a drawing with snap-to-grid on, its position is snapped to the grid point nearest the object's upper left corner. When an object is resized, the dimension(s) being moved is(are) snapped to the grid.

To turn on the Snap to Grid function, from the Main Menu select **Drawing Settings>Snap To Grid**. When this item is unchecked, snap-to-grid is not in use; when checked, it is used.

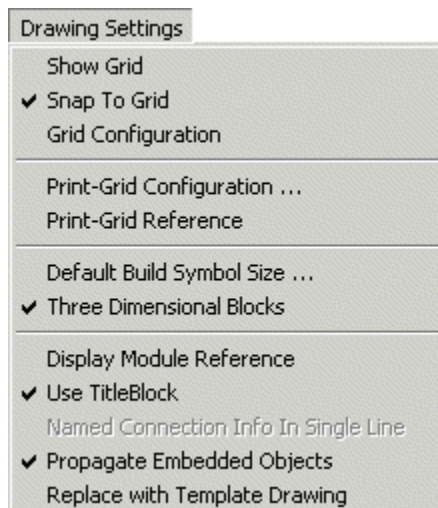
Show Grid

Display of the grid is toggled by **Drawing Settings>Show Grid**. When this item is unchecked, the grid is not displayed; when checked, it is displayed. Snap-to-grid can be turned on without being displayed, but the grid display is available only when snap-to-grid is on.

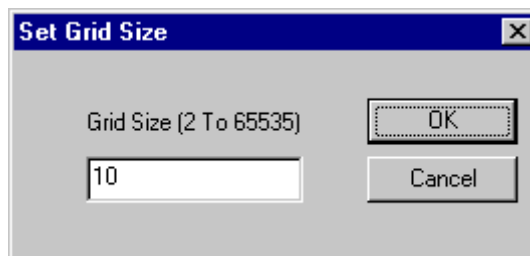
Grid Configuration

Grid size is set by selecting **Drawing Settings>Grid Configuration**. This action displays a dialog box that accepts the grid spacing in pixels. A small grid size of about 7 provides enough resolution to work well with a typical drawing while providing enough coarseness to aid in aligning objects.

The Grid Configuration dialog is accessed from the Drawing Settings menu:



This is the Grid Configuration dialog box:



8.10 Basic Connection Procedures

Types of connections in drawings

Three types of connections can be distinguished based on the relationship of connected blocks through a page and a drawing:

1. on-page
2. off-page
3. off-drawing

On-page connections are between two blocks on the same page of a drawing.

Off-page connections are between two blocks on different pages of the same drawing.

Off-drawing connections are between two blocks that are not on the same drawing.

On-page connections

On-page connections are displayed either as connection lines or as named connections. On-page connections can also be displayed in tables.

Connection tables

A connection line or a named connection can be converted to a table. A single connection table lists all the connections for a particular connector. Although the connection for a single connector can be converted to a connection table, this would rarely be useful. Connection tables are useful when a single parameter is read by multiple points or is written to by multiple points. In these cases, the number of connection lines and names in a drawing can be reduced by collecting the multiple connections in a table and exposing only one pin for the connector instead of multiple pins.

Off-page connections

Off-page connections behave the same way as on-page connections when a drawing is open. However, when a drawing is previewed or printed, the off-page connections drawn as lines are temporarily converted to named connections. In addition, off-page connections display the page number and the print-grid reference (optional) for the connected block.

Off-drawing connections

Off-drawing connections are displayed as named connections or in a connection table. When shown as a named connection, the module name of the connected block is displayed beneath the connection name. When listed in a connection table, the module name is part of the connection.

Connection Menu

The following figure shows the connection menu:



The following menu items are toggles; consequently only one of these modes can be activated at a time. When any one of the four modes is activated, it can be deactivated by clicking on the drawing background.

1. Connect
2. Delete Connection
3. Convert Line
4. Convert Table

Use Map

When a drawing is open, **Connection>Use Map** toggles the use of auto routing (see preceding topic “Auto layout/Autorouting”).

Edit map

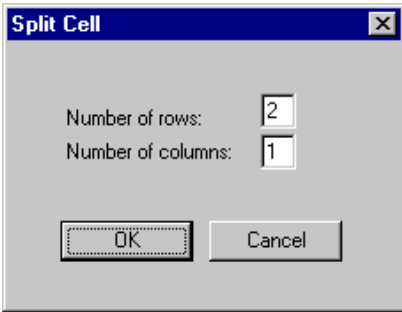
When it is necessary to resize a block, map cells can be edited to accommodate the larger and smaller blocks. Cells are enlarged by merging, and reduced by splitting. To resize map cells, use the following procedures:

Begin/Quit Edit

When a drawing is open, and the drawing map is displayed, it is not in edit map mode. **Begin Edit** turns on the edit map mode, displaying the cells. **Quit Edit** turns off edit map mode. These features are on the right-mouse button menu.

Split cell

To split a cell, use the following procedure:

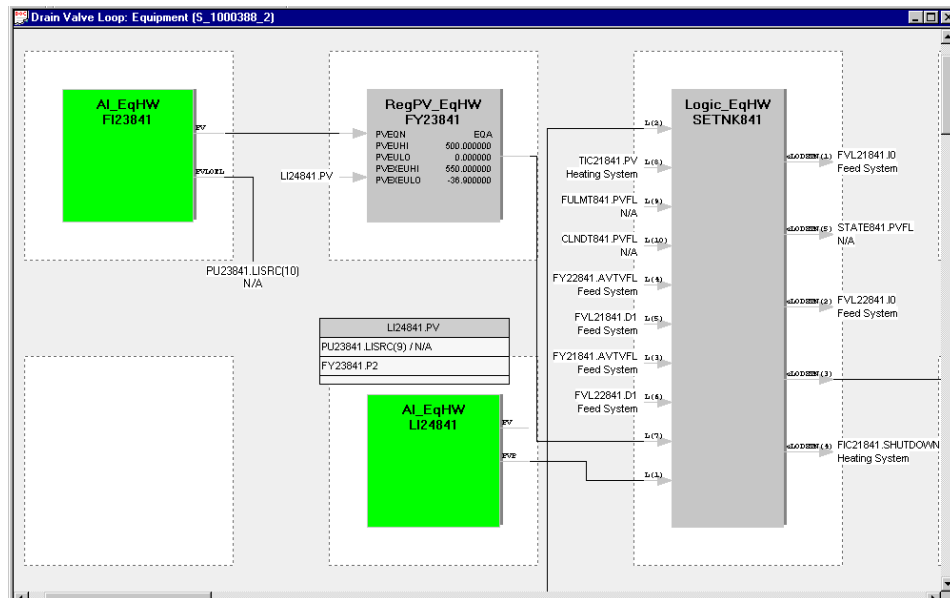
Step	Action	Result
1	From the main menu, select Connection>Edit Map>Edit Map .	Cells are outlined in blue; dashed lines appear more vivid.
2	If you want to split a cell, select a single cell.	The cell shows that it is selected by its "handles".
3	Right-click> Split Cell .	The Split Cell dialog box appears.
		
4	Enter the number of rows/columns and click OK .	The cell splits accordingly.

Merge cells

To merge cells (with the map open) use the following procedure:

Step	Action	Result
1	From the main menu, select Connection>Edit Map >Edit Map .	Cells are outlined in blue; dashed lines appear more vivid.
2	Hold down the Ctrl key and select the cells you want to merge. You can merge adjacent cells vertically or horizontally.	The handles for the cells appear.
3	Right-click. Click the menu item Merge Cells .	The cells are merged. See the following figure in which cells have been merged.

The following figure illustrates an example of a cell that has been merged vertically to accommodate the number of inputs in a logic block. Avenues have been widened to accommodate the view of the inputs and outputs. Parameters have been added to show values for the RegPV block. Note that some passive outputs have been converted to tables.



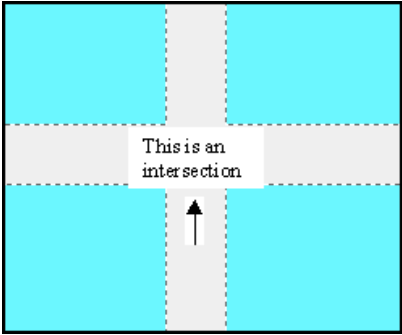
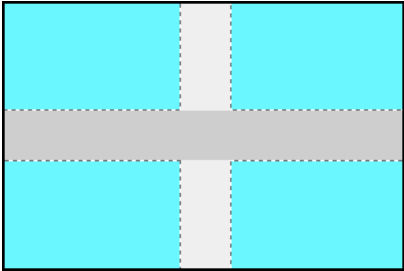
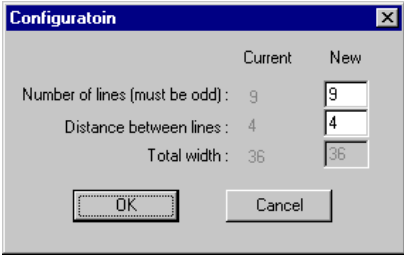
8. Working with Drawings

8.10. Basic Connection Procedures

When a build symbol (block) is moved or resized such that it extends beyond its boundaries, its connections are converted to named connections.

Edit avenues

To edit the avenues, with the map open, use the following procedure. The distance between the lines is in **pixels**.

Step	Action	Result
1	From the main menu, select Connection>Edit Map .	Cells are outlined in blue; dashed lines appear more vivid.
2	Select the avenue to be edited. Do not select an avenue at an intersection.	It turns a darker hue of gray.
		
3	Right click.	The Configuration dialog box appears.
		
4	Make the desired edits and click OK .	Cells bordering the lanes edited grow/shrink accordingly.

Drag a connection or use Switch command

Lines can be routed with more line crossings that are desired. In such a case, line segments can be moved to a different lane to reduce the number of intersections. A line segment can be moved in either of two ways:

- With the map in edit mode, a line can be dragged to an available lane.
- The switch command can be used to swap the positions of parallel line segments. They do not have to be adjacent.

Use the Switch command

Step	Action	Result
1	From the main menu, select Connection>Edit Map .	Cells are outlined in blue; dashed lines appear to delimit the avenues.
2	Hold the Ctrl key down and select both line segments.	They change color to show they are selected.
3	Right-click to display the Switch command, and select it.	The two line segments switch locations.

New Map

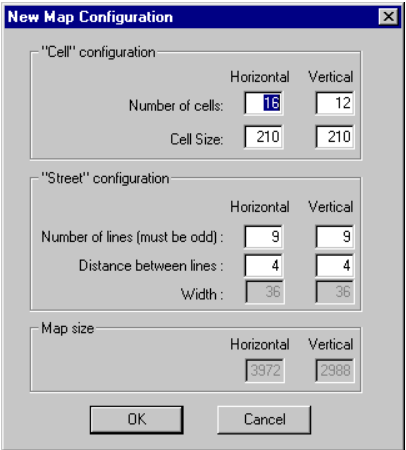
If the current map is not adequate, a new map can replace it. The New Map dialog box enables you to replace an existing map with a new map. You can specify the number of cells in the matrix (horizontally and vertically), the cell size, and the dimensions of the avenues. When the dialog box is closed, the new map replaces the current map. This dialog box is best used for developing a module template.

Use the following procedure to replace an existing map with a New Map.

Step	Action	Result
1	Open the module drawing, and select View>Map .	The module build symbols (blocks) are displayed in cells and connections are displayed in the avenues between cells.

8. Working with Drawings

8.10. Basic Connection Procedures

Step	Action	Result
2	From the main menu, select Connection>New Map .	The New Map Configuration dialog box appears. 
3	Define the cell and line configuration as desired and click OK .	The new map is applied.

Reconnect All

In the course of rearranging blocks, blocks may not be placed entirely within a cell, causing connection lines to be automatically converted to named connections. Once blocks are rearranged, the named connections can be reconverted to connection lines by invoking **Reconnect All**. **Reconnect All** runs auto routing on the entire drawing, so any moved connection line is redrawn.

Connect blocks within a module using autorouting

The lines depicting the connections in a module drawing can be automatically positioned by TPS Builder. This is called “autorouting.” To make connections using this feature, follow these steps:

Step	Action
1	If you have not yet done so, read Drawing Connection Preparation in this section before making connections.
2	Enter Connection mode: From the Connection menu, choose Connect. The cursor will become a cross, +.

Step	Action
3	- Click on a connector. - Click on the connector to which you want to connect.
	Result: A line will join the connectors, and configuration data will be added to the appropriate parameter configuration forms.
4	Repeat for other connections.
5	When finished, click anywhere in the "background" of the drawing (that is, places other than on a drawing symbol or connection line) to end Connection mode.
6	If needed adjust the position of the drawing symbols for a good layout of the drawing symbols and connection lines.
7	Save your drawing.

Once the connection is made, just as in previous versions of TPS Builder, a connection line segment can be moved by dragging one of the segment's endpoints. (However, once a line segment is moved manually, its coordinates no longer match what is in memory, so other autorouted connection lines may not cater for its new position.).

Connect blocks within a module with manual routing

To make connections in a module drawing by manually setting the line segments, follow these steps:

Step	Action
1	If you have not yet done so, read Drawing Connection Preparation in this section before making connections.
2	Check drawing symbol spacing. Make sure that a space exists between the "handles" of each symbol, as shown. (This is where connection lines can be placed.)

The diagram illustrates the required spacing for manual routing between two blocks. On the left is a square block labeled 'AI'. On the right is a square block labeled 'RegCtl'. Between them is a gap. Above this gap, the word 'Space' is written with a downward-pointing arrow. Small squares, representing handles, are positioned at the top and bottom of both blocks and at the ends of the gap. Two horizontal arrows point from the right handle of the 'AI' block to the left handle of the 'RegCtl' block, passing through the gap.

3	From the Connection menu, choose Connect to enter Connection mode. The cursor will become a cross, +.
4	- Click on a starting connector, then - Click at the point you want the first line segment to end. - Click at the point you want the next line segment to end. - Repeat for other segments, ending by clicking the ending connector. Result: The line as defined will join the connectors and configuration data will be added to the appropriate configuration forms.
5	Repeat for other connections.
6	When finished, click anywhere in the "background" of the drawing (that is, places other than on a drawing symbol or connection line) to end Connection mode.



ATTENTION

After you manually route connections, turn off Auto Line if you want to subsequently move blocks connected to them and preserve as many segments as possible. Otherwise, they will become autorouted connections.

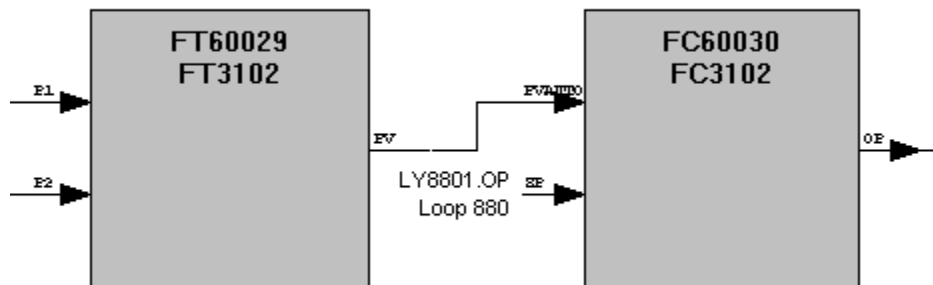
Connect blocks across drawings

You can make connections across module drawings in the same project. For example, you may want to connect a logic output in one module to a regulatory control input in another module.

To make connections across modules, use the configuration form procedure. Refer to the Parameter Configuration section, Configure connections using the configuration form.

Display Module Reference

The Display Module Reference item on the Drawing Settings menu allows you to toggle the display of the module name in off-drawing named connections. The module name is displayed by default, but can be turned off using this menu item. The following figure is an example displaying the module reference item as well as the point referenced.



8. Working with Drawings

8.10. Basic Connection Procedures

Use hyperlinks for navigation

If you make connections across modules, the associated named connection in each module's drawing is a hyperlink. Follow these steps to use the hyperlinks:

Step	Action
1	In a drawing, locate the named connection that represents a connection to a block in another drawing.
2	Double-click the named connection.
3	If the connection is to a block that exists in only one module in one view, then that drawing will open and become the active drawing.
4	If the Open Drawing dialog box appears, it means that the connection is to a block that exists in modules in different views (for example, Equipment and Hardware views). In this case, double-click the module whose drawing you want to open. The corresponding drawing will open and become the active drawing.
5	To return to the first drawing, double-click the hyperlinked connection in the active drawing.



ATTENTION

If a hyperlinked connection is to a block that exists in modules in different views, only one of the module drawings can be open at one time.



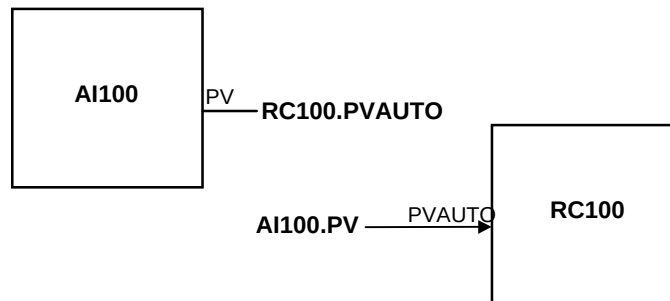
TIP

Note that when you move the cursor over a hyperlink, it changes to a "hand."

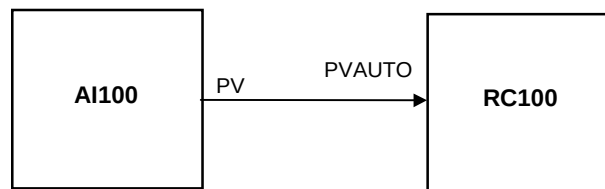
Convert connections

Converting connections is changing the appearance of a connection to or from a (graphical connection) line from/to a named connection.

Named connection:



Line Connection:



Follow these steps to convert a connection in a drawing:

Step	Action
1	From the Connection menu, choose Convert Line.
2	Click on the line (or named connection) to be converted. The connection will change from its current state to the alternative state (line-to-named or named-to-line).



TIP

If you use a named connection, you can move the name if desired, for better drawing layout.

A line will extend from the drawing symbol to the name.

Delete connections

Connected template or strategy blocks, or modules for which project-up parameters have been set and connected, can be disconnected by using one of the following methods.

Line connections

Follow these steps to delete line connections:

Step	Action
1	In your drawing, choose Connect > Delete Connections. The cursor will become a cross, +.
2	Locate the connection you want to delete and click on it. The line is removed and applicable parameter configuration forms cleared.
3	Click on the line. The line will be removed and applicable parameter configuration forms cleared.

Named connections

Follow these steps to delete named connections:

Step	Action
1	In your drawing, choose Connect > Delete Connections. The cursor will become a cross, +.
2	Locate the connection you want to delete and click on it. The line is removed and applicable parameter configuration forms cleared.
3	Click on one of the connection names. The names will be removed and the applicable parameter configuration forms cleared.

Convert named connections to table

A connection table can be used to replace passive connectors, especially when a number of connections make the drawing appear cluttered.

A connection table has at least two lines. The first line is shaded and names the connector for which the table is displayed. Subsequent lines name the connections.

The following table shows the formats for different types of connections. The second line shows the format of an on-page connection; the third line, an off-page connection; the fourth line, an off-drawing connection; and the last line, an off-page connection when the connected block is not contained in a module.

<point>.<parameter>[(<index>)]
<point>.<parameter>[(<index>)] / <page number>
<point>.<parameter>[(<index>)] / <module name>
<point>.<parameter>[(<index>)] / N/A

A connection table can be moved and resized just like a build symbol. The default size of the table is the same as the default size of the build symbols.

The font of the text in a connection table can be configured as a property of the connection table. On the connection table's context menu, select **"Properties...."** Select the Fonts tab and set the attributes for ConnectionTableFont. This dialog can also be opened by double-clicking on a connection table.

Entries in a connection table cannot be edited directly. To delete a connection listed in a table, open the parameter form for the connected block and edit the connection through the parameter form. To add a connection to the table, open the parameter form of the control to be connected and add the connection named in the table's first line. Table connections can also be edited by converting the table back to line and named connections, then editing the connections individually. Deleting a connection table (by selecting the table and then pressing the Delete key) removes the table and toggles all the connections back to their normal mode.

Convert to table

To convert a connection to a table, use the following procedure:

Step	Action	Result
1	Open the drawing and select View>Map .	The drawing is displayed.
2	Click the pins you want to convert to a table. NOTE: you can convert only passive connectors to tables.	A table appears in the place where the pins were with the connections as listed in the parameter form.
3	Resize the table and move it to the desired place in the drawing. It should be placed in one of the drawing cells. An example of a table with named connection converted to tables is shown in the section "Optional Print Grid" preceding this topic.	

To convert a table back to a connection, follow this procedure:

Step	Action	Result
1	Placed the cursor on the table and select it.	"Handles" appear on the table.
2	Delete the table using the Delete key.	The table disappears.
3	Click on the connector pins. You can make the connections as lines or named connections.	The connections reappear.

Parameter configuration forms

Connections in drawings will be removed if you delete connection settings in the configuration forms of the connected strategies.

8.11 Propagating Template Drawing Changes

Description

Using the **Drawing Settings > Replace with Template Drawing** menu item is an alternative method for propagating template drawing changes.

When the Replace with Template Drawing option is selected, derived drawings are overwritten with the template drawing. Any changes specific to the derived drawing are lost, so additional symbols must be repositioned.

When this option is not selected, template drawing changes are propagated to derived control drawings, preserving whatever changes have been made specifically in each derived control drawing. Also, connection lines in the derived drawings are autorouted.

Deciding which propagation to use

Deciding which option to use depends on how the control drawings were created:

- **Change propagation option:** Use when autorouting is used for the connection lines and the strategy drawings have more components than the template drawing.
- **Replacement option:** Use when autolayout and autorouting are used only during template drawing generation: Connection > Auto Line and Connection > Use Map are deselected after template drawings are generated and the drawings are extensively revised.

In the Replacement option, the strategy drawings will look exactly like the template drawings. This option works best when the derived drawings have few controls in addition to what is in their respective templates.

Prevent propagation of embedded objects

When template drawing changes are propagated, the embedded objects are propagated when their position in the updated drawing is changed. This may produce duplicate copies of embedded objects. To prevent propagation of embedded objects, deselect Propagate Embedded Objects from the Drawing Settings Menu.

8. Working with Drawings

8.12. Printing Drawings

8.12 Printing Drawings

What can be printed

You can print the drawing of a template or strategy module. TPS Builder places drawings on a page with a border (see the following example), and basic drawing information.

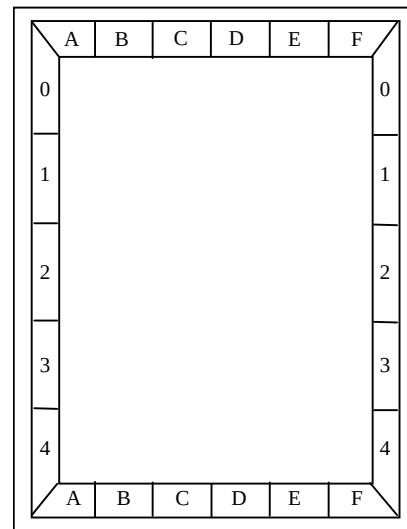
Printing can be done on a black-and-white or color printer (for drawings that use color).

Border (grid configuration) example

This example shows a drawing border of six columns (A - F) and five rows (0 - 4).

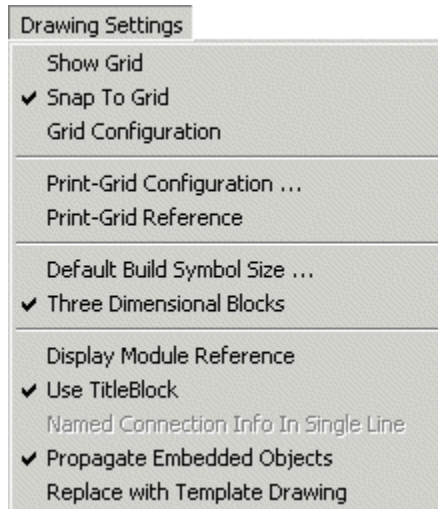
The labeling results in off-page connections labeled with both the page number and column/row intersection, for example, 5, A1.

Print grid is an optional choice under the Drawings Settings menu.



Optional Print Grid

The print-grid option provides a means of referencing objects in the drawing, particularly in off-page connections. Use of this grid reference is optional. It is turned off and on by toggling **Print-Grid Reference** on the Drawing Settings menu. Note also, that the configuration for this grid is not on the File menu, it is on the Drawing Settings menu as **Print-Grid Configuration**.

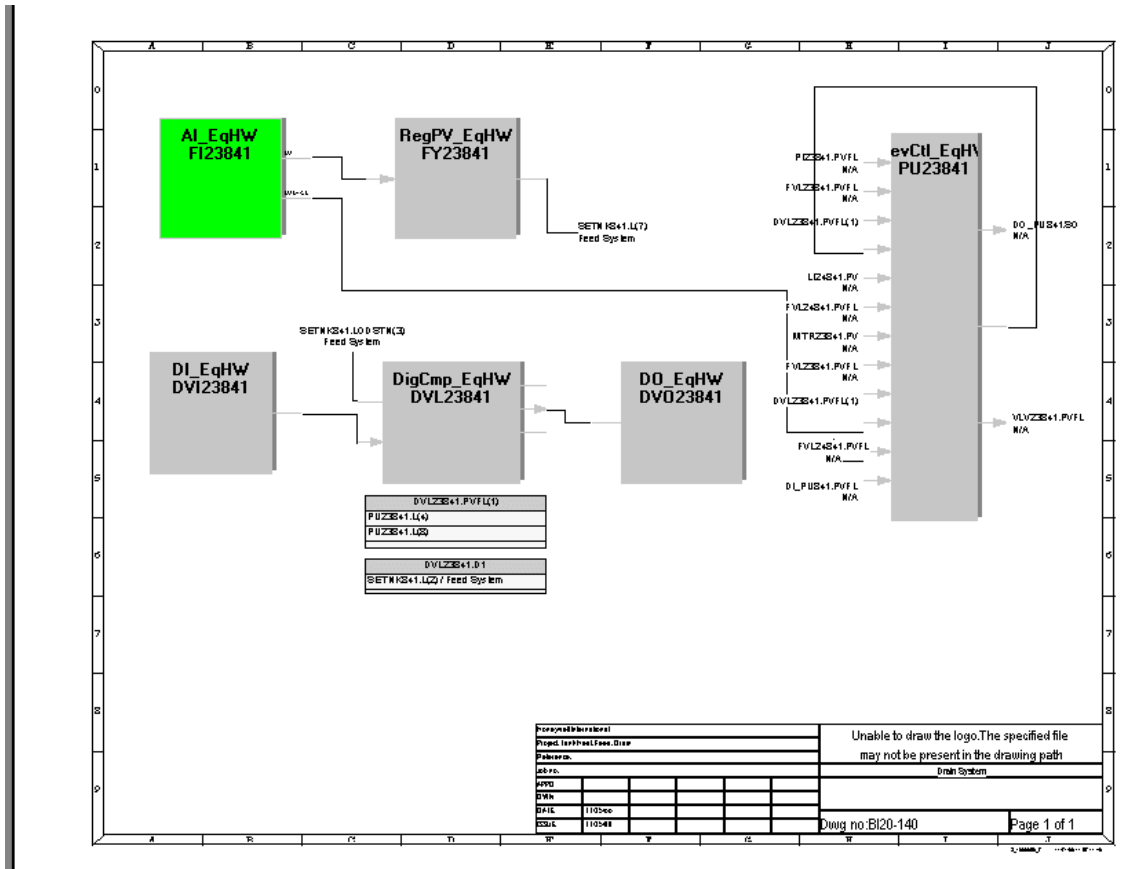


8. Working with Drawings

8.12. Printing Drawings

Print-Grid Reference

The following figure is a landscape view with the **Print-Grid Reference** turned on. You can use the Page Preview before you decide to print the grid reference.

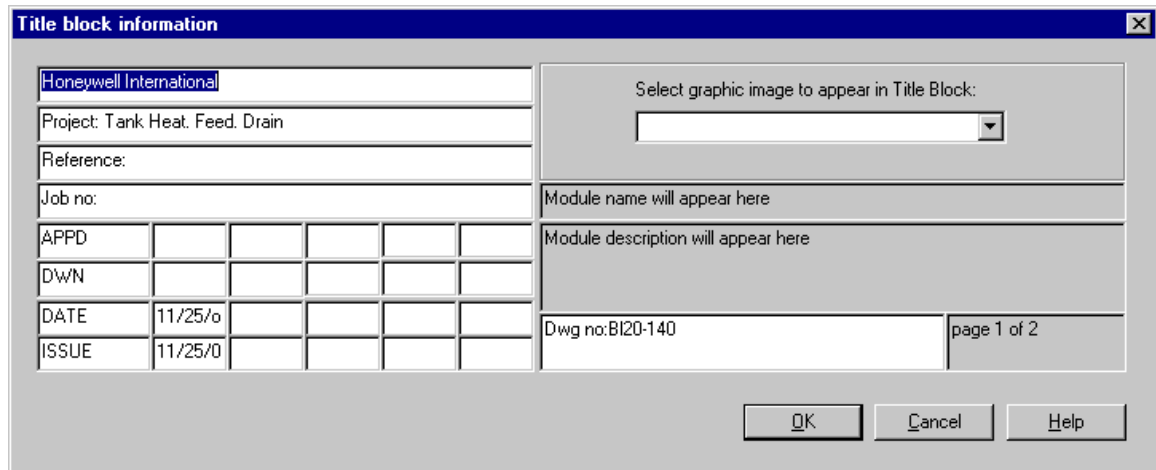


Optional Title Block

Introduction

Use **TitleBlock** on the Drawing Settings menu toggles between the Title Block and the footer. A standard title block can be used instead of the drawing footer in TPS Builder control drawings. The title block is used by default. You can switch to using the footer by toggling Use TitleBlock on the Drawing Settings menu. The means of entering the data for the title block is also different than was used for the footer. The values entered using this dialog apply to all the drawings in the project or library.

The following figure is an example of a Title block information dialog box:



The dialog box titled "Title block information" contains the following fields and controls:

- Company name: Honeywell International
- Project: Tank Heat. Feed. Drain
- Reference:
- Job no:
- Module name will appear here
- Module description will appear here
- Dwg no: B120-140
- page 1 of 2
- Buttons: OK, Cancel, Help

APPD					
DWN					
DATE	11/25/0				
ISSUE	11/25/0				

The Title Block can be used as an option (see the Drawing Settings figure following "Optional Print Grid" preceding this topic).

The means of entering the data for the title block is also different than the data used for the footer. The title block (approximately 5.5" wide x 1.1" high) is placed in the lower right corner of a drawing for print preview and printing. The Title Block Information dialog replicates the layout of the title block fields. This dialog is invoked from the context menu for the project or library icon in a project or library window. The editable fields contain values that apply to all modules in a project or library. Some of the fields have labels by default, but these are treated as part of the field values. These default labels can be edited as part of the field value. The top right field is allocated for an image and the combo box provides for selecting a graphics file. The graphic file should be placed in the data directory for the current repository.

8. Working with Drawings

8.12. Printing Drawings

The graphic file should be placed in the data directory for the current repository. It can be of any of the following graphic formats.

- MS Windows device-independent bitmaps (*.bmp and *.dib)
- Joint Photographic Experts Group file interchange format (*.jpg and *.jpeg)
- Graphics Interchange Format (*.gif)
- MS Windows icon (*.ico)
- MS Windows Enhanced Metafile (*.emf)

The image is sized to fit the field in the title block.

TPS Builder enters the data for three other fields at the time of print preview or printing.

1. Module name is the name of the module as displayed on the tree control, in the parameter form, and in the Identification dialog. It can be edited in any of these three places.
2. Module description is the description of the module as displayed in the Identification dialog. It is an 80-character field that can be edited in this dialog. It is printed as two lines of up to 40 characters each.
3. Page number is calculated by TPS Builder.

In addition to the information printed in the title block, two fields are printed below the title block. These are the drawing file name and the date/time the drawing was last modified.

To see an example of the title block on a drawing, refer to the picture in the subsection “Optional Print Grid” preceding this topic.

How to print a drawing

Follow these steps to print a drawing:

Step	Action
1	Open the drawing you want to print.
2	Check the drawing layout. If needed position drawing symbols, objects, and connections. For assistance, make page breaks visible by choosing Page Breaks from the View menu.
3	Decide on the appearance of the drawing border, for use in identifying off-page connections: • Number of columns and rows: One to ten each. Columns generally match the number of drawing symbols across. • Column and Row labeling: Alphabetical or numerical.
4	From the File menu, choose Configure Grid.
5	Complete the Grid Configuration dialog box in accordance with step 3 and click OK.
6	To check the appearance of your drawing before printing it, choose File Print Preview.
7	To print the drawing, do one of the following: • If in Print Preview, click Print in the preview display. • If in the drawing, Choose File > Print.
8	In response to the print request, the Drawing Information dialog appears. Complete this dialog box as needed and click OK. The Printer dialog box appears.
9	Check the settings in the Printer dialog box and click OK to print the drawing.

Default Print Settings

When Print Setup is used to set the page size and when no drawing is open (but any tree view of the project or library is opened), the page size and orientation settings are stored as default values in the registry.

When neither the tree view nor any drawing is open in TPS Builder, the Print Setup option is disabled in the File menu.

8. Working with Drawings

8.12. Printing Drawings

Print Settings for Individual Drawings

Page size and orientation values can be saved with the drawing in either of the following ways:

- Open the drawing, invoke Print Setup, and save the drawing; or
- Open the drawing for printing, set the values in the Print dialog, print the drawing and then save it.

These settings are saved and stored only for the drawing that is currently open.

If a drawing does not have page size and orientation values stored in its drawing file, the default values will be used from the registry.



TIP

It is good idea to open every drawing at least once prior to batch printing to ensure that the correct settings are stored, unless the default settings will suffice.

8.13 Contained Strategy Module Selection

Introduction

The ability to select contained strategy modules was added to facilitate batch printing. However, its use is not limited to batch printing: once strategies are selected, any operation for multiple strategies is valid, for example moving to another position in the project hierarchy.

You can select all the drawings in a sub-tree by clicking the left mouse button on the top drawing while holding down the Ctrl and Alt keys.

You can select all of the drawings in the project by clicking the left mouse button on the project icon while holding down the Ctrl and Alt keys.

Contained modules can be selected by using **Ctrl+Alt+click** on a module or a project icon. This action selects the module or project, expands the tree control within the selected module or project, and selects the contained modules. This works to four levels of nesting. An additional level can be selected by repeating the Ctrl+Alt+click on the originally selected node. Each additional use of this selection adds one level to the selection.

The following statements are true, regardless of whether a multiple selection was made as just described or by successively Ctrl+clicking on module icons.

Using Ctrl+click on a module has the same effect as adding it to the selection if it was not selected, and deleting it from the selection if it was selected.

All drawings, except the last selected, are deselected when you click the left mouse button in the project window without the Ctrl key.

Clicking on a + or – sign in front of a module object will expand (+) or collapse (–) the branch within the module without losing the current selection.

8. Working with Drawings

8.13. Contained Strategy Module Selection

9. PM-Family Connectivity Maps

9.1 Overview

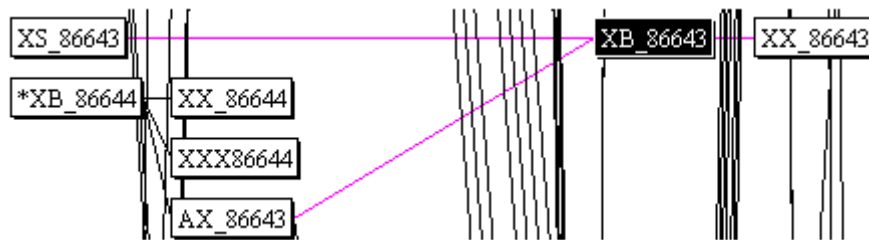
Description

A connectivity map displays TPS system entity strategies as a network diagram of strategy names and interconnecting lines that represent configured connections. A connectivity map can be generated for all selected Process Manager (PM) family entity strategies within a particular project. Maps are generated on a Universal Control Network (UCN) basis, using all configured PM-family nodes, or a range of nodes.

The complexity of a map depends on the number of connections in the project and on the connectivity options chosen. For example, choosing to process only Regulatory Control points will result in a less complex map than choosing to process all point connections in a project.

Example

The following is a small excerpt from a complex PM-family connectivity map. Point XB_86643 is selected, which causes its direct connections to be displayed in light magenta. Point XB_86644 has an asterisk because it names AX_86643 as an input source.



Maps in the TPS Builder workspace

A connectivity map is presented in a separate window in the TPS Builder workspace. Thus, basic window functions are available, as explained in this document's Using the TPS Builder Workspace section, "Basic workspace operations." More than one map can be displayed simultaneously.

9. PM-Family Connectivity Maps

9.1. Overview

Map windows can be minimized. To quickly identify the contents of a minimized map window, click once on its icon. Information about the window contents will display as follows:

projectname maptype Ntwk nn : node range [analysis options]

where *projectname* = the project name

maptype = Connectivity or Peer

Ntwk *nn* : #-# = *nn* is the network number, #-# is the range of nodes selected

[analysis options] = The point(s) chosen for analysis

Peer connections

"Peers" are PM-family points that are connected, but are assigned to different nodes. Peer communication is achieved by a connection through a UCN. Peer connections can be indicated by color using the Connectivity Map dialog box.

In connectivity maps, the color of the connections signifies peer and non-peer connectivity. For detail about peer relationships, you can generate a connectivity map solely of peer connections (often called a peer map).

Note that points named in connections, but not in the project, are represented with red lines connecting them because their addresses are treated as Network 0, Node 0.

CL file analysis

PM Control Language (CL) files can also be analyzed for connection relationships. If this option is chosen, bound data points (Process Module) are mapped to the declared EXTERNAL references within processed CL file(s).

Connectivity maps usage

Use connectivity maps to evaluate the connection aspects of PM-family points in your project. In addition, connectivity maps can be helpful if you are unfamiliar with the connection relationships of points you are backbuilding into TPS Builder. A connectivity map can help you determine a module structure that will best represent these relationships.

9.2 Generating a Connectivity Map

Description

The connectivity map function is quite straightforward, as explained in the following paragraphs.

Planning

Connectivity maps require that address assignment is completed.

In addition, a little planning will help you make the most of the connectivity map function. Planning includes the following tasks:

Step	Action
1	Connectivity maps are based on a specific project. Ensure that the strategies you want to evaluate are in the same project.
2	Be prepared to answer the following questions: • For which network address do you want to view the connectivity? • Do you want to include all nodes on that network, or a range of nodes? • Do you want to view a map of all point types at once, or some subset of the following connection types? – Regulatory Control – Regulatory PV – Digital Composite – Logic – Device Control – CL External References • Do you want to view only peer connectivity?
3	If you have several connections to analyze, ensure that you have adequate time for the procedure, which can be time consuming. The more densely connected the network is, the more time it will take. Of course, you can begin the procedure and leave your computer running; or you can put the connectivity map (TPS Builder) in the background and work in another application.
4	Consider a color scheme for the connectivity map. You can set colors of your choice for • Background • Entity (point) names • Configured connections • Peer connections • Invalid peer connections

9. PM-Family Connectivity Maps

9.2. Generating a Connectivity Map

Step	Action
5	If you want to analyze CL files, ensure that they are in the same directory and that this directory is accessible from your computer. It is recommended that TPS Builder-related CL files be kept in the <i>Projectx</i> \program directory, where <i>Projectx</i> is the name of your project. (The complete directory path depends on your TPS Builder installation.)

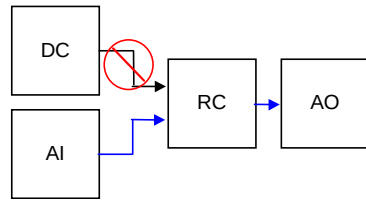
Procedure

Follow these steps to generate a connectivity map:

Step	Action
1	Complete the planning discussed previously.
2	With at least one project, library, or drawing in the TPS Builder workspace, choose PM Connectivity Map from the Tools menu.
3	Complete the Connectivity Map dialog box, choosing from several options regarding: • Selection criteria on the General tab • CL files to be analyzed (if any, see next step) • Map color scheme Refer to the example following this procedure for details on the result of certain analysis options.
4	If CL files are to be included, choose the CL External References option on the General tab, and the CL files themselves on the CL Files tab.
5	When the Connectivity Map dialog box is complete, click the Generate button. The process will begin, and a progress indicator will display. When processing is complete, the connectivity map displays.
6	Assess the map as explained in "Evaluating a Connectivity Map."

Connectivity options example

This diagram shows connections configured from digital composite (DC) and analog input (AI) points to a regulatory control (RC) point, which is connected to an analog output (AO) point:



Here, choosing *only* Regulatory Control in the Connectivity Map dialog box would result in a connectivity map showing the AI and AO connection relationship to the RC point, but not the Digital Composite connection. The reason for this is that the connection is defined in the digital composite, not in the regulatory control point.

Halt map processing

You can halt processing by clicking the Cancel button on the progress indicator. An incomplete map will display.

9.3 Evaluating a Connectivity Map

Maps in the TPS Builder workspace

For a discussion of map presentation in the workspace, refer to this section's Overview.

Map layout

A map is drawn such that

- inputs are on the left,
- outputs are on the right, and
- signals always flow to the right.

Occasionally, a point on the left receives an input from a point to the right. In these cases, the connection is not drawn, but the point receiving the input is displayed with an asterisk (*) in front of its name.

Evaluation tasks

You can perform the following tasks while a connectivity map is the active window:

- View a list of the current map's points (entities)
- View individual point connections
- Configure parameters
- Regenerate a map
- View peer statistics

Each of these tasks is discussed in the following paragraphs.

View Entity List

To view the list of entities for the generated map and locate an entity, perform the following steps:

Step	Action
1	Right-click anywhere in the map. On the resulting menu, choose Entity List: This Map.
2	In the resulting alphanumerically sorted Entity List for the map, locate the entity of interest and double-click it. The map will reposition so that the entity is selected and near the middle of the window.
3	When finished, close the entity list by clicking X in the upper right corner of the dialog box.

View individual entity connections

An entity can be selected in a map by clicking it. When an entity is selected, its direct connections are shown in light magenta so that they can be readily distinguished.

If you cannot locate the entity on the map, view the entity list as described in the previous procedure.

Connection details can be viewed by opening the entity's configuration form, as discussed next.

9. PM-Family Connectivity Maps

9.3. Evaluating a Connectivity Map

Configure parameters from a connectivity map

Most of the time you will be interested in modifying connection-related parameters when in a connectivity map, although any parameters can be modified. To do so:

Step	Action
1	Select (single-click) the entity on the map.
2	Right-click the selected entity and choose Configure Parameters on the resulting menu. The corresponding Configuration Form will display.
3	Modify parameters as needed and when finished, click OK in the Configuration Form.
4	Repeat if needed for other entities.



TIP

To evaluate the effect of changes to connection-related parameters, regenerate the map as discussed next.

Regenerate a map

Maps can be regenerated (1) to display updated connections (refer to previous paragraph), or (2) to change map options, such as selected entities or map colors. To regenerate the map:

Step	Action
1	Right-click in the map window and on the resulting menu, choose Regenerate Map.
2	If desired, make changes in the Connectivity Map dialog box.
3	When all options are as desired, click Generate. The map will be regenerated.

View peer statistics

Peer relationships are discussed in this section's Overview.

To view peer statistics for a generated connectivity map:

Step	Action
1	Right-click anywhere in the map.
2	Choose View Peer Statistics on the resulting menu. The Peer Statistics dialog box will display.
3	Evaluate the information presented: • Node: The number of each node represented in the generated connectivity map. • Inputs, Outputs: The numbers of inputs and outputs to/from each node. • CL: The number of points externally referenced in CL. • Invalid: The number of invalid peer connections. If the map shows any invalid peer connections, a warning to this effect is displayed.
4	When finished, click Close .

9. PM-Family Connectivity Maps

9.4. Peer Maps

9.4 Peer Maps

Description

"Peers" are connected PM-family entities that are assigned to different nodes. Peer communication is achieved by a connection through a Universal Control Network (UCN). For more details, refer to Overview.

Generate a peer map

To analyze peer relationships, follow these steps:

Step	Action	Reference
1	Generate a connectivity map, making sure to choose the Peer Connectivity option in the Connectivity Map dialog box, General tab. Also, check the color scheme for peer connections on the Colors tab, and change if desired.	Generating a Connectivity Map
2	Evaluate the map.	Evaluating a Connectivity Map
3	If desired, evaluate peer statistics	See the following paragraph.

Peer statistics

You can obtain a summary of peer statistics for any connectivity or peer map by right-clicking in the map and choosing View Peer Statistics on the resulting menu.

10. Working with Logs

10.1 Overview

TPS Builder logs

TPS Builder records data changes and problems in a variety of logs. In addition to activities, three types of problems are logged, those found:

- in an EB, JS, or other file during a backbuild;
- in the database during checking; and
- in relation to slot assignment.

Windows XP Event log

Errors that occur due to hardware, memory, program, or data problems are recorded in the Windows XP Event Log. Refer to Windows Help (**Start > Help**) for information on this log.

What do you want to do?

If you want to...	Then go to...
Enable logging, manage log size	<i>TPS Builder Installation, Implementation, and Administration Guide</i>
• Access or print a log • Find an entry in a log • Delete entries from a log	Basic Log Procedures in this section
Modify the appearance of the log	Adjusting Log Columns or Rows in this section
Display only certain data in a log	Filtering Log Contents in this section
Display log contents in a particular order	Sorting a Log in this section
Place log data into a spreadsheet, word processor, or other program	Copying Log Contents to Another Application in this guide

Additional information

Details on using the Logs dialog box, as well as information about each type of log in TPS Builder, are provided in the TPS Builder online Help.

10.2 Basic Log Procedures

Access a log

Follow these steps to access any TPS Builder log:

Step	Action
1	With any project, library, or drawing in the TPS Builder workspace, choose Report > Logs .
2	In the Logs dialog box, click the tab of the log you are interested in.

Find an entry in a log

You can search for a particular entry in any single column in a log. You cannot search across columns. The Find function searches up or down for one pass, from the point it started. It does not "wrap" around in the column.

To find an entry in a log:

Step	Action
1	Click Reports > Logs .
2	Click the tab of the log of interest.
3	If desired, filter or sort the contents before performing the Find.
4	Click in the first cell of the column in which you want to search (or, to search above or below a particular point, click in the cell where you want to begin the search).
5	Click Find .
6	In the Find dialog box, type the needed term or phrase. If desired, choose the Match Case option to limit the search to the exact capitalization used.
7	Choose a search direction (Up or Down; Down is the default).
8	Click OK . The first instance of the term or phrase, in the column selected, will be highlighted.
9	Click Find Next to find the next instance. Repeat as needed. When no more instances are found, the system will beep.

Print a log

To print a log, follow these steps:

Step	Action
1	Choose Reports > Logs .
2	Click the tab of the log of interest.
3	If desired, filter or sort the log as needed.
4	Click Print .
5	Click OK . The Print dialog box appears.
6	Complete the Printer dialog box and click OK .



ATTENTION

A log can be many pages in length. To minimize the amount of material you print, you may want to copy only the needed rows to another application (such as a spreadsheet or word processor), and print from that application.

Refer to "Copying Log Contents to Another Application."

Delete selected entries from a log

You can delete rows, but not columns or individual cells.

To delete rows, follow these steps:

Step	Action
1	Click Reports > Logs .
2	Click the tab of the log of interest.
3	If desired, filter or sort the log as needed.
4	To select the entire contents of the log, click the upper leftmost square of the log (leftmost in the heading row) and go to step 7.
5	To select a portion of the log, locate the first row to be deleted and click the leftmost square of the row. The row will be highlighted.

10. Working with Logs

10.2. Basic Log Procedures

Step	Action
6	For a contiguous group of rows, locate the last row of the group you want to delete and Shift+click the leftmost square of that row. The group will be selected. For a non-contiguous portion of the log, Ctrl+click the leftmost square of each row you want to delete.
7	When all rows are selected, press the Delete key.

Delete all entries in a log

To delete the entire contents of a particular log, follow these steps:

Step	Action
1	Click Reports > Logs .
2	Click the tab of the log of interest.
3	Click Clear All .
4	In response to the confirmation message, click Yes .

Close a log

To close the Messages Log, simply click the **X** in the upper right corner of the dialog box.

10.3 Adjusting Log Columns or Rows

Description

During any single log session, you can adjust the log's appearance to suit your needs. You can:

- Adjust column widths
- Move columns
- Adjust row width



ATTENTION

The changes will remain in effect only as long as the Message Logs dialog box is open; once it is closed, the default arrangement of each log returns.

Adjust column widths

To adjust the width of columns for a single log session, follow these steps:

Step	Action
1	Move the cursor so that it is between two column headings until the cursor changes to  .
2	Press and hold the mouse left button and drag left or right to size the window.

Move columns

To move columns from one location in a log to another for a single log session, follow these steps:

Step	Action
1	To select the column, click anywhere in a column heading except on the column title.
2	If you want to move more than one contiguous column, then do not release the mouse button, and drag across the additional column(s) to be moved.
3	When all columns are selected (highlighted), release the mouse button. Keep the cursor in the column heading.

10. Working with Logs

10.3. Adjusting Log Columns or Rows

Step	Action
4	When the cursor changes to an arrow,  , press the mouse left button and drag the column. Note: The cursor changes to  as you drag, and each new column location (between columns) highlights with a red dotted line.
5	When the desired location for the column is reached, release the mouse button. The column(s) move to the new position

	TIP To move a column to the log's rightmost position, move the column to the second-to-rightmost column position. Then, move the rightmost column elsewhere in the log.
---	---

Adjust row spacing

To adjust the amount of space in rows for a single log session, follow these steps:

Step	Action
1	Move the cursor so that it is between two rows in the leftmost column of plain blocks. 
2	When the cursor changes to  , press and hold the mouse left button, then drag down or up to increase or decrease the row size.

10.4 Filtering Log Contents

Description

The default for Message Logs is to show all event information available, in chronological order from earliest date to most recent date. You can limit the display of messages, or change the sort order.

Filter by specific date

To filter a log so that data for only a specific date or range of dates displays, follow these steps:

Step	Action
1	Open the Message Logs dialog box and choose the log of interest.
2	Click the Filter button.
3	In the View From group, click the Events On button, then enter a specific date and time, in the format set for your Windows environment, as presented in the log (mm/dd/yy hh:mm:ss). For example, 08/28/97 08:00:00.
4	In the View Through group, click the Events On button, then enter a specific date and time, in the same format as presented in the log. For example, 08/28/97 16:30:00. Note: Note that the examples in the previous steps show how to specify a single shift.
5	Click OK.



TIP

To see all messages for a specific day, enter the day you want as the starting day, and the next day as the ending day. You do not need to enter hours, etc.

Filter by message type

To filter a log so that data for only a specific type of message displays, follow these steps:

Step	Action
1	Open the Message Logs dialog box and choose the log of interest.
2	Click the Filter button.
3	In the Types group, click the desired message type.
4	Click OK.

Filter by task, operation, or user

To filter a log by specific task, operation, or user, follow these steps:

Step	Action
1	Open the Message Logs dialog box and choose the log of interest.
2	Locate a row containing the task, operation, or user you want the log filtered upon. Click in the task, operation, or user cell.
3	Click the option desired: Selected Task, Selected Operation, or Selected User.
4	Click OK.

Filter by multiple criteria

To filter a log by more than one criteria, follow these steps:

Step	Action
1	Open the Message Logs dialog box and choose the log of interest.
2	If you want the filter to include a selected task, operation, or user, then set up that filter first. Do not click OK or close the Filter dialog box.
3	If you want the filter to include date criteria, then set up that filter and either click OK or go the next step to set up message criteria.
4	To include type-of-message criteria, then set up that filter.
5	When all filter criteria are set up, click OK.

Remove filters

To return all data to the log, click **Filter**, then click **Clear**, and **OK** to close the Filter dialog box.

10.5 Sorting a Log

Description

A log can be sorted in ascending or descending order by column.



ATTENTION

- Blank cells are placed ahead of cells with data for ascending order, or after cells with data in descending order.
 - Sorting is shown by the sorting symbol in each column's heading.
 - No symbol indicates default sorting (or release of sorting preference for a column).
 - ▼, a "down" triangle, indicates user-request for ascending sort order for the column (lowest to highest, 0 - 9 and A - Z).
 - ▲, an "up" triangle, indicates user-request for descending sort order for the column (highest to lowest, 9 - 0 and Z - A).
-

Sort by a single column

To sort a log by a single column, follow these steps:

Step	Action
1	Position the cursor in the desired column heading until the cursor becomes a down arrow, ↓.
2	Right-click once. The log will be sorted in ascending order per the contents of that column, the sorting symbol will become ▼, and the column will be highlighted.
3	To change the order to descending, right-click in the column heading again.
4	To return to default sorting preference, right-click once again.

Sort by multiple rows

Multiple sorts can be accomplished by first ordering the columns from left to right, to reflect your preference for the sort order (primary, secondary, tertiary, and so on). The primary sort column must be the leftmost column of the group to be sorted.

To sort multiple columns, follow these steps:

Step	Action
1	Determine the order in which you want the columns sorted (primary, secondary, tertiary, etc. sort). If needed, see the examples following this procedure.
2	Ensure that the columns to be sorted are positioned from left to right according to their sorting order. That is, if the primary column is not the leftmost of the group to be sorted, move it. Secondary to the left of tertiary, and so on. Note: These columns do not need to be side-by-side, just in left-right order.
3	Sort the primary column.
4	Sort the secondary column as explained previously.
5	Continue sorting other columns as needed.

10. Working with Logs

10.5. Sorting a Log

Sorting examples

You might want to sort messages and users. Notice the results in the following two examples. In the first example, the message column is placed to the left of the user column, and in the second example, the user column is placed to the left of the message column.

Example 1: Message = primary sort; User = secondary sort

Open drawing Unit 1	...	Peaper
Open drawing Unit 1	...	Smith
Open drawing Unit 2	...	Houston
Open drawing Unit 2	...	McConnell
Open drawing Unit 2	...	Peaper
Open drawing Unit 3	...	Andry
Open drawing Unit 3	...	Peaper

Example 2: User = primary sort; Message = secondary sort

Andry	...	Open drawing Unit 3
Houston	...	Open drawing Unit 2
McConnell	...	Open drawing Unit 2
Peaper	...	Open drawing Unit 1
Peaper	...	Open drawing Unit 2
Peaper	...	Open drawing Unit 3
Smith	...	Open drawing Unit 1

10.6 Copying Log Contents to Another Application

Description

With TPS Builder R130, you cannot save a log configuration (particular sorting and filtering, column arrangement).

You can, however, copy of a log's contents, as sorted, filtered, and arranged and paste it into any file of the appropriate format, such as in the file of a

- spreadsheet,
- text editor, or
- word processor.

What can you copy?

The following guidelines apply to copying of log contents:

- You can copy rows of log data.
- You can filter a log, and thereby copy only the filtered contents.
- You cannot copy individual or groups of columns (even if you arrange the onscreen log to display only certain columns).

In a spreadsheet, the data will be appropriately placed in cells. In a text or word-processor file, the data will be tab-delimited; formatting will be required to make the data fit on the page and align correctly.

10. Working with Logs

10.6. Copying Log Contents to Another Application

What is copied?

The TPS Builder log includes

- Column headings,
- Log data, and
- Hidden data.

When you copy a log, the column headings are not copied. The log, including hidden data, is copied. Therefore, if you copy the Activity log, the first seven columns might appear as follows:

1	3	5002	Create project Project1	14	Create project	10/07/97 15:12:01
1	0	0	Log Activity is enabled.	46	Set Log Activity	10/07/97 15:11:42
1	0	0	Log Activity is enabled.	46	Set Log Activity	10/10/97 07:50:37
1	0	0	Log Activity is enabled.	46	Set Log Activity	10/13/97 17:01:38

In this example, the first, second, third, and fifth columns contain hidden data that can be deleted. If you want column headings, you need to add them yourself. An edited version of the previous example would be similar to the following:

Message	Operation	Time
Create project Project1	Create project	10/07/97 15:12:01
Log Activity is enabled.	Set Log Activity	10/07/97 15:11:42
Log Activity is enabled.	Set Log Activity	10/10/97 07:50:37
Log Activity is enabled.	Set Log Activity	10/13/97 17:01:38

How to copy log contents

Follow these steps to copy log contents to another file:

Step	Action
1	Click Reports > Logs .
2	Click the tab of the log of interest.
3	If desired, filter, sort, or arrange the log as needed.
4	Select all or part of the log (in whole rows).
5	When all rows to be copied are selected, press Ctrl+C to copy them.
6	Open the file in which you want to paste the log contents.
7	Press Ctrl+V to paste the contents.

Time presentation in spreadsheets

Two issues exist with time presentation in spreadsheets, as discussed in the following table.

Issue	Discussion
Column width	In a spreadsheet, the initial presentation of time may appear as number signs, "#####." This presentation occurs because the column is not wide enough to display all information. If your spreadsheet presents the time this way, widen the column. The time data should display correctly.
Time format	Some portion of the time may truncate, depending on the format set for date/time in your spreadsheet. For example, the seconds may truncate. If this occurs, and the truncated information is needed, use spreadsheet options to change the date/time format. For example, in Microsoft Excel, choose Format > Cells. Choose the Number tab, and select the format desired. Refer to your spreadsheet Help or documentation for more information.

10. Working with Logs

10.6. Copying Log Contents to Another Application

11. Working with Reports

11.1 Overview

Description

Reports make information regarding TPS Builder projects or libraries, as well as overall statistics and other details, available for viewing, printing, and other activities.

Reports are formatted for printing, and include borders, column headings, and identification blocks typical to engineering documentation.

How reports are generated

They are generated by Seagate Crystal Reports software, the runtime version of which is provided with TPS Builder.

Create your own reports

Crystal Reports is available for purchase from Seagate. If you purchase a copy of Crystal Reports, you can modify the reports provided, or define your own reports and add them to those accessed from the TPS Builder Reports menu.

For more information, refer to the Seagate Crystal Reports website:

<http://www.crystalinc.com/crystalreports/>

What do you want to do?

If you want to...	Then go to...
Access or print a report	Basic Report Procedures
View a report online	Previewing a Report
Transfer report data to another application, such as a spreadsheet	Exporting a Report

Additional information

Details on using the Reports dialog box, as well as information about each type of report in TPS Builder, are provided in the TPS Builder online Help.

11.2 Basic Report Procedures

Access a report

Follow these steps to access any TPS Builder report:

Step	Action
1	To print a report dealing with a specific project or library, make sure the window of the project or library you are interested in is open and active.
2	Choose Reports from the Reports menu.
3	Choose one of these activities to view the report: • Print a report • Previewing a Report • Exporting a Report

Print a report

Every report can be printed. You can print more than one report at a time if desired.



ATTENTION

All reports supplied by TPS Builder are designed for 8.5-by-11 inch paper.

Follow these steps to print a report:

Step	Action
1	To print a report dealing with a specific project or library, make sure the window of the project or library you are interested in is open and active.
2	From the Report menu, choose Report.
3	In the Report dialog box, click in the print-selection box,  , of each report you want to print. An "X" in the box indicates a report is selected for printing. To clear a selected report, click in the print-selection box again.
4	When all reports are selected for printing, click Print .
5	Complete the Print dialog box and click OK .

11.3 Previewing a Report

How previewing works

Every report can be previewed onscreen. Previewing a report launches Crystal Reports in a separate window.

Using the preview window

Note the following:

- The report window remains open as long as the Reports dialog box is open in TPS Builder. When the Reports dialog box closes, the report window also closes.
- The report window can be sized using standard window controls.
- The report window has its own set of controls, as described in the TPS Builder online Help.

Report previewing steps

Follow these steps to preview a report:

Step	Action
1	If you want to view a report dealing with a specific project or library, make sure the window of the project or library you are interested in is open and active.
2	From the Report menu, choose Report.
3	In the Report dialog box, click to highlight a report name of interest. (Do not click in the print-selection box, )
4	Click the Preview button. The report will appear in a separate window.
5	Size and position the report window as desired, and use the report controls to navigate throughout the report, or to perform other actions, as described in "Preview controls."
6	If desired, you can preview additional reports. A separate Crystal Report window will open for each report selected. You cannot, however, change the project or library selected in step 1.
7	To close a report window, either use the X button in the upper right corner of the window, or close the Reports dialog box in TPS Builder.

11.4 Exporting a Report

How exporting works

Exporting transfers report data from Crystal Reports to another file. Every report can be exported to a file.

Supported file formats

A number of supported file formats are available, for example:

- Character-, comma-, or tab-separated values
- Different versions of Microsoft Excel
- Different versions of Lotus 1-2-3
- Crystal Reports
- Data Interchange Format
- Several different ODBC formats, such as dBase, MS Access, or Paradox
- Several different text formats, such as ODBC text, Rich Text Format, tab-separated text, or Word for Windows

Report exporting steps

Follow these steps to export a report:

Step	Action
1	If you want to export a report dealing with a specific project or library, make sure the window of the project or library you are interested in is open and active.
2	From the Report menu, choose Report .
3	In the Report dialog box, click to highlight a report name of interest. (Do <i>not</i> click in the print-selection box,  .)
4	Click Export .
5	In the Export dialog box, choose a format and destination for the TPS Builder data. Click the arrow button to view all available formats and select one. Destination can be a file in the location of your choice, or an Exchange or Microsoft Mail (MAPI) file.

11. Working with Reports

11.4. Exporting a Report

Step	Action
6	When the format and destination are chosen, click OK .
7	Depending on the format chosen, additional dialog boxes may present additional options. For example, if you choose character-separated values, you will be asked to define the characters to be used as separators (defaults are provided). Accept the defaults, or type in another separator character and/or quotation mark character, and click OK .
8	For many formats, the number and date format can match the report, or you can choose a new format(s). To keep the TPS Builder formats, check the appropriate boxes in the Number and Date Format dialog box when it appears, then click OK .
9	Once all options are completed for a disk file destination, the Choose Export File dialog box will appear. In this dialog box, choose a directory path and file for the exported report.
10	Click Save and the report will be exported as specified.

11. Working with Reports

11.5. Including a Custom Report in TPS Builder

11.5 Including a Custom Report in TPS Builder

Custom reports and TPS Builder

If you have Seagate Crystal Reports, you can create custom reports and add them to the TPS Builder Reports dialog so that they are accessible from within TPS Builder.

The version of Crystal Reports you have must be compatible with the version of the MS Access or other database you are using with TPS Builder.

The MS Access tables and queries can be used directly in the reports (source: cs95.mdb or cs97.mdb), using the ODBC data source configuration process outlined in Crystal Reports documentation.

Create the report

Follow these steps to create a report you want to include in TPS Builder:

Step	Action
1	Determine the TPS Builder data you want included in the report.
2	Use Seagate Crystal Reports to create a custom report. Follow instructions provided with that product. Ensure the report name includes the extension of ".rpt."
3	If you want your custom report to reside where TPS Builder standard reports reside, then save the report to the TPS Builder Report folder. (Default location is: Program Files\Honeywell\Tps\Tools\Builder\Report\.)

Include the report in TPS Builder

Follow these steps to make a custom report accessible in the TPS Builder Reports dialog box:

Step	Action
1	On the Reports menu, choose Reports.
2	In the Reports dialog box, click Configure.
3	In the Configure Reports dialog box, click the ... button to locate the folder containing the report you want to add.

11. Working with Reports

11.5. Including a Custom Report in TPS Builder

Step	Action
4	When you locate the folder, double-click it, then click OK. The reports available in that folder will be listed in the Configure Reports dialog box.
5	In the left column, click the report you want to add.
6	For Report Descriptor, enter the name you want to appear in the Reports dialog box.
7	Click Add. The report is added to the end of the list of included reports (right column).
8	If desired, in the right column, drag the report name to the location in the list where you want it to appear.
9	Repeat steps 5 - 8 for each additional report you want to include.
10	When finished, click OK. The report(s) will appear in the Reports dialog box.

11. Working with Reports

11.5. Including a Custom Report in TPS Builder

12. Importing TPS Entity Data from TPS Files (Backbuilding)

12.1 Overview

What is backbuilding?

In general terms, backbuilding is the process of transferring control system data away from the control system, usually into an off-line database, for the purposes of reconfiguration or documentation.

Within TPS Builder, backbuilding is the importing of TPS system entity data (point and node), from Exception Build (EB), JS, or other files, into the TPS Builder database. These entities appear in the form of TPS Builder strategy blocks (and in certain cases, strategy modules), within a selected project.

Files that can be backbuilt

File types that can be backbuilt into TPS Builder include:

EB files: Text files that define TPS Network entities by listing their parameter names and values. EB backbuild capability exists when you install TPS Builder's LCN/UCN libraries.

JS files: Text files that define Logic blocks of Multifunction Controllers or Advanced Multifunction Controllers, by listing their number, algorithm, inputs, and outputs. Backbuilding of JS files is possible when you purchase and install the Hiway Gateway libraries.

Discussion

The following table identifies different types of backbuilding and discusses each one.

Backbuilding...	Discussion
Into a new project	For new projects, backbuilding is straightforward. The data is simply written to the database using TPS Builder's Application Program Interface (API). As part of the backbuild process, a system template is associated with each backbuilt point according to the point's type.

12. Importing TPS Entity Data from TPS Files (Backbuilding)

12.1. Overview

Backbuilding...	Discussion
Into an existing project	For an existing project, TPS Builder checks the project (including all modules within the project) to see if the entity exists. If it does already exist, then you can choose to overwrite each of its parameters. If you do, TPS Builder a) Checks to see if the file's parameter value is different from the project database parameter value and if so, then b) Checks to see if the existing parameter is template defining. If the parameter is not template defining, then the value will be written to the project database; if it is a template-defining parameter, then you will receive an error message and it will be logged in the Backbuild Log. Note that if backbuilding node strategies, overwrite will occur. (It is not optional.) The backbuild for points containing custom data segments is a bit more complex, as explained in this section, Backbuilding Custom Data Segments.
Node entities	A single template is included for each node type supported by TPS Builder. For the UCN, nodes must be defined by two entities: \$NMxxNyy, where xx is the network number, and yy is the node number \$NMxxByy, where xx is the network number, and yy is the node number When node entities are backbuilt, all parameters from these two entities are merged into one TPS Builder node strategy. If the entities are backbuilt into a new project, the TPS Builder strategy is named UCN##node## where UCN## identifies the network number. node## is the node abbreviation (for example, HPM), followed by the node number. This single TPS Builder strategy represents both UCN entities.
Custom data segments	To backbuild Application Module (AM) points that have custom data segments (CDSs), special steps must be taken, as explained in this section, Backbuilding Custom Data Segments.

Order of parameters in the EB file

When backbuilding EB files, if parameters are not ordered correctly for exposure, some parameters may not backbuild correctly the first time.

Example: For a Regulatory PV point with Summer algorithm, PVEQN should precede PISRC(1) and PISRC(2). If PVEQN follows PISRC(2), then PISRC(2) is not exposed, so it will not be backbuilt. A message is written to the Backbuild Log to that effect. In this case, the point can be backbuilt a second time and PISRC(2) will be built correctly.

Therefore, when the Backbuild Log opens after a backbuild and indicates that some parameters were not found, then you can try backbuilding a second time. Also, you can check the EB file and see if some parameters that control exposure were not ordered correctly. For the second backbuild, you may choose only those entities affected by parameter order. Choose the Overwrite Existing Entities option.



ATTENTION

Regulatory PV points are particularly sensitive to the issue or parameter order. Plan on backbuilding files twice to resolve all problems.

Release-specific details

Regarding TPS network releases, note that if the conversion is from a pre-R510 project, then parameter values in the earlier release that do not apply in R510 and later are treated as follows:

When upgrading from an R400 series project,

- R400 APM digital composite and device control points: The MAXTIMx parameter will be excluded.
- R400, R410, R420, or R430 digital composite points: The ALMOPT parameter will be excluded.

When upgrading from any R400 series project, the values in ALPRIOR will not be transferred to the appropriate alarm priority parameters.

In all cases, entries will be made in the Backbuild log identifying the parameters that have been excluded.

If numerous points are involved, you can transfer the ALPRIOR values to the appropriate alarm priority parameters using the TPS Builder bulk edit function.

12. Importing TPS Entity Data from TPS Files (Backbuilding)

12.1. Overview

Backbuild log

This log records a message for any transaction that is not committed during the backbuild process. If messages are written to the log, it displays them at completion of the backbuild.

Refer to Working with Logs in this document.

Backbuild processing time

The amount of time it will take to complete a backbuild depends on several factors, such as

- the size of the file to be backbuilt,
- the complexity of the entities being backbuilt (for example, device control or logic points take longer),
- whether you are building into a new or existing project,
- whether you are building into a new or existing database (and its size), and
- your personal computer's configuration.

For an empty TPS Builder database, backbuild rate is approximately 1 entity per second on a minimum platform computer (Pentium Pro 200 with 64 MB RAM). If you are backbuilding into a large, established database, or are building complex points such as logic and/or device control points, this rate may be reduced significantly.

You can use other applications while the backbuild runs.

Increasing backbuild performance

A few tips will help decrease the time it takes for backbuilding:

- Backbuild I/O points first, then backbuild control points. Using this order enables connections to resolve immediately (eliminating "second pass" backbuilding, or resolving connections separately).
- Do not open the window of the project you are backbuilding to. This eliminates the time it takes TPS Builder to refresh the screen as points are added to the project.
- If you have fair confidence in the file(s) you are using, turn off logs (from the Options menu, choose Enable Logging to toggle it off). This eliminates the time it takes TPS Builder to write errors to the Backbuild log. Of course, if you take this step and have problems, you will have no record of the problems.
- Be mindful of the space backbuilt points take on your hard drive, and compact files regularly. For example, in one case 11,000 backbuilt points equaled a project of nearly 500 MB. After compression, the same project was about 43 MB.

For details, refer to the Database Administration section of the *TPS Builder Installation, Implementation, and Administration Guide* Compact the Database.

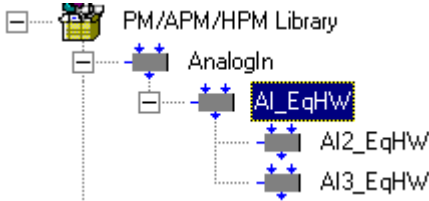
12. Importing TPS Entity Data from TPS Files (Backbuilding)

12.1. Overview

How to choose a template for backbuilding

A template is assigned to every entity backbuilt into TPS Builder. As you specialize templates, they become available as the basis for backbuilding. The template used for backbuilding, by default, is the specialized "Eq/HW" template provided by Honeywell, for example, AI_Eq/HW."

To choose a specialized template for backbuilding:

Step	Action
1	Locate the specialized template you want to use and determine its corresponding system template. For example, AnalogIn is the system template for all analog input templates: 
2	Right-click the system template and choose Identification in the resulting menu.
3	In the Identification dialog box, choose the template you want to use for the next backbuilding session.
4	Click OK . The selected template will be used for backbuilding of all related point types until you make another selection.

12.2 Backbuilding EB or JS Files

Planning

Planning is key to successful backbuilding. Planning includes the following tasks. These tasks are applicable to:

- the backbuilding of EB files if you have installed the TPS Builder LCN/UCN libraries;
- the backbuilding of JS files if you have purchased and installed the TPS Builder Hiway Gateway libraries.

Follow these steps before backbuilding EB or JS files:

Step	Action
1	Determine (and create if necessary) the TPS Builder project(s) in which you want the backbuilt entities to reside. Once the entities are in a project, you cannot move them to another project. (If you change your mind, you would need to delete the entities from the project they were added to, and repeat the backbuild process for the desired project.)
2	Locate the EB or JS files and ensure they are accessible from your computer. (For example, if they are located on the network, make sure you can access them.) One approach is to copy these EB or JS files to the TPS Builder default directory, because the directory path will be entered into the Backbuild dialog box automatically. The default directory is <i>Projectx\build</i>, where <i>Projectx</i> is the name of your project. (The complete directory path depends on your TPS Builder installation.)
3	Research the EB or JS files to determine whether you will backbuild all, or some of the entities within each file.
4	Research the EB files to determine whether they include Application Module (AM) Regulatory, Custom, or Switch points that have custom data segments (CDSs). If so, complete "Backbuild Custom Data Segments" before you backbuild these points.
5	Research the EB files to determine whether they include logic block data points. If so, the information about the algorithms configured must be backbuilt from corresponding JS files. Plan to backbuild both the EB and JS files.
6	If you have many entities to backbuild, ensure that you have adequate time for the procedure. See "Backbuild processing time."

12. Importing TPS Entity Data from TPS Files (Backbuilding)

12.2. Backbuilding EB or JS Files

Step	Action
7	Consider the strategy module structure that you want for these entities, and create empty modules within the project. You do not need to do this before backbuilding, but you will need to do it at some point to organize the backbuilt entities.
8	If you have many entities to backbuild, decide on the process you want to use: • Create the module structure, backbuild entities for a particular module, generate that module, then continue with entities for the next module, and so on. Or, • Backbuild all entities and then generate the needed modules. Determine the process you want to follow before you begin.

How to backbuild the data

Follow these steps to backbuild TPS Network entity data from EB or JS files into a TPS Builder project:

Step	Action
1	Complete "Planning."
2	If you have any Application Module points that have custom data segments (CDSs), make sure that you complete the CDS backbuild procedure before continuing. Refer to Backbuilding Custom Data Segments in this section.
3	With at least one project, library, or drawing in the TPS Builder workspace, choose Backbuild from the Tools menu. (You may want to open the project that will receive the backbuilt entities, but it is optional.)
4	Complete the Backbuild dialog box, choosing the project, EB or JS file(s), and overwrite options. • If a single file is selected and only some of the entities are to be backbuilt, then choose those entities. • If you are backbuilding AM points with CDSs, for which strategies have already been created make sure you choose the Overwrite Existing Entities option. • If you are backbuilding strategies that name yet-to-be built strategies as connections or references, deselect the Resolve Connections option.
5	When the Backbuild dialog box is complete, click Backbuild . The process will begin, and a progress indicator will display the progress for each entity. Once the entities are backbuilt, the progress indicator will note that connections are being resolved.

12. Importing TPS Entity Data from TPS Files (Backbuilding)

12.2. Backbuilding EB or JS Files

Step	Action
6	If any errors are encountered during the backbuild, the Backbuild Log will display. Evaluate the messages, and make corrections as needed, as discussed after this procedure.
7	Use module generation as required to add the new strategy blocks to strategy modules. Refer to Generating a Module Automatically in the Working with Strategies section of this document.

Considerations regarding existing entities

When backbuilding entities that are already in a target project (that is, entity names and types in the EB file match strategy names and types in the project), you may choose to overwrite entities in the project. When overwrite is chosen, the parameter values (EB file and project) are compared and an overwrite takes place when the two values differ.

No action is taken for a parameter in the project if it does not appear in an EB file. Consequently, if only some of the parameters for an existing entity are backbuilt with the overwrite option, the other parameters for that entity remain as they were before the backbuild. Therefore, all parameter values to be changed, including those to be removed, must be backbuilt.

EB files backbuilt from the TPS system are complete. EB files produced by configuration tools, however, often do not list all entity parameters. Therefore, care must be taken when overwriting entities using EB files produced by these tools.

If in doubt about the effects of overwriting entities during a backbuild, the integrity of a backbuilt point can be ensured by deleting the questionable entities from the target project before backbuilding.

How to resolve backbuild errors

Some errors, particularly those related to names, are not identified by TPS Builder in R110:

- Tagnames are checked for length only, and other types of errors, such as spaces in the name, are not identified.
- If backbuilding a CDS, PKGNAME

Take care to ensure these names are correct. For other errors, view the Backbuild Log, and resolve the errors noted.

12. Importing TPS Entity Data from TPS Files (Backbuilding)

12.3. Resolving Connections

12.3 Resolving Connections

Description

TPS Builder can check all connection-related parameters, to determine if the points named as connections exist. If they do exist, the TPS Builder database records the information accordingly. If any do not exist, then a plain text string is entered for the connection parameter.

How to resolve connections

Follow these steps to resolve connections after a backbuild, or any time following configuration:

Step	Action
1	Open the project to be checked.
2	Select the project name.
3	From the Connection menu, choose Resolve Connections

TPS Builder checks to see if any unresolved connections exist. If not, no action takes place. If unresolved connections exist, then a progress dialog box displays. When the process completes, the dialog box disappears.

Canceling

You can cancel connection resolution by clicking the Cancel button in the progress dialog box.

12.4 Backbuilding Custom Data Segments

Description

Application Module (AM) Custom, Regulatory, or Switch points can include custom data segments (CDSs). For a review of CDSs, you may want to see the .

CDS data on one of these points can be backbuilt into TPS Builder by following the steps presented in the following paragraphs.

What the process does

The process essentially:

- creates custom parameters as defined by parameter names in the CDS,
- creates a parameter group named with the name of the CL file containing the CDS, and
- places the custom parameters in the parameter group.

What the process does not do

The process does *not*:

- create the strategies of the AM points or
- add any parameter configuration data to the strategy.

Process overview

To backbuild AM points with CDSs, the following steps are required:

- Template preparation
- Strategy preparation
- Backbuild point data

12. Importing TPS Entity Data from TPS Files (Backbuilding)

12.4. Backbuilding Custom Data Segments

Refer to the following paragraphs.

Template preparation

Step	Action
1	Specialize an AM template that most closely meets your needs (view membership, template-defined parameters, and so forth). If desired, move the new template to a library other than the one in which it was specialized.
2	Backbuild the CL file that contains the CDS you want to include.
3	From its library, right-click the specialized AM template you created in step 1.
4	On the resulting menu, choose Parameter Groups.
5	Make sure that the Load Custom Parameter Groups Only option is selected. In the Parameter Groups, locate and select the parameter group with the name of the CL file you backbuilt.
6	Click the ADD ---> button to add the new CDS parameter group to the template. Click OK .
7	If the Honeywell-provided AM library is not open, open it. Select the system template upon which the specialized template is based. For example, if the template is for an AM Regulatory point, right-click the Regulatory template in the AM library.
8	In the Identification dialog box, choose the new template as your Backbuild Template.

The specialized template is now ready to be used to create strategies that can accept CDS data during a regular backbuild.

Strategy preparation

Open or create the project that is to contain the strategies with CDSs (the "target" project).

How to backbuild point data

Complete a normal backbuild of the EB file containing the AM points with the CDSs to capture all configuration data, including that of the CDS's custom points. Make sure the specialized template created in the previous steps is chosen as the backbuild template.

When backbuilding, make sure the Overwrite option is set.

Refer to "Backbuilding EB or JS Files."

12. Importing TPS Entity Data from TPS Files (Backbuilding)

12.4. Backbuilding Custom Data Segments

How to deal with multiple CDSs

If the EB file includes different points with different CDSs, then follow these steps, either

- split the EB file into multiple files, each one containing only points with like CDSs, and then backbuild each file separately, with a separate, applicable template chosen for each file, or
- perform multiple backbuilds as explained in the following table:

Step	Action
1	Specialize as many templates as needed, with the needed parameter groups added. For example, if you had three points, with CDSs as follows: • CP1 with CDS A • CP2 with CDS A • CP3 with CDS B
2	Choose the first template for backbuilding, and in the Backbuild dialog box, select only those strategies with the corresponding CDS. In our example, this would be the template containing parameter group A, and you would select only CP1 and CP2 for backbuilding.
3	Choose the next template for backbuilding, and in the Backbuild dialog box, select only those strategies with the corresponding CDS. In our example, this would be the template containing parameter group B, and you would select only CP3 for backbuilding.
4	Repeat as needed until all backbuilding is completed.

Check names carefully

Refer to “How to resolve backbuild errors.”

12. Importing TPS Entity Data from TPS Files (Backbuilding)

12.5. Backbuilding CL Files

12.5 Backbuilding CL Files

Description

CL files containing custom data segments and parameter lists can be used to create custom parameter groups, custom parameters, and custom enumerations in TPS Builder. The process of converting CL files to TPS Builder components is another facet of "backbuilding."

How to backbuild CL files

To backbuild a CL file, follow these steps:

Step	Action
1	Will the custom parameters and parameter group to be created be new? If not, delete the existing parameters and parameter group. (You cannot overwrite these items with a CL file backbuild.)
2	Locate the CL file containing the CDS or parameter list and ensure it is accessible from your computer. (For example, if it is located on the network, make sure you can access it.) It is recommended that all TPS Builder-related CL files be kept in <i>Projectx\program</i> , where <i>Projectx</i> is the name of your project. (The complete directory path depends on your TPS Builder installation.)
3	With at least one project, library, or drawing in the TPS Builder workspace, choose Parameter Groups from the Parameter menu.
4	In the resulting dialog box, click Backbuild CL File . This action opens the Windows file browser so that you can locate the directory containing the file you want to backbuild.
5	Locate the needed CL file, select it, then click Open .

What is created

In accordance with the CL file, TPS Builder creates the custom parameters, enumerations, if included, and a parameter group.

Errors

If any syntax errors exist in the CL file, the backbuild will not create anything. An error message will display in this event.

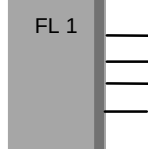
12.6 Cleaning Up Backbuilt Logic Drawings

Description

When you backbuild logic or device control points, modules including all logic blocks are created, including the module drawings. In most cases, you will need to perform a "clean up" of the drawings.

How to clean up backbuilt logic drawings

To clean up logic drawings:

Step	Action
1	If not familiar with drawings in TPS Builder, refer to the "Working with Drawings" section of this document.
2	Move all blocks into the area you want to use, for example, a single page.
3	For numerics and flags with many connectors, typically FL1 and FL2, make the block bigger, so that connectors are clearly separated To do so, click to select the drawing symbol, then position the cursor over a handle in the middle top or bottom, and when the cursor changes to a vertical double arrow, drag up or down. This step also fixes "wacky" lines you may see when you first open a backbuilt logic drawing.
	Before  After 
4	The TPS Network default for unconnected pins is LI1. Prior to TPS Builder, you had to explicitly configure connections to FL1 (logical off) and FL2 (logical on) for "unused" pins. In TPS Builder the defaults are • FL2 (logical on) for AND and NAND gates; • FL1 (logical off) for other gates. This means that you can safely delete the connections from FL1 and FL2 to the gate inputs if they are unused.
5	Put all blocks that do not have connections at the top or bottom of the drawing.
6	Move gates, numerics, and flags so that you have as many lines as possible going from left to right.

12. Importing TPS Entity Data from TPS Files (Backbuilding)

12.6. Cleaning Up Backbuilt Logic Drawings

Step	Action
7	Reposition connection lines that overlap and cross other blocks.
8	Repeat steps 6 and 7 to improve the drawing further.

Tips

These tips will help you clean up the drawing:

- If multiple lines overlap, start with one line that you can see a corner for, move it out of the way, and continue doing this until everything is clean.
- If you want to see where a line is going, click it. All segments of the line will have handles. Follow the handles to see the line's path.
- When moving lines, depending on your needs, turn Auto Line (in the Connections menu) on or off. With Auto Line on, TPS Builder will autoroute all lines connected to a block that you move. With Auto Line off, TPS Builder will preserve as much of the current line as possible while a block is moved.

Sometimes screen refresh problems occur that affect the appearance of your drawing (for example, connections do not line up nicely or pieces are missing). The easiest way to force a refresh of the screen is to slightly move a block, with Auto Line off.

13. Creating an EB or EL File

13.1 Overview

Description

This overview discusses the process of creating text file, named with extensions of .EB or EL, of TPS system entity (node and point) parameter data in a TPS Builder project. The EB file follows the syntactic and format specifications required by the TPS system platform.

Two functions are involved:

- Setting forward building options, such as choosing the entities to be included and EB file specifications.
- TPS Builder performs the forward building task and creates the EB file(s).

Forward building can also apply to the creation of other types of files, depending on the TPS Builder libraries that you have purchased and installed. If you have purchased and installed the Hiway Gateway libraries, JS files can be forward built. In that case, refer to the applicable online Help. From the Help menu, choose Contents, Hiway Gateway Library Help, then the applicable topic.

Output

Each entity is output to the file with a header comment that includes the entity name. For example, {SYSTEM ENTITY RC100}.

Following all parameter data for each entity, a blank footer line is added, { }.

EB file size constraints

TPS Builder considers the TPS system limitations on the size of EB files. Namely, files cannot exceed the following constraints:

- | | |
|--------------------------|--------|
| • Number of points/file: | 250 |
| • Number of lines/file: | 5000 |
| • Number of bytes/file | 90,000 |

If these limits are exceeded, TPS Builder automatically generates multiple files, named as explained next.

13. Creating an EB or EL File

13.1. Overview

File naming

TPS Builder automatically adds a suffix to the name you specify for the EB file. This allows TPS Builder to automatically create additional files if any of the previous limits is exceeded. If the initial file name

- is less than seven characters, the suffix will begin with "AA" (myfileAA.eb)
- is seven characters, the suffix will begin with "A" (thefileA.eb)

When a file reaches maximum size, the new file(s) created is named with the next logical name, incrementing through the alphabet to compose the suffix (myfileAB.eb, myfileAC.eb, and so on).

Should the forward build require more file names than can be automatically generated, a dialog box will prompt you to enter a new file name.



ATTENTION

To overwrite or append to an existing EB file, do not include the suffix in the name you specify in the Forward Build dialog box. For example, if the first forward built file is myfileAA, and you want to overwrite the file, specify just "myfile" in the dialog box.

If you have purchased and installed the Hiway Gateway libraries, refer to the associated TPS Builder online Help for information about JS files.

13.2 Basic Forward Building Procedures

Description

Forward building is the process of creating an Exception Build (EB) file from TPS Builder configuration data.

Planning

Planning is key to successful forward building. Planning includes the following tasks. These tasks are applicable to the forward building of EB files if you have installed the TPS Builder LCN/UCN libraries.

Before forward building:

Step	Action
1	Decide which entities are to be forward built: • All strategies within the project. • A particular subset of the TPS system strategies (such as all AM points, or all HPM Analog Input points). • Specific strategies that you select in the TPS Builder project, or from an entity list available from the Forward Build dialog box. • Strategies as named in an entity list (EL) file(s).
2	Decide these aspects of the EB file that will be created: • How will the entities be sorted (by point name or hardware or TPS Loading Order *)? • In which directory will the EB file reside? • The name of the EB file? • If the EB file already exists, do you want to overwrite it, or append it?

* TPS Loading order functionality controls the order in which the points are built so that the loading errors are minimized. On selecting the **TPS Loading Order** sort criteria, the AI, AO, DI, DO, Flag, Numeric, Timer, Process Module, and Array points will be forward built first, followed by the Regulatory PV, and Digital Composite points. Next, the Regulatory Control points are forward built and lastly the Logic, and Device Control point are forward built.



ATTENTION

If a download failure occurs, select the points manually and download them.

Forward build the EB file

To create an EB file from TPS Builder project data:

Step	Action
1	Complete planning as discussed previously.
2	Open the project containing the entities to be included in the forward build (or make its window active if it is already open).
3	If all entities are to be forward built, then select the project name in the project window. If some entities are to be chosen, you can: • select them in the project, use Shift+click or Ctrl+click methods to select multiple items, or • wait and select the entities when you complete the Forward Build dialog box in step 5.
4	From the Tools menu, choose Forward Build.
5	Complete the Forward Build dialog box.
6	Click OK to create the EB file. In response to OK, the process will begin, and two progress indicators will display sequentially. You can halt processing by clicking Cancel on either progress indicator. In this case, an incomplete EB may be created.
7	When the progress indicators disappear, you can repeat this procedure to forward build additional entities; or you can click the Cancel buttons to close the Forward Build dialog box.

View EB file contents

You can view the contents of an EB file with any text editor. You can associate your favorite text editor with EB files by following the Windows procedure for associating files of a given type (.eb in this case).

Modify EB file contents

If you choose to modify an EB file:

- Make sure that the file is saved as a text file.
- Make sure you follow the syntactic and format specifications required by the TPS system platform.



ATTENTION

Changes to an EB file will not be reflected in your TPS Builder project unless you backbuild the modified files back into TPS Builder.

13.3 Creating an Entity List File

Description

You can create an entity list (EL) file for use with such functions as checking or forward building in TPS Builder. An EL file is a text file, named with an extension of .EL, containing names of TPS system entities in a TPS Builder project. Each entity name is presented on a separate line in the file.

Planning

Before creating an EL file, follow these steps:

Step	Action
1	Decide which entity names are to be included in the file: • All entities within the project. • A particular subset of the TPS system entities (such as all AM points, or all HPM Analog Input points). Specific entities that you select in the TPS Builder project.
2	Decide these aspects of the EL file that will be created: • How will the entities be sorted (by point name or hardware or TPS Loading Order *)? • In which directory will the EL file reside? • The name of the EL file? If the EL file already exists, do you want to overwrite it, or append it?

13. Creating an EB or EL File

13.3. Creating an Entity List File

Step	Action
	* TPS Loading order functionality controls the order in which the points are built so that the loading errors are minimized. On selecting the TPS Loading Order sort criteria, the AI, AO, DI, DO, Flag, Numeric, Timer, Process Module, and Array points will be forward built first, followed by the Regulatory PV, and Digital Composite points. Next, the Regulatory Control points are forward built and lastly the Logic, and Device Control point are forward built.



ATTENTION

If a download failure occurs, select the points manually and download them.

Procedure

To create an EL file from TPS Builder project data:

Step	Action
1	Complete planning as discussed above.
2	Open the project containing the entities to be included in the entity list (or make its window active if it is already open).
3	If all entities are to be included in the EL file, then select the project name in the project window; or if some entities are to be included, and you want to select them in the project, see Select/Deselect Items in a Window. Alternative: you can wait and select the entities when you complete the Build Entity List File dialog in step 6.
4	From the Tools menu, choose Build Entity List File.
5	Complete the Build Entity List File dialog box.
6	Click OK to create the EL file (or click Cancel to close the dialog without creating an EL file). In response to OK, the process will begin, and two progress indicators will display sequentially. You can halt processing by clicking the Cancel button on either progress indicator. In this case, an incomplete EL file may be created (depending on when the cancel was invoked).
7	When the progress indicators disappear, you can repeat this procedure to create additional EL files or you can click the Cancel buttons to close the Build EL Files dialog.

Viewing EL file contents

You can view the contents of an EL file with any text editor. You can associate your favorite text editor with EL files by following the Windows procedure for associating files of a given type (.el in this case).

Modifying EL file contents

If you choose to modify an EL file, make sure that the file is saved as a text file.

Note that changes to an EL file will not be reflected in your TPS Builder project unless you modify the project accordingly.

13.4 Downloading Entities

Description

If TPS Builder is running on a Global User Station (GUS) node, or on a node that has a connection to a GUS node, the Download function is available. Use this function to load configuration data from TPS Builder to the entities on TPS Network (TPN):

- PM-family node and point configuration data: PM, APM, HPM, and EHPM
- Network Interface Module (NIM/ENIM) configuration data
- Fieldbus-related configuration data: For TPS Builder R130, the fieldbus link can be downloaded, which includes
 - HPM Control Module (CM) point data
 - Fieldbus link (device, function block, and schedule) data

Refer to the Fieldbus Implementation Guide for details on downloading fieldbus data.

Process overview

In order to download configuration data, TPS Builder performs the following activities:

- Creates an Exception Build (EB) file for selected entities.
- Loads the EB file to the History Module.
- Creates and uses Execution Command (EC) files and the EC Trigger to activate the load on the TPS Network.

13. Creating an EB or EL File

13.4. Downloading Entities

Preparation for download

Before downloading configuration data, complete the following steps:

Step	Action
1	Identify the strategies to be downloaded. You can download strategies from one project at a time.
2	Complete configuration of the identified strategies in TPS Builder.
3	Ensure that the following actions are complete: • If the points already exist, they have been set to INACTIVE. • The primary domain controller is configured and the GUS you are using for the download is part of that domain. File transfer is in service on the GUS. • The History Module to which the download is to occur has been mapped as a network drive on your computer. • Native Window is connected and running. • The Engineering main menu, or the Command Processor page of the Engineering menu, is displayed in the Native Window. • The TPN clock and GUS clock match to the minute. If needed, refer to the appropriate TPS Network documents.
4	In TPS Builder, set the online operations path.

How to download a single strategy

To automatically download the configuration data of a single strategy, to the TPN:

Step	Action
1	Complete preparation discussed previously.
2	Open the project in the needed view.
3	Locate the strategy whose data are to be downloaded.
4	Right-click the strategy and choose Load on the resulting menu. The download will complete automatically. Progress and errors are reported as discussed below.

How to download multiple strategies

To download configuration data of multiple strategies within a project, to the TPN:

Step	Action
1	Complete preparation discussed previously.
2	Open the project in the needed view.
3	Locate the strategy whose data are to be downloaded.
4	Click one of the needed strategies, then Ctrl+click additional strategies. Or, you can choose them in the Forward Build/Download Entities dialog box (step 6).
5	Choose Tools > Forward Build/Download Entities .
6	Complete the Forward Bid/Download Entities dialog box. If you did not select strategies in step 4, then choose them from this dialog box.
7	Click OK . The download will complete, with progress and errors reported as discussed below.

Progress indicator

A progress indicator shows specific progress of each of the strategies being downloaded.

Download errors

Errors are recorded in the TPS Builder Error log.

Errors are also documented in the TPN error file (.ef file). Refer to appropriate documents in your electronic or paper TPN bookset for details on this type of error file. For example, refer to the *Data Entity Builder Manual*.

13.5 Setting the Online Operations Path

Description

In order to download configuration data from TPS Builder to nodes and points on the TPS Network (TPN), online operations data must be specified for TPS Builder.

Preparation

Before setting the online operations data in TPS Builder:

Step	Action
1	Identify the History Module (HM) volume and directory into which configuration data files will be downloaded.
2	On your GUS node, identify the drive designated for the HM. If the HM has not yet been mapped as a network drive on the GUS, then map it now.
3	Determine a name for the EC file that will be created. Use a name of five characters. Do <i>not</i> include the extension, ".ec." (TPS Builder will do this.)
4	Identify the TPS Builder project-related folder in which you want Exception Build (EB) files created. (TPS Builder creates the EB files for your project first, and then loads them to the HM in a separate step). The default location for these files is <i>Projectx\lbuild</i> , where <i>Projectx</i> is the name of your project. (The complete directory path depends on your TPS Builder installation.) You may choose another location if desired, however.
5	Note whether there is any reason you would <i>not</i> override existing entity data with the data of your current download.



ATTENTION

A TPS Builder download will fail unless you choose to override existing entity data. If you cannot override existing data, then do not use the TPS Builder download function.

How to set the online operations path

To set online operations data for TPS Builder:

Step	Action
1	Complete preparation discussed previously.
2	On the Options menu, choose Online Operations Paths.
3	Complete the Default Paths for Online Operations dialog box.
4	Click OK .

These settings will be used for any download of configuration data to the TPN.

13.6 Operations by Selected Entities

Description

You can create an entity list (EL) file for use with such functions as checking or forward building in TPS Builder. An EL file is a text file, named with an extension of .EL, containing names of TPS system entities in a TPS Builder project. Each entity name is presented on a separate line in the file.

Planning

Before creating an EL file:

Step	Action
1	Decide which entity names are to be included in the file: • All entities within the project. • A particular subset of the TPS system entities (such as all AM points, or all HPM Analog Input points). • Specific entities that you select in the TPS Builder project.

13. Creating an EB or EL File

13.6. Operations by Selected Entities

Step	Action
2	Decide these aspects of the EL file that will be created: • How will the entities be sorted (by point name or hardware or TPS Loading Order *)? • In which directory will the EL file reside? • The name of the EL file? • If the EL file already exists, do you want to overwrite it, or append it? * TPS Loading order functionality controls the order in which the points are built so that the loading errors are minimized. On selecting the TPS Loading Order sort criteria, the AI, AO, DI, DO, Flag, Numeric, Timer, Process Module, and Array points will be forward built first, followed by the Regulatory PV, and Digital Composite points. Next, the Regulatory Control points are forward built and lastly the Logic, and Device Control point are forward built.



ATTENTION

If a download failure occurs, select the points manually and download them.

Procedure

To create an EL file from TPS Builder project data:

Step	Action
1	Complete planning as discussed previously.
2	Open the project containing the entities to be included in the entity list (or make its window active if it is already open).
3	If all entities are to be included in the EL file, then click the project name in the project window. If some entities are to be selected, and you want to select them in the project, use Shift+click or Ctrl+click to make selections, or you can wait and select the entities in the Build Entity List File dialog box (step 5).
4	From the Tools menu, choose Build Entity List File .
5	Complete the Build Entity List File dialog box.
6	Click OK . The process will begin, and two progress indicators will display sequentially. You can halt processing by clicking Cancel on either progress indicator. In this case, an incomplete EL file may be created (depending on when the cancel was invoked).
7	When the progress indicators disappear, you can repeat this procedure to create additional EL files or you can click Cancel to close the Build EL Files dialog box.

Viewing EL file contents

You can view the contents of an EL file with any text editor. You can associate your favorite text editor with EL files by following the Windows procedure for associating files of a given type (.el in this case).

Modifying EL file contents

If you choose to modify an EL file, make sure that the file is saved as a text file.



ATTENTION

Changes to an EL file will not be reflected in your TPS Builder project unless you modify the project accordingly.

13. Creating an EB or EL File

13.6. Operations by Selected Entities

14. PHD Configuration

14.1 Overview

Description

Honeywell's Process History Database (PHD) provides for the real-time data capture, historization, and intelligent retrieval of process data required for support of plant information systems.

You can use TPS Builder to provide basic PHD configuration support for LCN/UCN and Data Hiway points. TPS Builder will generate tag files based on the configuration, which you will then have to load to PHD using PHD tag loader.

Creating PHD configuration files

PHD configuration files are created from TPS Builder using the following procedure:

Step	Action
1	Define parameters to be historized for templates within the LCN/UCN and Data Hiway and CLM libraries. Strategies created from the templates will have the defined parameters eligible for historization.
2	Strategies (points) themselves become eligible for historization when the TPS Builder-specific parameter, Historization, is set to "Yes."
3	The point.parameter data will be included in PHD configuration files when you build the files from within TPS Builder. These files may contain data for individual strategies, for any set of strategies selected within a single TPS Builder project, or for templates associated with the selected strategies ("parent tag" data).

14. PHD Configuration

14.1. Overview

PHD file types

Using TPS Builder, you can build the following types of files:

- Tag file: The basic PHD configuration file, containing the names of specific TPS Builder strategies (LCN/UCN and/or Data Hiway points), and parameters that have been defined for historization. The exact strategies included depend on two factors. First, these strategies must have the Historization parameter set to Yes. Second, these strategies must be selected when the configuration file is built.
- Engineering Units (EU) file: This file contains details about the Engineering Units defined for the strategies included in the Tag file.
- Parent Tag file: This file contains details about the TPS Builder templates used to create the strategies that are included in the Tag file. It is presented in terms of PHD's Parent tag.

Tag file example

When a Tag file alone is built, it will be similar to the following example:

```
TAGNAME,DESCR,PARENT_TAGNAME,UNITS,SRC_TAGNAME,SRC_ATTRIB,SRC_INDEXA,SRC_UNITS,HI_EXTREME,LO_EXTREME,QUANT
UM
AI_EQHW1000001.PV,,AI_EQHW.PV,*,AI_EQHW1000001,PV,0,*,105.800002,0.000000,-1
AI_EQHW1000001.PVCALC,,AI_EQHW.PVCALC,*,AI_EQHW1000001,PVCALC,0,*,116.380002,-10.580000,-1
AI_EQHW1000001.LOCUTOFF,,AI_EQHW.LOCUTOFF,*,AI_EQHW1000001,LOCUTOFF,0,*,110.000000,-10.000000,-1
AI_EQHW1000001.TF,,AI_EQHW.TF,MINUTES,AI_EQHW1000001,TF,0,MINUTES,66.000000,-6.000000,-1
DIGCMP_EQ1000002.LMSRC,,DIGCMP_EQHW.LMSRC,,DIGCMP_EQ1000002,LMSRC,0,,8192.000000,-819.200000,
DIGCMP_EQ1000002.NODINPTS,,DIGCMP_EQHW.NODINPTS,,DIGCMP_EQ1000002,NODINPTS,0,,3.000000,-0.100000,-1
DIGCMP_EQ1000002.NODOPTS,,DIGCMP_EQHW.NODOPTS,,DIGCMP_EQ1000002,NODOPTS,0,,1.500000,-0.060000,
```

EU file example

Following is an excerpt from an EU file built by TPS Builder:

%(PHDStdEU)

MINUTES

Parent Tag file example

When you build a Parent Tag together with the Tag file, the contents of the two files will be similar to the following examples:

Tag file

```
TAGNAME, DESCR, PARENT_TAGNAME, UNITS, SRC_TAGNAME, SRC_ATTRIB, SRC_UNITS, HI_EXT  
REME, LO_EXTREME, QUANTUM
```

Parent Tag file

```
TAGNAME, SRC_TAGTYPE, SRC_ATTRIB, SRC_SCANSECS, QUANTUM  
AI_EQHW.PV, F4, VALUE, 60, -1
```

14.2 Build PHD Configuration Files

Description

You can use TPS Builder to build basic configuration files for use with the Process History Database (PHD). The files you can create are the .csv files, required by PHD. The file types include:

- Tag files
- EU files
- Parent Tag files

Refer to the PHD Configuration section for a description of these files.

Preliminary actions

Execute the following steps to ensure that you have the information you need to build the files, and that the PHD configuration file you are going to build will contain the required details:

Step	Action
1	Read the PHD Configuration section to get a basic understanding of the TPS Builder process.
2	Ensure that the parameters you want to historize have been so defined. Parameters to be historized are defined on the templates from which the strategies are created. Refer to Configure Historization Attribute for information.

14. PHD Configuration

14.2. Build PHD Configuration Files

Step	Action
3	Set the Historization parameter to "YES" on template or individually on all the strategies to be historized.
4	Identify the project containing the strategies to be included in the file.
5	Decide which entities are to be included in the PHD configuration files: • All entities within the project. • A particular subset of the TPS system entities (such as all AM points, or all HPM Analog Input points). • Specific entities that you select in the TPS Builder project, or from an entity list available from the Build PHD Configuration File dialog. • Entities as named in an entity list (EL) file(s).
6	Make the following choices for the file(s) to be created: • Will you build the Tag file alone, or Tag file plus the EU file and/or Parent Tag file? • Where will the files reside (complete directory path)? • What will the file name(s) be? • If the Tag and EU files already exist, do you want to overwrite or append them? • For which type of Real-Time Data Interface (RDI) is the Tag file being built (TPN_LXS, TPN_OPC, or TPN_CM50S)? • Do you want the enumerations to be ordinal, string, or both?

Build the PHD configuration files

To create PHD configuration files based on TPS Builder project data, execute the following steps:

Step	Action
1	Complete planning as discussed in the previous section.
2	Open the project containing the entities to be included in the PHD configuration files (or make its window active if it is already open).
3	If all entities are to be included, then select the project name in the project window. If you only want to include some of the entities, select/deselect them in the window (or you can wait to select the entities when you complete step 5).
4	From the Tools menu, select Build PHD Config File...
5	Complete the Build PHD Configuration File dialog box. Refer to the Build PHD Configuration File topic in the online Help for assistance.
6	Click OK to create the PHD configuration file(s) (or click Cancel to close the dialog without creating any files). The process will begin, and a progress indicator will appear. You can halt processing by clicking the Cancel button on the progress indicator. Note, however, that if you do, incomplete files may be created.
7	When the progress indicator disappears, you can repeat this procedure to build additional PHD configuration files; or you can click the Cancel button to close the Build PHD Configuration File dialog box.

Viewing contents of a PHD file

You can view the contents of the files using a spreadsheet application, text editor, or other application capable of reading a comma-delimited file.

Modifying PHD file contents

If you choose to modify one of the PHD configuration files, make sure that

- the file is saved as a .csv file, and
- you follow the syntactic and format specifications required by the PHD.

Note that changes to a PHD configuration file will not be reflected in your TPS Builder project unless you make equivalent changes in TPS Builder.

14.3 Configure Historization Attribute

Description

TPS Builder allows you to create Tag Load files for use with the Honeywell Process History Database (PHD). In TPS Builder, the Historization attribute applies to LCN/UCN or Data Hiway templates. This attribute allows you to

- define which parameters for the chosen point type will be historized for the Process History Database (PHD), and
- to set the following history-related attributes:
 - Scan rate: Defines how often values are collected,
 - Quantum compression: Specifies the methodology to be used to compress F datatype values, and
 - High and low extremes: Refer to the following section.

High and low extremes

High Extreme (%) and Low Extreme (%): Values for floating point (referred to as "F" or "R") and integer (I) data types are historized only if they are within the range specified by the extremes (greater than the low extreme and less than the high extreme). When a floating point value is not in this range, it is clamped to the defined extreme. When either data type, floating point or integer, falls outside its range, a confidence of 0 is assigned.

Hi Extreme (%) and Low Extreme (%) fields are available when a parameter has a defined range of values; otherwise these fields are grayed out. Parameter value ranges are defined by TPS and are recorded in the TPS Builder database. Some ranges are defined absolutely (for example, the range for HPM Timer PV is 0-32000) and others are defined by reference (for example, the range of PV is the value of PVEXEULO to the value of PVEXEUIH).

The Hi Extreme (%) and Low Extreme (%) values are percentages of a parameter's range of values. The default values for both Hi Extreme (%) and Low Extreme (%) are 10%.

In TPS Builder the values for PHD HI_EXTREME and LO_EXTREME parameters are calculated as follows.

$$\text{HI_EXTREME} = \text{maximum_value} + \text{Hi Extreme (\%)} * (\text{maximum_value} - \text{minimum_value})$$
$$\text{LO_EXTREME} = \text{minimum_value} - \text{Low Extreme (\%)} * (\text{maximum_value} - \text{minimum_value})$$

For example, the PV selected for historization has PVEXEULO = 0 and PVEXEUHI = 102.9. Also, if Hi Extreme (%) = 10 and Low Extreme (%) = 5. Then,

$$\begin{aligned}\text{HI_EXTREME} &= \text{PVEXEUHI} + \text{Hi Extreme (\%)} * (\text{PVEXEUHI} - \text{PVEXEULO}) \\ &= 102.9 + 10\% * (102.9 - 0) \\ &= 113.19\end{aligned}$$

$$\begin{aligned}\text{LO_EXTREME} &= \text{PVEXEULO} - \text{Low Extreme (\%)} * (\text{PVEXEUHI} - \text{PVEXEULO}) \\ &= 0 - 5\% * (102.9 - 0) \\ &= 5.145\end{aligned}$$

When the PHD configuration file (tag.csv) is written, it will contain these HI_EXTREME and LO_EXTREME values.

TPS Builder	Example
a) Computes the high-low range of values by subtracting the minimum from the maximum values defined for the point.	$100 - (-20) = 120$
b) High extreme: Multiplies the range by the high extreme percentage specified in TPS Builder.	$120 \times 10\% = 12.$
c) High extreme: Adds the resulting value to the maximum value.	$100 + 12 = 112$
d) Low extreme: Multiplies the range by the low extreme percentage specified in TPS Builder.	$120 \times 10\% = 12$
e) Low extreme: Subtracts the resulting value from the minimum value.	$-20 - 12 = -32$

Preliminary actions

Before beginning, execute the following steps:

Step	Action
1	Determine the LCN/UCN and/or Data Hiway templates for which you want to define historization.
2	For each template, determine which parameters are to be historized.

14. PHD Configuration

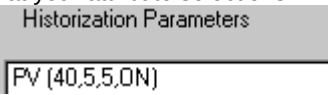
14.3. Configure Historization Attribute

Step	Action
3	Determine the values, per parameter, for the following attributes. Refer to previous discussions and the Control Attributes Dialog Box topic in the online Help for information. • Scan rate (in seconds): The default is 60 seconds. A scan rate of 5 to 30 seconds is considered high. • High extreme value (%): The default is 10%. • Low extreme value (%): The default is 10%. • Quantum compression: Decide whether you want to compress F datatype values.

Procedure

To configure the historization attribute for a selected template, execute the following steps:

Step	Action
1	Complete the preparation described above.
2	In the appropriate library right-click the template for which control attributes are to be configured.
3	In the resulting menu, choose Control Attributes.
4	In the Control Attributes Dialog Box, select the Historization tab. Refer to the Control Attributes Dialog Box topic in the online Help for information.
5	Based on your preparation (described above), in the Parameters list, click a parameter that is to be historized.
6	If the parameter represents an array, enter a valid Array Index number.
7	Enter values for Scan Rate (seconds), High Extreme (%), and Low Extreme (%).
8	If applicable to the parameter, select Quantum Compression if the data is to be compressed.

Step	Action
9	Click ADD →. The parameter name will be placed in the Historization Parameters list. Note that your attribute selections will be presented in parentheses in this list. 
10	Repeat as needed for additional parameters.
11	If you need to modify the parameters after they have been defined you can do so. Changes you make will take effect on all strategies that have been created using the template.
12	If needed, when finished with the Historization tab, click another tab in the dialog for further control attribute configuration.
13	Click OK when all control attributes are configured.

Modify the historization setting

You can make changes on the Historization tab by selecting the needed parameter in the appropriate column, clicking the ADD or REMOVE button, and then clicking OK when you're done.

You can also change the options for a historized parameter (scan rate, high extreme, low extreme, or quantum). Select the desired parameter in the Historized Parameters list, modify the options as needed, and then click OK.

Changes will affect strategies already created from the template, and will affect subsequent PHD configuration file builds.

14. PHD Configuration

14.3. Configure Historization Attribute

15. Parameter Configuration

15.1 Overview

Description

"Parameter configuration" is the process of defining all system-loadable characteristics of a particular strategy. In addition, some characteristics are set that are not directly loadable to the TPS system. For example, configuration includes setting the Historize parameter for a strategy. This parameter is used for TPS Builder purposes, in conjunction with PHD functions, and is not loaded to the TPS system.

All TPS Builder templates (except "empty" modules) have associated configuration forms. The forms for TPS system strategies will be familiar to anyone who has configured TPS system nodes and points.

Empty modules can have parameters only if they have been set as project-up (see "Related topics" in the next section).

The following topics are included. Click a specific topic, or scroll through all topics by using the scroll bar and >> button.

Related topics

The following topics are in this document:

- Basic Configuration Procedures
- Project-Up Fundamentals
- Decreasing Configuration Time
- Logic and Device Control Points
- UCN Node Configuration
- PHD Configuration

The following topics are in the online Help:

- Configuration Forms
- Configuration Form Options

The online Help also contains configuration information about optional subsystems you may have purchased and installed, such as the Data Hiway libraries.

15. Parameter Configuration

15.1. Overview

What can be configured

Depending on the subsystems you purchase and install, TPS Builder allows you to configure **TotalPlant** Network strategies, Universal Control Network (UCN) nodes and Data Hiway boxes. In addition, Communication Link Module (CLM) node and interface points can be configured.

Support for the Plant History Database (PHD) is also provided.

Identification details

TPS Builder presents identification details (some of which are not loadable) from parameter configuration. The following information is defined in the Identification dialog box:

- Name (template or strategy name). When representing a TPS Network point, this is the same as Tagname, which is a loadable parameter.
- Description
- Author
- Version

15.2 Basic Configuration Procedures

View default values

Every configuration form includes default values. These are values that will be included unless you change them. To view default values:

Step	Action
1	Open a library of interest.
2	Right-click the system template of interest that has not been modified, and on the resulting menu, choose Parameters .
3	Scroll on each tab of the configuration form, and click different tabs if needed until you find a parameter(s) of interest; or to quickly find a specific parameter(s), use Options > Parameter Configuration > Alphabetical Flat List for a single, alphabetized list of parameters.
4	See also "Recall a default value."

For TPS system entities, the defaults match those assigned by the system, as defined in the TPS system parameter reference dictionaries.

Recall a default value

If you change a default value in a template, you can recall the default value by placing your cursor in the parameter field and pressing **Ctrl+D**. (If no default exists, the pressing Ctrl+D will not change the entry. For example, no default exists for PTDESC. The field is initially blank. If you add something to this field, pressing Ctrl+D will not remove it.)

Basic parameter configuration

The basic configuration procedure is as follows:

Step	Action
1	Choose a format for the configuration forms you will be using. Refer to the <i>TPS Builder Installation, Implementation, and Administration Guide</i> , Software Configuration section, .
2	Create or select a strategy or template (usually specialized), or the associated drawing symbol in a TPS Builder drawing.
3	Right-click the selected item.
4	On the resulting menu, choose Parameters.

15. Parameter Configuration

15.2. Basic Configuration Procedures

Step	Action
5	In the resulting configuration forms, for parameters with default values, evaluate the defaults and change if needed. Refer to the appropriate reference manual for details about specific parameters.
6	Evaluate the Tag Name. In TPS Builder R110, complete tag name checking is not performed. The name is checked for length, but other items are not checked (for example spaces in the name). Correct the name if needed.
7	Evaluate all template-defined parameters to ensure that their values are correct for your current needs. • If not, locate the correct template, create a new strategy or specialize a new template, and then continue. (You will probably delete the first one, although you may determine a use for it.) • In some cases, you may need to change the template-defined parameter on the template on which it is defined.
8	Once template-defined parameters are verified, you may want to hide them in your configuration forms to speed up configuration. From the Options menu, choose Parameter Configuration > Hide Read-Only.
9	For parameters with no default values (generally indicated by a zero, or a blank field), determine the value required and enter it.
10	If an entry is not required, or a special entry, such as dashes is required, leave the field blank. Refer to the discussion of blank values in the TPS Builder online Help topic "Configuration Forms."
11	Evaluate all parameters. If using the tabbed dialog box, use the scroll bar to evaluate all parameters contained on the tab. Click all tabs in the form to check all parameters.
12	When finished, click OK to apply the final parameter entries and close the dialog box.

Check parameter configuration

You can check a single entity if desired, or wait and check part or all of a project.

In this document, the Working with Projects section, refer to "Check a Project."

Configure logic and device control points

Refer to the Logic and Device Control Points section in this document.

15.3 Configuring Connection Parameters

Working in a drawing

You can most easily configure connection parameters by making connections in a module drawing. This method allows you to set the connections and create your drawing at the same time, and it works for strategy and template modules. Refer to Drawing Connection Overview in the Working with Drawings section of this document.

Working in the parameter configuration form

For strategies, you can also define connection parameters in the parameter form (such as entering point.parameter for CISRC).

If the entity referenced is on the related drawing, an auto-routed connection will be drawn. Otherwise, the connection will be represented on a drawing as a named connection.

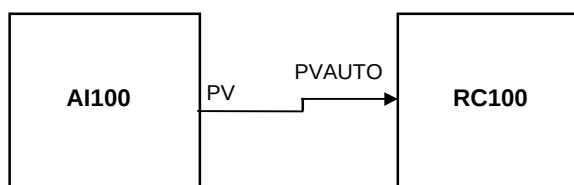
Result of connection configuration

Regardless of the way connections are made, TPS Builder establishes the same data flow connections and they can be viewed in the parameter configuration form or in a module drawing.

Examples

In a drawing

This example shows a normal connection line:

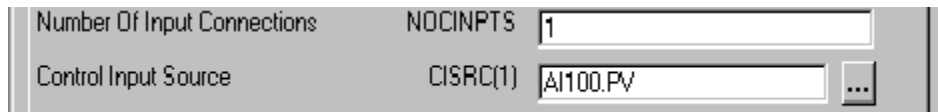


15. Parameter Configuration

15.3. Configuring Connection Parameters

In a parameter configuration form

This is a portion of RC100's parameter configuration form, based on the previous example.



The screenshot shows a portion of a parameter configuration form. It has two rows. The first row is labeled 'Number Of Input Connections' and has a value of '1' in a text box next to the label 'NOCINPTS'. The second row is labeled 'Control Input Source' and has a value of 'AI100.PV' in a text box next to the label 'CISRC(1)'. There is a small button with three dots to the right of the second text box.

Configure connections using the configuration form

To set connection parameters on configuration forms:

Step	Action
1	In a project or drawing, locate the strategy for which you want to set a connection parameter.
2	Right-click the strategy, and on the resulting menu, choose Parameters.
3	In the parameter configuration form, locate the connection parameter, for example CISRC(1). If the parameter is not visible, complete other aspects of configuration that will expose the needed parameter (for example, NOCOPTS must be set before an appropriate number of CODSTN fields will be available).
4	Click the  button next to the connection parameter field to access the Strategy Browser dialog box.
5	In the Strategy Browser dialog box, locate the strategy to or from which the connection is to be made. When you first open the Strategy Browser, contents of modules do not appear. Expand the module needed, by clicking the "+" (plus) sign to the left of the module name.
6	Click the strategy to view its parameters. Scroll in the parameter list to locate the parameter to be used for the connection.
7	Click the needed parameter, and complete the Strategy Browser dialog box as required.
8	Click OK to enter the connection parameter into the configuration form.
9	Complete the configuration as needed, and click OK in the configuration form.

In a module drawing containing the strategies for which you have configured connection parameters, connection lines are added between strategies in accordance with the configuration.

If connections are made across modules, then:

- In the drawings, named connections are shown.
- The named connections are hyperlinks. That is, you can double-click either name in either drawing to open and/or make the corresponding drawing active.
- If the drawings of both modules are open when you make the connection, the drawing to which you are making a connection will not show the named connection until after you save it, close it, and then reopen it.

15. Parameter Configuration

15.3. Configuring Connection Parameters

16. Logic and Device Control Points

16.1 Overview

Description

In TPS Builder, PM-family logic and device control points are represented as strategy modules (not blocks). Logic and device control blocks represent the inputs, outputs, flags, alarms, numerics, and gates.

To configure these two types of points, refer to the guidelines explained in the TPS system reference documents, such as the Section 14: *Logic points* and Section 48: *Device Control point* of the *HPM Control Theory* manual. Also refer to the guidelines in this online Help for using modules and drawings in TPS Builder, and to the following information.

Libraries

Templates for logic and device control points are contained in the PM/APM/HPM library.

The individual blocks representing inputs, outputs, flags, numerics, and so forth, are contained in individual TPS Builder libraries, one for each of the two point types:

- APM/HPM Device Control Blocks
- PM/APM/HPM Logic Blocks

16. Logic and Device Control Points

16.1. Overview

Block names in drawings

The name that appears on a block in a logic or device control drawing includes a function abbreviation and an index number (1, 2, 3, and so on), for example "FL 1," "LI 1." The abbreviations for logic blocks are as follows. Device control blocks use similar abbreviations.

- Logic alarm = ALM
- Logic flag = FL
- Logic gate = NULL before an algorithm identifier (LOGALGID) is selected. Once the identifier is selected, the name changes accordingly. For example,
 - AND
 - OR
 - NOT
 - Delay
- Logic Input = LI
- Logic Numeric = NN
- Logic Output = LO

The numbers used are consecutive for each block type: FL 1, FL 2, and so forth. For gates, this means that if you have four different gates, they are numbered consecutively regardless of the type of gates chosen: AND 1, OR 2, AND 3, NOT 4.

If you change the index value for a block on its parameter form, the change is made to the drawing symbol name.



ATTENTION

In TPS Builder R110, checking of the correctness of index numbers in a logic drawing is incomplete. For example, inputs are checked against the absolute maximums allowed, rather than against the logic mix for the point.

Make sure you follow correct configuration guidelines when configuring a logic point.

Block names in projects

A logic or device control block's name, as it appears in a project, is the same name that appears in both the Identification and the parameter configuration dialog box.

You can use any name desired in the Identification/parameter configuration dialog boxes (which is displayed in your project). The drawing symbol name, however, will remain as described previously.

One suggestion for block names is to reference both the point name and the function abbreviation. For example, for a motor control point, MC100, a logic input block might be named MC100_LI1.

16.2 Configuring a Logic Point

View membership requirements

To create a logic point strategy, the view memberships of the templates for the logic point and the logic blocks must match *exactly*. For example:

- If the view memberships for the point include both Equipment and Hardware, then the view memberships for the logic blocks must also include both Equipment and Hardware. The logic blocks cannot have only one of the view memberships.
- Similarly, the view membership of the logic point cannot be only Equipment, if any of the logic blocks' view memberships is Equipment and Hardware.

Before starting your configuration, make sure that the view memberships of the templates you plan to use match exactly.

How to configure a logic point

To configure a logic point:

Step	Action
1	Open the project that is to contain the logic point, selecting the needed view.
2	Open the PM/APM/HPM library, and the Logic Blocks library (or user libraries if you have placed specialized logic and logic block templates in them).
3	Size and position the project and library windows in a manner that will be convenient for dragging and dropping templates from the libraries to the project.
4	In a library, locate the logic template. The view membership of the template must include that of the project.
5	Drag the logic template to the project, highlighting either the project name, or the name of a module in which you want the logic point to reside.
6	Name the resulting logic point in the dialog box provided.
7	Right-click the new logic point and choose Parameters. Complete the parameter configuration form for the point, including LISRC(1...12) and LODSTN(1...12). Click OK to accept the configuration and close the form.
8	In the project, double-click the logic point module to open its drawing (or continue to work in the project, if desired).

Step	Action
9	According to the point configuration, add logic blocks to the logic module as follows. For each of these logic blocks, TPS Builder sets BLOKINDX automatically. Change BLOKINDX only if needed. (There is usually no reason to do so.) • For each LISRC, add a LogicInput block. • For each LODSTN, add a LogicOutput block. • For each numeric, add a LogicNumeric block. For each block, set the initial NN value. The NN parameter of LogicNumeric block #1 becomes the NN(1) parameter of the logic point. • For each flag, add a LogicFlag block. For each block, set the initial FL value if BLOKINDX is 7.. 12. The FL parameter of LogicFlag block #1 becomes the FL(1) parameter of the logic point. On flags, there may be a reason to change the BLOKINDX because FL1 .. FL7 are defined by the TPS system. • For each gate, add a LogicGate block. For each block, configure the parameters on the form. • The LOGALGID(1) parameter of LogicGate block #1 becomes the LOGALGID(1) parameter of the logic point. • For each custom alarm, add a LogicAlarm block. For each block, configure the parameters on the form. Note that the CXPR parameter of LogicAlarm block #1 becomes the C1PR parameter of the logic point.
10	Make connections between the logic blocks either in the logic drawing or using the parameter configuration forms. The active connection parameters are: • LOSRC and LOENBL of the LogicOutput blocks • CXSRC of the LogicAlarm blocks • S1 through S4 and R1, R2 of the LogicGate blocks.

16.3 Configuring a Device Control Point

View membership requirements

To create a device control point strategy, the view memberships of the templates for the device control point and the device control blocks must match *exactly*. For example:

- If the view memberships for the point include both Equipment and Hardware, then the view memberships for the device control blocks must also include both Equipment and Hardware. The device control blocks cannot have only one of the view memberships.
- Similarly, the view membership of the device control point cannot be only Equipment, if any of the device control blocks' view memberships is Equipment and Hardware.

Before starting your configuration, make sure that the view memberships of the templates you plan to use match exactly.

How to configure a device control point

To configure a device control point, first complete these steps:

Step	Action
1	Open the project that is to contain the device control point, selecting the needed view.
2	Open the PM/APM/HPM library, and the Device Control Blocks library (or user libraries if you have placed specialized device control and device control block templates in them).
3	Size and position the project and library windows in a manner that will be convenient for dragging and dropping templates from the libraries to the project.
4	In a library, locate the device control template. The view membership of the template must include that of the project.
5	Drag the device control template to the project, highlighting either the project name, or the name of a module in which you want the point to reside.
6	Name the resulting device control point in the dialog box provided.
7	Right-click the new point and choose Parameters.
8	Complete the parameter configuration form for the point.

Step	Action
9	On the Input Connections tab: • enter the Number of Logic Inputs, then • press the tab key. The LISRC and associated fields matching the number of inputs you entered will be added to the form. Complete these fields.
10	On the Output Connections tab: • enter the Number of Logic Outputs, then • press the tab key.
11	The LODSTN fields matching the number of outputs you entered will be added to the form. Complete these fields, then close the form.

Complete the configuration

To complete the logic module, complete these steps:

Step	Action
1	Double-click the logic module to open its drawing (or continue to work in the project, if desired).
2	According to the control strategy for the point, add device control blocks to the device control module as follows: • For each LISRC: – Add a DevLogInput block, a PI gate and an SI gate. – Connect the L parameter of the DevLogInput block to the PISRC0 parameter of PI gate. – Connect the PIDSTN parameter of the PI Gate to the SI parameter of the SI gate. Note that if the Control Attributes View Labels option is set for the blocks, – The connector for PISRC0 is labeled "src1." – The connector for PIDSTN is labeled "dst." – The connector for SI is labeled "in." • For each PI gate, configure the PIALGID, PINN, PIDEADBAND parameters. • For each SI gate, configure the SIALGID and SIDLYTIM parameters. • For each LODSTN (1 ... 2), add a DevLogOut block. • For each Numeric to be used, add a DevLogNum block. Note that the NN parameter of DevLogNum block #1 becomes the NN(1) parameter of the device control point. • For each logic flag to be used, add a DevLogFlag block if BLOKINDX is 7 .. 12. Set the BLOCKINDX and FL initial value. Note that the FL parameter of DevLogFlag block #1 becomes the FL(1) parameter of the device control points. • For each primary gate, add a P Gate. Configure the PGALGID and PGPLWTH parameters. • For each secondary gate, add an S Gate. Configure the SGALGID and SGPLWTH parameters.

Step	Action
3	Add a DigitIn block for each D1 and D2 if it is the source for an output connection.
4	Add a SecVar block if it is the source for an output connection.
5	Add a DigitOut block.
6	Make connections between the device control blocks either in the device control drawing or using the parameter configuration forms. The active connection parameters are: • PISRC, labeled "src2," of the PI Gate • SIDSTN, labeled "dst," of the SI Gate • PGDSTN, labeled "dst," of the P Gate • SGDSTN, labeled "dst" of the S Gate

The PG parameter of the P Gate is exposed six times for up to six connections, and is labeled "in."

16. Logic and Device Control Points

16.3. Configuring a Device Control Point

17. Logic Simulation

17.1 Overview

Description

This TPS Builder function allows you to simulate the logic flow of a TPS Network Logic or Device Control point, represented in TPS Builder as a template or strategy module. You can set logical inputs, flags and numerics. You can run a simulation on a fully configured point, or begin with simple logic configured, run the simulation, add more to the logic, run the simulation again, and so on.

You view the results of those settings throughout the point's configured logic, as shown in its module drawing.

Monitoring a simulation

One point simulation can be monitored at a time. For Logic and Device Control points, a configurable refresh rate, pause function, and color indications for the on, off, and numeric states are provided for monitoring the simulation. Inputs are manipulated using a Simulator Inputs dialog box.

Monitoring Device Control points

A special display allows you to monitor and set Device Control point Process Variable (PV) and Output (OP) values. This display is based on the graphic for the TPS Network Group display for the Device Control point. By attempting to select various states for the Output and checking the response, you can determine the effect of Permissives and Overrides on the ability of the Output to accept those states.

17. Logic Simulation

17.1. Overview

Simulation activities

Main simulation tasks include:

- Setting Logical Inputs, Flags, Numerics
- Setting Simulation Display Colors
- Setting Refresh Rate of Simulation
- Starting Simulation
- Evaluating Logic Simulation
- Finishing Simulation

17.2 Setting Logical Inputs, Flags, Numerics

Basic procedure

For a TPS Network logic or device control point being simulated, follow these steps to vary inputs so that the logic configuration can be studied under various conditions:

Step	Action
1	Open the drawing of the Device Control or Logic point to be simulated, or if the drawing is already open or in simulation, go to the next step.
2	From the Tools menu, choose Simulation to access a submenu of simulation activities.
3	On the Simulation submenu, choose Start. This activates the simulation and allows TPS Builder to identify the number of logical inputs, flags, and numerics of the point being simulated. You can leave the simulation running, or pause it.
4	From the Tools menu, choose Simulation > Simulator Inputs to access the Simulator Inputs dialog box. The dialog box initially shows default settings of "Off" for all logical values and zero (0) or all numeric values.
5	In the dialog box, set values as needed for the simulation. Any changes you make will take effect in the simulation after it is started/restarted, or next refreshed. If the simulation was paused, resume it. If the dialog box blocks the drawing, drag it to another location, or minimize it. Note that it may minimize to a location outside of the TPS Builder workspace.
6	Evaluate the results of the settings.

Making new settings

If desired, access the Simulator Inputs dialog box and make new settings. Access it as follows:

- If it remained on the screen, simply click in the field you want to modify.
- If it was minimized, maximize it.
- If the dialog box was closed, access it from the Tools > Simulation menu.

17.3 Setting Simulation Display Colors

Description

You can set colors to indicate the on, off, and numeric conditions in the simulation of a TPS network Device Control or Logic point. You can set or change colors at any time the drawing for one of these points is open.

The colors you set apply only to a module drawing in simulation mode, and do not affect drawing property colors.

Procedure

To set colors for simulation, follow these steps:

Step	Action
1	Open the drawing of the Device Control or Logic point to be simulated, or if the drawing is already open or in simulation, go to the next step.
2	From the Tools menu, choose Simulation > Set Logic Colors.
3	In the Set Logic Colors dialog box, click the button of the state for which the color is to be set: On, Off, or Numeric. The standard Windows Color dialog box appears.
4	In the Color dialog box, choose a color of choice. For help on using this dialog box, click the ? (question mark) in the upper right corner of the dialog box, then click on an item of interest. Click anywhere to dismiss the help information.
5	Click OK in the Color dialog box to assign the color to the selected condition. In the Set Logic Colors dialog box, your choice appears.
6	Repeat these steps for other conditions. When finished, click OK to close the Set Logic Colors dialog box.

When colors take effect

The colors set will take effect at the next start/restart, or refresh of a simulation.

17.4 Setting Refresh Rate of Simulation

Description

The refresh rate for TPS network Logic or Device Control point simulations affects how often the simulation is updated.

Procedure

To set the simulation refresh rate, follow these steps:

Step	Action
1	Open the drawing of the Device Control or Logic point to be simulated, or if the drawing is already open or in simulation, go to the next step.
2	From the Tools menu, choose Simulation > Set Frequency.
3	In the Set Frequency dialog box, enter a refresh rate of 1 to 30 (seconds).
4	Click OK to set the new rate and close the Set Frequency dialog box.

When the rate takes effect

The rate set will take effect at the next start/restart, or refresh of a simulation.

17.5 Starting Simulation

One simulation at a time

One TPS Network Logic or Device Control point simulation can be run at a time.

Starting a simulation

To start the simulation of logic flow in a TPS network Logic or Device Control point, follow these steps:

Step	Action
1	Open the drawing of the Device Control or Logic point to be simulated, or if the drawing is already open, go to the next step.
2	From the Tools menu, choose Simulation to access a submenu of simulation activities.
3	On the Simulation submenu, choose Start. The simulation starts, using default or previously set values for logic colors and refresh rate (inputs are reset to defaults after every session).
4	If desired, change settings (refer to Setting Logical Inputs, Flags, Numerics in this section).

Interrupting a simulation

The following actions can be taken on a simulation once it is running:

Action	Description
Pause	An active simulation can be paused. From the Tools menu, choose Simulation > Pause. The simulation will not be refreshed after Pause is selected. You can change settings while the simulation is paused. Activate a paused simulation by either restarting or resuming it.
Restart	From the Tools menu, choose Simulation > Restart to activate a paused simulation and reset all delay and/or pulse width gate settings.
Resume	From the Tools menu, choose Simulation > Resume to activate a paused simulation without resetting delay and/or pulse width gate settings.

If you make changes

Changes to simulation inputs, logic colors, and/or refresh rate will take effect whether you restart or resume a paused simulation.

17.6 Evaluating Logic Simulation

Description

A logic simulation allows you to judge the validity of a particular TPS network Logic or Device Control point configuration. It enables you to study the effects of different inputs on the configured point. With different colors set for on, off, and numeric conditions, you can vary inputs and watch the state changes as a result of the configured logic.

Also, with each refresh in simulation mode:

- Each Delay gate displays the time (in seconds) remaining before it activates.
- Each Pulse Width gate displays the time (in seconds) it will remain active.

Accessing simulation activities

All simulation activities are accessible from the Tools > Simulation menu. For a list of all related tasks, refer to the Logic Simulation Overview.



TIP

Only one point can undergo simulation at a time. Always finish one simulation before attempting to start another one.

17. Logic Simulation

17.6. Evaluating Logic Simulation

Typical evaluation steps

The simulation function is very flexible, allowing you to perform steps in several different ways. One typical evaluation approach might be as follows:

Step	Action
1	Open the drawing of the point to be simulated and familiarize yourself with its logic configuration.
2	Start the simulation and note the refresh rate and colors in effect. If desired, adjust them to better suit your needs.
3	Open the Simulator Inputs dialog box and position it in a convenient location on the screen, so that you can see both the dialog box and the drawing of the point to be simulated. (If needed, adjust the size of the workspace and the drawing for a better view.)
4	In the Simulator Inputs dialog box, change logical inputs, flags, and numerics to represent various input conditions you want to study.
5	Watch the colors in the simulation change in response to your input changes. In addition, watch delay and pulse-width gate times. If the drawing is large and/or complex, you may want to pause the simulation so that you can study it in detail.
6	If the Simulator Inputs dialog box blocks the drawing, you can drag the dialog box out of the way (even "off of the screen," as long as its title bar remains visible). Point in the title bar and drag to move the dialog box; or you can minimize it. (Note that it is placed outside of the TPS Builder workspace.)
7	After pausing a simulation, restart or resume it when ready. You may want to change inputs again before reactivating the simulation.
8	If you are evaluating a Device Control point, you can set various digital composite states using the Device Control Simulator Display.
9	If the simulation indicates that configuration changes need to be made, then Finish the simulation. Once the simulation is ended, make changes as necessary (parameter settings, connections, and so forth) to the blocks in the logic drawing.
10	After configuration changes are made, start the simulation again to reevaluate the effects of your changes.
11	If the Simulation Inputs and/or DC Simulator dialog boxes remain open after you have ended, close them.

17.7 Finishing Simulation

To end Simulation mode and stop all simulation activity:

- Choose Simulation > Finish from the Tools menu, or
- Click the **X** (close) button in the upper right corner of the drawing to stop simulation and close the drawing.

If any dialog box (Simulation Input, DC Simulator) remains open, close it by clicking the **X** (close) button in the upper right corner of the dialog box.

17. Logic Simulation

17.7. Finishing Simulation

18. UCN Node Configuration

18.1 Overview

Description

Configuring a Universal Control Network (UCN) node entails specifying the characteristics of a node within a project. Major tasks include the following. Before beginning node configuration, read this overview.

Node types

You can configure the following nodes using TPS Builder Universal Control Network (UCN) Library:

- Logic Manager/Safety Manager (LM/SM)
- Network Interface Module (NIM/ENIM)
- Process Manager family of nodes (PM, APM, HPM, and EHPM)

Reference documents

Information on node configuration is contained in node-specific documents in the TPS bookset, for example, refer to the HPM Implementation Guidelines *manual*.

Configuration-related parameter information is in parameter reference dictionaries and control functions/algorithms documents, such as

- HPM Parameter Reference Dictionary HPM Control Hardware Theory of Operation

18. UCN Node Configuration

18.2. Prepare for UCN Node Configuration

18.2 Prepare for UCN Node Configuration

Procedure

A node can be configured as a strategy within a project, or specialized as a template within a library. Before configuring a node, perform the following tasks:

Step	Action
1	Determine node configuration information. Note that you do not need to determine precise node loading, because TPS Builder will help you with this task. For redundant nodes, refer to the redundancy guidelines below.
2	Open the UCN Configuration library and identify the node template to be used.
3	Create or open the TPS Builder project or library of which the node will be a member, or open the corresponding drawing. Ensure that the project or drawing view is the same as that of the template to be used. The most common view for node configuration is the Hardware view.
4	Drag and drop the node template from the UCN Configuration library to the identified project or drawing; or specialize a particular node template. A node of the specified template type is now a part of the project or library. The network and node numbers are normally blank, but if the template being used includes them, they will be inherited and listed.
5	A single TPS node strategy represents the two entities used in the system (\$NMxxNyy and \$NMxxByy, where xx is the network number, and yy is the node number in both cases).
6	The recommended naming convention for each TPS node strategy is UCN##node## where UCN## identifies the network number. node## is the node abbreviation (for example, HPM), followed by the node number.
7	If you need to configure a NODETYP parameter as NotConfig for UCN configuration, refer to the discussion below.

Redundancy guidelines

When nodes are configured as redundant pairs, the following guidelines apply:

- Partners must use consecutive UCN addresses (NODENUM). The first node uses an odd UCN address, the partner uses the next even UCN address.
- Both nodes must be created in TPS Builder. Fully configure and assign points to the first node (odd-numbered UCN Address). Configure only UCN node parameters for the second node (even-numbered UCN address).
- When a node with an even-numbered UCN address is selected, parameters appearing on the node parameter configuration form that are not required will not be exposed, and the Slot Allocation, HPM Link Units, I/O Module Layout, Ctrl Processor Display, and Print mouse right-button menu options will not be available.

NotConfig limitation

In TPS Builder R110, the NotConfig option is not available for UCN configuration. This parameter value allows you to deconfigure a node when changing from redundant to nonredundant node type.

If you need to use NotConfig for UCN configuration, use one of two workarounds:

1. Configure the particular node from the Parameter Entry Display (PED).
2. Using a text editor, write an Exception Build file that specifies the network and node numbers and the node type of NotConfig. The file would be similar to the following:

```
{ }  
&T  NODE  
&N  $NM01N01  
NTWKNUM=01  
NODENUM=01  
NODETYP=NotConfig  
{ }  
&E
```

Note that the address shows up in both the entity name and the parameter values.

18.3 Define UCN Node Parameters

Procedure

For the node being configured, follow these steps:

Step	Action
1	Complete Prepare for UCN Node Configuration in this section. Pay particular attention to the redundancy guidelines discussed in that topic.
2	If the project or drawing containing the node is not opened, open it.
3	Right-click the node icon, or the node symbol in the drawing.
4	On the resulting menu, choose Parameters to access the node parameter configuration form.
5	On this form, the Node Configuration tab, enter the Network Number and Node Number.
6	Enter the remaining parameters located on the tabs within the form. The parameters and entries required depend on exposure rules, the template in use, and your choices as explained for the node parameter form.



ATTENTION

With TPN R684, TPS Builder supports EUCN configuration for ENIM, EHPM and SM.

The **\$DEVIDX** parameter has two purposes:

- For the ENIM, it is known as the **Authentication Group**.
- For the EHPM and SM, it is known as the **FTE Device Index**.



REFERENCE - EXTERNAL

For more information on the **\$DEVIDX** parameter, refer to the Parameter Reference Dictionary.

To assign the Authentication Group to ENIM or the FTE Device Index to EHPM and SM

- | | |
|---|--|
| 7 | Configure the Device index value in \$DEVIDX .

This parameter is defaulted to zero (0), and must be zero for the legacy nodes. |
|---|--|
-

Step	Action
	TIP • Device index value ranges from 0 to 509. • Ensure that the device index is an odd value.
8	When all the entries are complete, click OK to save the changes and close the dialog box.
	ATTENTION For the secondary node, the \$DEVIDX parameter appears by default after you enter the details for the primary node. For example, if you enter the \$DEVIDX parameter for the primary node, the device index parameter for the secondary node is reserved.

18.4 Control Processor Slot Allocation

Description

In UCN node configuration, this function enables you to allocate the number of normal (and fast slots as applicable) for each type of point (both I/O and control) in the node being defined.

TPS Builder calculates and displays the processor and memory loads as you enter the information in the Control Processor Slot Allocation dialog box. For HPM nodes, TPS Builder also calculates the number of I/O link units required, which you can review and change if desired.

Exceeding the maximum load

If you exceed the maximum loads for the node being configured, TPS Builder will alert you with color coding as follows (it does not prevent you from exceeding the maximums, however).

-  If 85% of the maximum load is exceeded, the loading displays on a yellow background.
-  When the maximum load is exceeded, the loading displays on a red background.

18. UCN Node Configuration

18.4. Control Processor Slot Allocation

Procedure

For the node being configured, follow these steps to allocate slots for each type of point being defined:

Step	Action
1	Complete "Prepare for UCN Node Configuration" and configure parameters on the Node Configure tab (and Node Specific Config tab if desired, which includes slot allocation parameters) on the parameter configuration form.
2	Right-click the node icon or symbol in its project or drawing window.
3	On the resulting menu, choose Slot Allocation to access the Slot Allocation dialog box.
4	If you did not complete the Node Specific Config tab of the parameter configuration form, complete the Slot Allocation dialog box by defining the absolute number of normal and fast slots for all control point types for the node. Refer to the appropriate parameter reference dictionary for details on each entry.
5	Evaluate the processing unit totals (and memory units when applicable) provided by TPS Builder at the bottom of the display, and adjust the entries as needed for proper node loading. Refer to "Slot allocation details" below.
6	When finished, click OK to save the slot allocation data and close the dialog box.

Slot allocation details

For information about slot allocation calculations and limits, refer to these topics in this section:

- PM Family Slot Allocation Details
- LM/SM Slot Allocation Details

18.5 PM Family Slot Allocation Details

Description

Slot allocation calculations and limits are discussed in this topic. They include:

- Slot Maximums by Point Type
- Maximum Node Loads
- Process Unit Calculation
- Memory Unit Capacity
- Memory Unit Constraints

Slot Maximums by Point Type

In control processor slot allocation, the absolute number of slots cannot exceed the maximum specified for each point type:

Point Type	PM	APM	HPM
Regulatory Control	160	160	250
Fast Reg Control	40 / NCTLSLOT	40 / NCTLSLOT	200 / NCTLFAST
Regulatory PV	80	80	400
Fast Reg PV	40 / NPVSLOT	40 / NPVSLOT	200 / NPVSLOT
Logic	80	80	400
Fast Logic	40 / NLOGSLOT	40 / NLOGSLOT	200 / NLOGSLOT
Device Control	N/A	160	400
Fast Device Control	N/A	160 / NDEVSLLOT	200 / NDEVSLLOT
Digital Composite	255	512	999
Fast Digital Composite	255 / NDCSLOT	255 / NDCSLOT	999 / NDCSLOT
Process Module	128 / 64 *	160 / 80 *	250 / 125 *
Semi Fast Process Module	N/A	N/A	100/NPMSLOT
Numerics	2047	16,384	16,384

18. UCN Node Configuration

18.5. PM Family Slot Allocation Details

Point Type	PM	APM	HPM
Arrays	N/A	256	500
Strings	N/A	16,384	16,384
Times	N/A	4096	4096

* Depends on the value of SEQPROC: 1_PU / 2_PU.

Maximum Node Loads

When allocating slots in node configuration, the maximum loads, per node type, are as follows:

	PM	APM	HPM
PU =>	160	160	800
MU =>	3,100	15,000	22,000
LU =>	N/A	N/A	1000

Process Unit Calculation

The formulas used for calculating the Process Units (PUs) that are reported during slot allocation are as follows:

$$\begin{aligned}
 \text{PM PUs} &= 4 \times (\text{nfastctl} + \text{nfastpv} + \text{nfastlog} + (.1 \times \text{nfastdc})) \\
 &\quad + \text{srm} \times (\text{nctlslot} - \text{nfastctl} + \text{npvslot} - \text{nfastpv}) \\
 &\quad + \text{ldcm} \times (\text{nlogslot} - \text{nfastlog} \\
 &\quad + .1 \times \text{ndcslot} - \text{nfastdc})) + \text{npmslot} \times \text{pmdp} \\
 \\
 \text{APM PUs} &= 4 \times (\text{nfastctl} + \text{nfastpv} + \text{nfastlog} + \text{nfastdv} + (.1 \times \text{nfastdc})) \\
 &\quad + \text{srm} \times \text{nctlslot} - \text{nfastctl} + \text{npvslot} - \text{nfastpv}) \\
 &\quad + \text{ldcm} \times \text{nlogslot} - \text{nfastlog} + \text{ndevslot} - \text{nfastdev} \\
 &\quad + .1 \times (\text{ndcslot} - \text{nfastdc})) + \text{npmslot} \times \text{pmdp} \\
 \\
 \text{HPM PUs} &= 4 \times (\text{nfastctl} + \text{nfastpv} + \text{nfastlog} + \text{nfastdv} + (.1 \times \text{nfastdc})) \\
 &\quad + \text{srm} \times \text{nctlslot} - \text{nfastctl} + \text{npvslot} - \text{nfastpv}) \\
 &\quad + \text{ldcm} \times \text{nlogslot} - \text{nfastlog} + \text{ndevslot} - \text{nfastdev} \\
 &\quad + .1 \times (\text{ndcslot} - \text{nfastdc})) + \text{npmslot} \times \text{pmdp} \\
 &\quad + (\text{nfastpm} \times \text{pmdp}) + (\text{nnumalm}/40)
 \end{aligned}$$

where

srm = Regulatory control and process variable scan rate multiplier; the fourth digit of SCANRATE
 ldcm = Logic and digital composite multiplier; the eighth digit of SCANRATE
 pmdp = Process module data point multiplier; the first character of SEQPROC

18. UCN Node Configuration

18.5. PM Family Slot Allocation Details

Memory Unit Capacity

HPMM capacity is measured in terms of Memory Units. For the point database and sequence programs, the maximum capacity is 22,000 Memory Units (MU) allocated as defined in the following table:

Slot Type	Memory Units Per Point
Digital Composite	5
Logic	15
Process Module	15
Regulatory PV	12
Regulatory Control	13
String	1/8 (Note 1)
Sequence Programs	1
Numerics	1/16 (Note 1)
Numerics (w/PV Alarming)	1/4 (Note 1)
Flags	0
Device Control	30
Array	8
Times	3/32 (Note 1)
Timer	0

Note 1:
8 strings = 1 MU
16 Numerics = 1 MU
4 Numerics (w/PV Alarming) = 1 MU
32 Times = 3 MU

Memory Unit Constraints

Any mix of the slot types can be used, but the mix is subject to the following constraints:

- Point mix used must not exceed 800 Processing Units or 22,000 Memory Units per HPM.
- Absolute number of each slot type per HPM cannot exceed the maximum shown in the following table (regardless of the available PUs).

18. UCN Node Configuration
18.5. PM Family Slot Allocation Details

Slot Type	Maximum Number	Comments
Digital Composite	999	
Device Control	400	
Process Module	250	
Regulatory PV	125	
Regulatory Control	250	
Logic	400	
Array	500	80 external max.
Numeric	16,384	First 2047 can have tags
Numeric w/PV Alarming	2047 or NNUMERIC	Whichever is less
String	16,384	
Time	4096	
Timer	64	
Flag	16,384	2047 can have tags

18.6 LM/SM Slot Allocation Details

Slot allocation calculations and limits for the Logic Manager and Safety Manager are discussed in this topic. They include:

- Slot Maximums by Point Type
- Maximum Node Loads
- Process Unit Calculation

Slot maximums by point type

In LM/SM slot allocation, the absolute number of slots cannot exceed the maximum specified for each point type:

Point Type	LM	PU Value per Point (srm)*	SM	PU Value per Point (srm)
Analog Input	127(AR1DT2)* 254(AR2DT2)	22 (AR1DT2) 11 (AR2DT2)	600 (AR2DT2) 1000 (AR1DT1, AR1DT2)	10.5 (AR1DT2) 5.25 (AR1DT1, AR1DT2)
Analog Output	482 (AR1DT2) 965 (AR2DT2)	5.8 (AR1DT2) 2.9 (AR2DT2)	1000	4.25 (AR1DT2) 8.5 (AR2DT2)
Digital Composite	304	9.2	652	11.1
Digital Input	1866	1.3	2000	11.25 (AR1DT1) 2.5 (AR1DT2, AR2DT2)
Digital Output	1950	0.7	2000	0.6 (AR1DT1) 1.2 (AR1DT2, AR2DT2)
Flags	1024	-	2000	-
Linkage	14	200	30	200
Numerics	1024	-	1000	-
Timer	700	4	1500	3.1

* Notes:

PU = Process Unit

srm = SCANRATE multiplier

AR1DT2, AR2DT2 = Scanrates; refer to your LM or SM Parameter Reference Dictionary.

Maximum node loads

When allocating slots in node configuration, the maximum loads, per node type, are as follows:

	LM	SM
PU =>	2800	6000

Process unit calculation

The formulas used for calculating the LM/SM Process Units (PUs) that are reported during slot allocation are as follows:

$$\text{LM or SM PUs} = \text{naislot} \times \text{srm} \times \text{ai} + \text{naoslot} \times \text{srm} \times \text{ao} + \text{ndislot} \times \text{srm} \times \text{di} + \text{ndoslot} \times \text{srm} \times \text{do} + \text{nlogslot} \times \text{srm} \times \text{log} + \text{ntimeslot} \times \text{srm} \times \text{tm}$$

where

$$\text{srm} = \text{SCANRATE multiplier}$$

18. UCN Node Configuration

18.7. Define Units for a Node

18.7 Define Units for a Node

Description

Units provide a basis for address assignment. Unit IDs are usually used to group points and associate these groups with nodes. Up to 100 units can be defined per node.

Procedure

For the UCN node being configured, follow these steps to define units for a node:

Step	Action
1	Complete preparation for the node being configured. Refer to Prepare for UCN Node Configuration in this section.
2	Define UCN node parameters using the UCN Node Parameter Configuration Form. Configure parameters on the Node Configure and Node-Specific tabs.
3	Choose the Units tab, and on this tab, enter two-character (alphanumeric) unit names for up to 100 units. Each unit name may be associated with only one node.
4	Click OK to save your changes and close the dialog box.

18.8 I/O Module Layout

Description

I/O Module layout is the assignment of input/output modules (IOMs) to card positions within the cabinet.

Procedure

For the UCN node being configured, follow these steps for I/O Module layout:

Step	Action
1	Complete preparation for the node being configured. Refer to Prepare for UCN Node Configuration in this section.
2	In the UCN Node Parameter Configuration Form, define at least the following parameter definitions: • Node type • Network number • Node number • Package option • For HPMS, HPM file option
3	Right-click the node icon or symbol in its project or drawing.
4	On the resulting menu, choose I/O Layout to access the I/O Module Layout dialog box.
5	Complete I/O Module Layout dialog box by assigning the node's I/O modules to card positions within the cabinet: a) Click an empty card position. (Only one position can be selected at any time.) b) Click the IOM Assign button. c) Complete the I/O Module Assignment dialog box, as required. Choose an IOM type and all additional parameter information. Refer to the appropriate parameter reference dictionary for details on each entry. d) Click OK to save and return to the I/O Module Layout dialog box.
6	Repeat step 5 for each IOM to be assigned.

18. UCN Node Configuration

18.8. I/O Module Layout

Step	Action
7	If desired, reserve certain card positions to prevent IOMs from being assigned to them: a) Left-, then right-click an empty card position to be reserved. b) In the resulting menu, choose Reserve. c) Repeat for other card positions to be reserved.
8	When finished, click OK to save the card file layout and close the dialog box.



TIP

A reserved card position can be unreserved. Refer to [Modify I/O Module Layout](#) in this section.

18.9 Modify I/O Module Layout

Initial steps

Once IOM assignments have been made, they can be modified, using these steps:

Step	Action
1	Open the project or related drawing that contains the node for which you want to modify IOM assignments.
2	Right-click the node icon or symbol in its project or drawing.
3	On the resulting menu, choose I/O Layout to open the I/O Module Layout dialog box.
4	Modify, move, or delete an IOM as discussed in the following paragraphs.

Modify an IOM

Follow these steps to change IOM assignment:

Step	Action
1	Access the I/O Module Layout dialog box as explained in Initial steps in this topic.
2	Click the IOM to be modified.
3	Click the IOM Assign button.
4	In the IOM Assignment dialog box, make necessary changes.
5	Click OK to return to the I/O Module Layout dialog box.
6	When finished, click OK in the I/O Module Layout dialog box to save all changes and close the dialog box.

Move an IOM

Follow these steps to move an IOM:

Step	Action
1	Access the I/O Module Layout dialog box as explained in Initial steps in this topic.
2	Delete the IOM to be moved from its current position. (See the next procedure.)
3	Make a new assignment in an empty, unreserved card position.

Delete an IOM

To delete an IOM, follow these steps:

Step	Action
1	Access the I/O Module Layout dialog box as explained in Initial steps in this topic.
2	right-click the IOM to be deleted.
3	On the resulting menu, choose IOM Delete. If the selected IOM is • nonredundant, it will be deleted (after you confirm the action); • the primary (A) member of a redundant pair, both members (A and B) will be deleted (after you confirm the action); • the redundant (B) member of a redundant pair, only the redundant (B) member will be deleted (after you confirm the action).
4	Delete other IOMs if needed.
5	When finished, click OK to save the IOM assignment changes and close the dialog box.

Release a reserved card position

To make a reserved card position available:

Step	Action
1	Access the I/O Module Layout dialog box as explained in "Initial steps."
2	Right-click the reserved card position.
3	On the resulting menu, choose Unreserve. An IOM can now be assigned to that position.

18.10 Address Assignment Overview

Description

Specific assignment of points to I/O Modules (for I/O points), or to slots within control processors (for control points and algorithms), can be done automatically using the TPS Builder Address Assignment function. This function can be applied to a complete project or to a selection of nodes. Once assignments have been made, they can be maintained, or revised by subsequent address assignment.

For LM and SM nodes, TPS Builder assigns both I/O points and Control points to slots within control processors (because these nodes do not have IO Modules and the points do not have a Module Number).

If your project includes Fieldbus strategies, they are ignored during address assignment. Address assignment assigns only non-Fieldbus TPS Network strategies to TPS Network I/O module types. (Fieldbus CM strategies, SECM's, PECM's and FBCM's, are assigned to slots on the Fieldbus IOP by the CM Mapping function, which is discussed in the Fieldbus-specific Help.)

18. UCN Node Configuration

18.10. Address Assignment Overview

How points are assigned

Address assignment can be based on either Unit IDs or node numbers. Both Unit IDs and node numbers provide a means of associating points with particular nodes, which the Address Assignment function uses to assign the points associated with a node to the slots associated with the node.

Address assignment based on...	Results in...
Unit	Each point is assigned a Network Number, Node Number, Module Number (if an I/O point of a PM-family node), and Slot Number.
Node numbers	Each point is assigned a Module Number (if a PM-family node I/O point) and Slot Number.

The function assigns points, in ascending name sequence, to the next available (unused) slot, except for fast slots.

Procedures

The following procedures are available:

- Assign Addresses by Unit ID
- Assign Addresses by Node Number
- Assign Points to Fast Slots
- Change PM-Family Point Address Assignments
- Change LM/SM Point Address Assignments

Log details

The following logs provide information about address assignment:

TPS Builder Log	Provides
Activity	• The number of each point type that has been assigned an address during the address assignment process. • The number of points (if any) that could not be assigned a slot for some reason.
Address Assignment	• The points that were not assigned during address assignment. • the reason that they were not assigned an address.
Error	• Information about unexpected problems related to address assignment.

18. UCN Node Configuration

18.11. Assign Addresses by Unit ID

18.11 Assign Addresses by Unit ID

Follow these steps to assign each point a network number, node number, module number (if a PM-family node I/O point), and slot number:

Step	Action
1	Open the project that contains the node(s) for which you want to assign addresses.
2	Ensure nodes to be used for address assignment are configured, including redundancy requirements.
3	Define units for the nodes to be used for address assignment.
4	Ensure all points to be addressed have the appropriate Unit ID on their parameter configuration forms.
5	From the TPS Builder main menu, choose Tools/Address Assignment.
6	Complete the Address Assignment dialog box.
7	Click the Process button. TPS Builder assigns Network Number, Node Number, Module (if a PM-family node I/O point), and Slot Number to all points chosen for address assignment.
8	Review the Activity Log to see the address assignment results.
9	If the log indicates points were not addressed, review the Error Log to identify the problems.
10	If necessary, correct problems and repeat this procedure.

18.12 Assign Addresses by Node Number

Follow these steps to assign each point a module number (if a PM-family node I/O point) and slot number:

Step	Action
1	Open the project that contains the node(s) for which you want to assign addresses.
2	Ensure nodes to be used for address assignment are configured, including redundancy options for PM-family nodes.
3	Ensure all points to be addressed have the appropriate NTWKNUM and NODENUM on their parameter configuration forms.
4	From the TPS Builder main menu, choose Tools > Address Assignment.
5	Complete the Address Assignment dialog box.
6	Click the Process button. TPS Builder assigns Module (if a PM-family node I/O point) and Slot Numbers to all points chosen for address assignment. Processing activities are recorded in the Address Assignment log.
7	When processing is complete, review the Address Assignment log.
8	If the log indicates points were not addressed, review the Error Log to identify the problems.
9	If necessary, correct problems and repeat this procedure.

18.13 Assign Points to Fast Slots

Description

Fast slots are available for regulatory control, regulatory PV, Logic, Device Control, and Digital Composite points assigned to a PM-family node.

The TPS Address Assignment function does not assign points to fast slots. You must use the following procedure to assign points to fast slots.

Procedure

To assign points to fast slots for a specific node, follow these steps:

Step	Action
1	Ensure that a number of fast slots has been defined (such as the NFASTCTL parameter in the node parameter configuration form).
2	For points to be assigned to fast slots, ensure that the NODENUM parameter is included in the points' parameter configuration forms; or you may want to complete address assignment. Points will be assigned to all but the fast slots.
3	In the TPS Builder project, right-click the node for which you want to assign points to fast slots.
4	On the resulting menu, choose Control Processor display.
5	In the Control Processor display, choose the point type to be assigned to fast slots and click the Display button. Points of the selected type, and their assignment within the selected node, are listed. Fast slots begin with slot 0001 and are initially labeled as "Spare."
6	Find a point to be assigned to a fast slot.
7	Click and drag the point to the desired fast slot position. A message asks you to confirm the move. If the action is as you desire, click OK.
8	Alternatively, you can double-click the point to open the point's parameter configuration form. Change the point's slot assignment (SLOTNUM parameter) on the parameter configuration form to the desired fast slot and click OK to save the change and close the parameter configuration form. The Control Processor display updates based on the change.
9	Repeat the procedure for all points to be assigned to fast slots.
10	When complete, click the Close button to close the Control Processor display.

18.14 Change PM-Family Point Address Assignments

Description

If necessary, the address assigned to a point in a PM-family node can be changed. You can

- access the parameter configuration form for a point and change its address on the form, or
- use the I/O Module Strategies display or Control Processor Strategies display to move points (described next).

Procedure

The I/O Module and Control Processor Strategies displays allow points to be moved within the same module (slot), or between modules (slots) by opening multiple displays.

To move points from one position (slot) to another, follow these steps:

Step	Action
1	Open the project or related drawing that contains the node for which you want to reassign points.
2	Right-click the node icon or symbol in its project or drawing.
3	For I/O Module points: - On the resulting menu, choose I/O Layout to access the I/O Module Layout dialog box. - Right-click the module containing the points you want to change. - On the resulting menu, choose IO/Module Layout. - In the I/O Module Layout dialog box, left-, then right-click a module. - In the resulting menu, choose IOM Display to access the I/O Module Strategies Display.
4	For Control Processor points: - On the resulting menu, choose Ctrl Proc Display. - In the Control Processor Point type dialog box, choose the type of point you want to change. - The corresponding Control Processor Strategies Display opens.

18. UCN Node Configuration

18.14. Change PM-Family Point Address Assignments

Step	Action
5	If you want to move points between modules, repeat steps 4 - 5 for the other modules. You will have a separate display for each module.
6	Find a point to be reassigned. If more than one display is open, click in the display containing the point and use the scroll bar to view all assignments.
7	Select and drag the point to the desired new position, whether it is in its current display or another one. When at the destination position, release the mouse. A confirmation message appears. See "Related topics" below for tips on reassigning points.
8	To move (or swap) the point(s), click OK in response to the message (or void the action by clicking Cancel).
9	Repeat the procedure for all points to be reassigned. As points are reassigned, the applicable I/O Modules or Control Processor Displays update to reflect the new assignments.
10	When point reassignment is complete, close the IOM or Control Processor Strategy display(s). If multiple I/O Module displays are open, you can close them all at one time by clicking the Close All button in the IO Module layout dialog box.

18.15 Change LM/SM Point Address Assignments

Description

If necessary, the address assigned to a point in a Logic Manager or Safety Manager can be changed. You can

- access the parameter configuration form for a point and change its address on the form, or
- use the IOP Display or Control Processor Display to move points.

Procedure

The I/O Module or Control Processor Display allows points to be moved within the same module (slot) or between modules (slots) by opening multiple displays.

To move points from one position (slot) to another:

Step	Action
1	Open the project or related drawing that contains the node for which you want to reassign points.
2	Right-click the node icon or symbol in its project or drawing.
3	On the resulting menu, choose I/O Module Display or Control Proc Display, depending on the type of point you want to change. • If IO Module Display is chosen, the IO Point Types dialog box appears. Choose the point type of the point(s) to be changed. Points of the selected type, and their assignment within the selected slot, are listed in the I/O Module Strategies Display. • If Control Proc Display is chosen, Control Processor Point Type dialog box appears. Choose the point type of the point(s) to be changed. Points of the selected type, and their assignment within the selected slot, are listed in the Control Processor Strategies Display.
4	Find a point to be reassigned. Select and drag the point to the desired new position. When at the destination position, release the mouse. A confirmation message appears.
5	To move (or swap) the point(s), click OK in response to the message (or void the action by clicking Cancel).
6	Repeat the procedure for all points to be reassigned. As points are reassigned, the applicable I/O Modules or Control Processor Displays update to reflect the new assignments.
7	When point reassignment is complete, close the I/O Module or Control Processor Strategies display.

18.16 Tips for Reassigning Points

Keep the following information in mind when changing address assignments.

Moving or swapping points

If the destination slot is unoccupied, TPS Builder assumes that the point is being moved. The message so indicates by the terms "Move" and "To."

If the destination slot is currently occupied, TPS Builder assumes that two points are being interchanged. The message so indicates by the terms "Swap" and "With."

Point reassignment errors

You cannot move a point between IOMs or control processors of different point types. If you do, TPS Builder responds with an error message.

Keeping address assignments

After points are assigned to specific I/O Modules and control processors, you may need to add further assignments using the Address Assignment function.

To maintain assignments made, choose one of the following options in the Address Assignment dialog box:

- Maintain Existing
- Maintain Fast Slots

A TPS Builder strategy is the configuration information for some element of the TPS system (or another system), intended for loading to the system. A strategy contains loadable data, such as point names, parameter values, and slot assignments.

Typical TPS system strategies include system and point configuration data.

18.17 Calculate Node Loading

Description

TPS Builder can help you balance the load across nodes in a project. On a project or node basis, you can calculate node load, in processing units. The calculations can be based on either

- point assignments that correlate to units assigned to the node, or
- actual point assignments (whether or not the units are assigned to the node)

The calculation results can be printed.

Procedure

To calculate the load for a project or node, follow these steps:

Step	Action
1	Open the project that contains the node(s) for which you want to calculate the load.
2	Complete parameter configuration for all points in the project or node of interest. Ensure that all points in the project are assigned to the appropriate units. If using a Node by Units calculation scope, ensure that unit IDs are defined.
3	From the TPS Builder main menu, choose Tools/Loading Calculation.
4	Complete the Node Loading dialog box to define the type of node loading calculation you want.
5	Click the Calc button to perform the selected calculation (or Cancel to void the selections and close the dialog box). When the calculation is complete, the Node Loading Report displays automatically.
6	Review the online. For details on this online report, refer to Node Loading Report in this section. If you calculated a load for multiple nodes, view the report for each node by using the navigation buttons.
7	Click the Print button to print the report for future reference. (This report is not saved after you close the Node Loading dialog box.)

18.18 Node Loading Report

Description

This report can be used to help you balance the point load across all nodes in a project. Once a balanced load is achieved, a final report can be run, and its results can be used to specify the number of control slots required for each node, as well as the I/O Module Layout for your project.

Report access

Refer to Calculate Node Loading in this section for details on how to access this report.

Report contents

The calculations made, and the information reported, varies depending on the type of report.

Report Item	Project Loading	Node by Units	Addressed Loading
Unit/Load list	For all nodes in the project, unit IDs used in point configuration are listed, with corresponding point loads, in processing units.	For each node, the unit IDs defined for the node are listed, with corresponding point loads, in processing units.	For each node, unit IDs used in point configuration are listed, with corresponding point loads, in processing units.
Subtotal CPU Load	--	The sum of processing units, excluding those associated with fast slots.	
Additional Load Due to Fast Slots	Not calculated for projects.	The load, in processing units, for each fast slot configured for the node in slot allocation.	The load, in processing units, for each fast slot used in point configuration.
Total CPU Load	The sum of all processing units, excluding fast slots.	The sum of all processing units, including fast slots.	
I/O Modules Required	The printed report also shows the total number of I/O Modules required to accommodate the calculated point counts.		

Actions

Print: Print the report. The printed report differs from the online report in that if more than 20 Unit IDs are included in the calculation, all will be printed (only 20 are displayed), and the total I/O Modules required to accommodate the calculated point counts are listed.

Close: Close the report and return to the Node Loading dialog box. (You can return to the report as long as the Node Loading dialog box remains open.)

For Node by Units and Addressed Loading reports:

: Go to first page of report.

: Go to previous page.

: Go to next page.

: Go to end of report.

18.19 HPM Link Units Overview

Description

Link Units address a constraint that arises in the HPM. Whereas node configuration in the PM and APM are constrained by Processing Units, the significantly increased processing capacity of the HPM means that the constraint on HPM performance can often shift to I/O link access.

Link access can be thought of as I/O bandwidth and link units as the measure of the capacity of I/O link access.

The TPS Builder Link Units function allows you to evaluate various configurations in terms of link units. You can then revise your HPM node configuration based on the HPM link units calculations.

Maximum link units

To avoid I/O link access overrun, the maximum link units allowed for an HPM is 1000. I/O link access depends on

- The point population of an HPM cabinet,
- Scan rate,
- I/O module type, and
- Any CL programs that may be present.

The portion of the I/O access load due to scanning can be calculated when an HPM is configured, but the I/O access load due to fetching from and storing to connections can only be estimated from the node configuration.

TPS Builder therefore provides an HPM Link Units Calculation function that shows the I/O access load and allows the load to be edited so that the effects of changes can be seen.

Reference

Refer to the Overrun Handling section of High-Performance Process Manager Control Theory, for details about HPM load capabilities.

18.20 Calculate HPM Link Units

Description

Use the following procedures to calculate link units, and if desired, save the calculations.

For details about this topic, refer to the HPM Link Units Overview in this section.

Calculate link units for a selected HPM

Link Units calculations can be performed at any time during configuration, but become more accurate as address assignments and peer-to-peer connections are made.

To calculate link units for a specific HPM, follow these steps:

Step	Action
1	Open the project or related drawing that contains the HPM for which you want to calculate the link units.
2	Click the HPM's icon in the TPS Builder Project Manager, or its symbol in the drawing.
3	From the TPS Builder Edit menu, choose HPM Link Units. Each time this dialog box is opened, TPS Builder automatically calculates the link units based on the HPM's current configuration; that is, the actual number of points and the actual number of I/O Modules assigned to the HPM.
4	Evaluate the information in the HPM Link Units dialog box. If desired, • calculate link units for various HPM configurations and/or • save the calculation.

18. UCN Node Configuration

18.20. Calculate HPM Link Units

Calculate link units for various HPM configurations

If parameter configuration and/or point assignment has not been completed for all points in an HPM, point totals can be estimated. You may want to make estimates to evaluate various configurations. To do so, follow these steps:

Step	Action
1	Open the project or related drawing containing the HPM.
2	Click the HPM's icon in the project, or its symbol in the drawing.
3	From the TPS Builder Edit menu, choose HPM Link Units. The calculation for the selected HPM is presented in the dialog box.
4	In the HPM Link Units dialog box, change the number of points and/or I/O Modules as desired. The link units calculation updates as you change values. This action does <i>not</i> change your actual configuration.
5	Evaluate the information.
6	If desired, save the calculation.

Save a calculation

Once you have calculated HPM link units, you can save the information in a file. For example, you might want to create different models. These files can be opened, revised, and saved from the HPM Link Units dialog box.

To save a particular calculation, follow these steps:

Step	Action
1	Establish the calculation you want to save.
2	In the HPM Link Units dialog box, click the Save As button.
3	In the Save dialog box, enter a file name for the calculation. Do not include the extension ".lu," as the extension is added by TPS Builder. For example, HPM03Case1.lu.
4	Click OK . The information is saved in the file, and the HPM Link Units dialog box remains open.
5	Perform other calculations, or click Close to close the dialog box.

Review a saved calculation

For the HPM for which you want to review the link units calculation:

Step	Action
1	Open the project or related drawing containing the HPM.
2	Click the HPM's icon in the project, or its symbol in the drawing.
3	From the TPS Builder Edit menu, choose HPM Link Units . The calculation for the selected HPM is presented in the dialog box.
4	Click the Open button.
5	In the Open dialog box, locate the file you want to review.
6	Double-click the file name. The HPM Link Units dialog box will change to show the saved calculation.
7	Review the information.
8	If desired, change information, then save it under the current file name (Save button), or save it with a new name (Save As button).

Delete a saved calculation

Use Windows file procedures to delete a saved calculation file. Locate the file in My Computer (or another file management window), and delete it.

18. UCN Node Configuration

18.21. Print Node Configuration Details

18.21 Print Node Configuration Details

What can be printed

An I/O Layout report and Node Loading report can be printed. Note that these reports are accessible only in conjunction with node configuration, and not from the Reports menu.

The I/O layout report is not applicable to LM/SM nodes.

I/O Layout report

Within a project:

Step	Action
1	Right-click a node icon, or the node symbol of a related drawing.
2	On the resulting menu, choose I/O Layout .
3	In the resulting dialog box, click Print .
4	Complete the Print dialog box and click OK. The I/O Layout report will be printed. It shows the information, in printed form that appears in the I/O Layout dialog box.

From the I/O Layout dialog box:

Step	Action
1	Click the Print button.
2	Complete the Print dialog box and click OK . The I/O Layout report will be printed.

Node Loading report

Within a project:

Step	Action
1	Right-click a node icon, or the node symbol of a related drawing.
2	Complete node loading (refer to "Related topics" below).
3	In the Node Loading Report, click the Print button.
4	Complete the Print dialog box and click OK. The Node Loading report will be printed.

18. UCN Node Configuration

18.21. Print Node Configuration Details

19. CLM Configuration

19.1 Overview

Description

Configuring a Communication Link Module (CLM) node entails specifying the characteristics of a node within a project. After importing CLM Library templates you can use TPS Builder to define CLM Interface Structures and create the hierarchy of these structures.

CLM Template Distribution

CLM Library templates for the different Platforms are special order templates distributed on a separate CD from Evergreen. Because additional documentation would be provided with the CD, this section only contains basic information on CLM functionality for TPS Builder.

For information on Library template details and Platforms supported, contact your Honeywell sales representative. Shipping of new and changed templates is independent of TPS Builder releases.

Reference Documents

The following documents provide additional information on CLM and importing templates into TPS Builder.



REFERENCE - EXTERNAL

More complete information on CLM platform is found in the following Manual:

CLM Platform Implementation Manual

For specific Platform information, refer to the Individual Platform Manuals



REFERENCE - INTERNAL

See "Database Administration," "Import TPS Builder Objects" in the *TPS Builder Installation, Implementation & Administration Guide* for specific information and procedures on importing templates and libraries.

19.2 Using the Templates

CLM Platform Library

The CLM Platform library would house the following CLM template types:

- CLMI
- PIT
- PITG

Other CLM libraries and templates

Other CLM templates that are provided in TPS Builder are distributed into different libraries based on the support to different kinds of networks.

Honeywell supplies you with the appropriate libraries for your specific CLM configuration. These libraries contain the templates for the CLM elements such as port, device, and image points.

Interface points

You can configure the following CLM interface points using CLM templates.

- Network Definition Table (NDT)
 - Including: CLMI, Ports, Devices)
- Point Interface Table Group (PITG)
- Point Interface Table (PIT)
- Point Interface Table Entry (PITE), or Image Points



ATTENTION

In TPS Builder, all CLM strategies are implemented as modules except PITEs, which are implemented as blocks.

19.3 CLM Template Functionality

Overview

Once the templates are imported into TPS Builder, you can use all existing TPS Builder features on the templates, including customization. Using TPS Builder simplifies the process of defining CLM Interface structures.

Resolving CLM Hierarchy

Unlike all other TPS Builder subsystems, CLM modules and points have a specific hierarchy. The CLM component hierarchy is as follows:

- CLMI is the highest level in the CLM hierarchy.
 - The CLMI element can only embed Ports.
- Ports form the second level in the hierarchy.
 - Ports can only embed Devices.
- Devices form the next level. Devices themselves have a hierarchy.
- PITs are next in the CLM hierarchy.
 - Any Device can embed PITs.
- Image Points (or PITEs) are last in the CLM hierarchy.
 - Only PITs can embed Image Points. Nothing can be embedded into Image Points.
 - Image Points in TPS Builder are implemented as blocks and this implementation doesn't allow any strategies to be embedded into the Image Points.

19. CLM Configuration

19.3. CLM Template Functionality

Forward Build CLM Entities

The “Forward Build CLM Entities” function allows you to forward build all CLM strategies into a ZI file to be used as File Input to the CLM Configuration Builder. EB files for the Image Points (PITEs) will be generated for all PITEs as well.

The Forward Build CLM Entities dialog provides an option to either select specific CLM strategies or build all the CLM strategies in a project.

- If you need to selectively build CLM strategies, you can select CLM strategies in a project and build all of them into a ZI file.



ATTENTION

All the selected CLM strategies are built into a ZI file in alphabetical order. You have to invoke the Resolve CLM Hierarchy function (**Tools>Resolve CLM Hierarchy**) to order the CLM strategies in accordance with CLM hierarchy.

Backbuild ZI Files

The Backbuild ZI Files function in TPS Builder only backbuilds CLM strategies from ZI files. From the Backbuild ZI Files dialog (**Tools>Backbuild ZI.Files**), you can choose to either select one or multiple ZI files to backbuild CLM entities.

All CLM strategies are built as root strategies (under the project). You have to invoke the Resolve CLM Hierarchy function (**Tools>Resolve CLM Hierarchy**) to order the CLM strategies in accordance with CLM hierarchy.

20. Improving Configuration Consistency and Efficiency

20.1 Overview

Description

TPS Builder allows you to set various attributes to increase configuration consistency and efficiency. You can set parameters to be:

- Template Defining
- Propagate Down
- Project-Up

These attributes are briefly defined as follows:

Attribute	Description
Template Defining	Applies only to templates. Template defining parameters can be configured only on the template on which they are so set.
Propagate-Down	Applies to only modules. Sets parameters so that they can be defined only at the module level, and all module components are configured with the value set for the module.
Project-Up	Applies only to templates or strategies contained in a module. Allows you to configure parameters from the (higher) module level, instead of opening the configuration form for every component of the module.

Possible conflicts



ATTENTION

Conflicts can occur between the template-defining and propagate-down attributes set in modules and blocks you want to add to the modules. Special attention to the relationship of these attributes will help you avoid problems.

For details, refer to this section, [Propagate-Down Fundamentals, Relationship between template-defining and propagate-down attributes.](#)

20. Improving Configuration Consistency and Efficiency

20.2. Template-Defining Fundamentals

Custom Help for libraries and templates

You can also improve efficiency by attaching WinHelp topics to libraries or templates. For details, refer to Using Custom Help with Libraries and Templates.

20.2 Template-Defining Fundamentals

Description

On a template, you can identify certain parameters to be configurable only on that template. They are set as "template defining." On any strategy or specialized template created from that template, these parameters will not be modifiable (they are "template defined").

Setting parameters as template defining is one way to minimize configuration time. Be sure that the parameter values you define in this way will apply in all cases.

Example

For example, you set a temperature parameter on Template X to be Template Defining, and define it as degrees Kelvin (°K). On *all* strategies and specialized templates created from Template X, that temperature parameter will be °K. If, on Template X, you change the parameter to degrees Fahrenheit (°F), then all related strategies and specialized templates will likewise be changed.

Can the attribute be changed?

You can remove the template-defining attribute at any time. When you do, the parameter(s) for which the attribute is removed become configurable on any related strategy or specialized template.

What the attribute applies to

The Template-Defining attribute applies only to templates. Parameters set as template defining can be configured only on the template in which they are so set.

Regarding template modules, note that template-defining applies only to parameters that are specific to the module itself, such as the parameters associated with a UCN node (not parameters that have been projected up).

20.3 Configuring the Template-Defining Attribute

Fundamentals



REFERENCE - INTERNAL

In the Fundamentals section of this document refer to [Templates](#), [Template-defining attribute](#).

In this section, refer to [Template-Defining Fundamentals](#).

Template-defining preparation

Before beginning, complete these steps:

Step	Action
1	Read the above-mentioned references.
2	Plan which parameters you want to be template defining.

Parameter configuration form method

This method is the fastest if you want to set template-defining parameters, but not other control attributes.

Step	Action
1	Complete the preparation described previously.
2	If in a library, right-click the template for which control attributes are to be configured. If in a drawing, left-, then right-click the template's drawing symbol.
3	In the resulting menu, choose Parameters .
4	In the Parameter configuration form, locate a parameter that you want to be template-defining.
5	Select or enter the value/setting you want for the parameter.
6	Press Ctrl+T . The value or setting will become italic font, indicating it is template-defining.
7	Repeat for other parameters that are to be template-defining.
8	Click OK when finished.

20. Improving Configuration Consistency and Efficiency

20.3. Configuring the Template-Defining Attribute



TIP

Ctrl+T has two uses in parameter configuration forms:

- For templates, it sets a value as template-defining.
 - For strategies, it resets the parameter value to the template value.
-

Control attributes form method

This method may be preferred if you are setting a variety of control attributes.

Step	Action
1	Complete the preparation described previously.
2	If in a library, right-click the item for which control attributes are to be configured. If in a template drawing, left-, then right-click the drawing symbol.
3	In the resulting menu, choose Control Attributes .
4	In the Control Attributes dialog box, choose the Template Defining tab.
5	Click a parameter name in the Parameters list.
6	If the parameter represents an array, enter a valid index number.
7	Click ADD → . The parameter name will be placed in the Visible Symbol Parameters list. If you chose to expose the label, the parameter will be marked as Labeled .
8	Repeat as needed for additional parameters.
9	If needed, when finished with the Template Defining tab, click another tab in the dialog box for further control attribute configuration.
10	Click OK when all control attributes are configured.

20.4 Propagate-Down Fundamentals

Description

The Propagate-Down attribute applies only to modules (template or strategy). This attribute allows you set parameters so that they can be defined only at the module level, and all module components are configured with the value set for the module.

Examples

Following are examples of propagate-down use.

- For TPS system parameters, only NODETYP, NTWKNUM, and NODENUM exist for both node modules and the points they might contain; setting of these parameters is a primary use of propagate-down.
- You can create a custom parameter group consisting of system parameters that you want configured the same for all members of the module.
- Custom parameters can be added to a module and the blocks it is to contain, for propagate-down purposes (for example, a user-defined parameter for a description of a piece of equipment that pertains to all points in a module).
- If you choose to align your equipment structure/view with your alarming hierarchy, you can do the following:
 - Create a custom parameter with the same name as a system parameter (for example, "Unit.")
 - Add that parameter to a module (as part of a custom parameter group).
 - Define that parameter as propagate-down.

In this example, the custom parameter Unit's value will be used for all elements of the module.

20. Improving Configuration Consistency and Efficiency

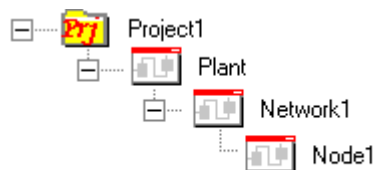
20.4. Propagate-Down Fundamentals

Propagate down for strategies

For strategy modules, propagate-down is accessible only if the module is within a module (see the example in the following figure).

Defining parameters as propagate-down makes them configurable only at the module level, and the value configured is used for all module components. This is another way to minimize the number of parameters to be configured and ensure accurate configuration. It also provides a method for checking certain parameters when adding pre-configured strategies to a module.

Consider a case of backbuilding or importing TPS Network data from an external database, and wanting to ensure that the network and node numbers are the same for all strategies.



Plant, a module holding all other modules for the project.

Network1, with NTWKNUM = 1 set as propagate-down

Node1, with NODENUM = 1 set as propagate-down

Propagate down for backbuilt or imported points

Backbuilt or imported points will be strategy blocks, not to be placed in any module. You must drag them into a module (or use module generation), such as Node1 in the previous example. When you do, TPS Builder checks the NTWKNUM and NODENUM values because they are set as propagate-down. Possible results are as follows:

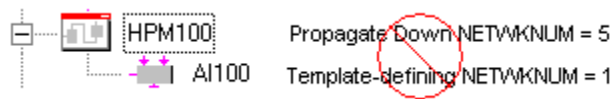
- If the parameters are not configured, then TPS Builder sets ("propagates down") the value for each and places the strategy in the Node1.
- If they have values, TPS Builder checks to see if they match the propagate-down values.
 - If they do match, TPS Builder places them in the module.
 - If they do not match, TPS Builder brings up a message of the values that will change and offers the option of undoing the move.

Propagate down for templates

When template module parameters are set as propagate-down, all strategy modules created from the template will include the propagate-down settings. Strategies created before the propagate-down setting is made are unaffected.

Relationship between template-defining and propagate-down attributes

You may have conflicts or unexpected results due to “contentions” between the template-defining and propagate-down attributes. The contention occurs because a template-defining attribute calls a parameter’s value from a template, and a propagate-down attribute calls the parameter’s value from a containing module. Consider the following example:



Can a parameter in HPM1 be set as propagate down, and the same parameter for AI100 template-defined? The answer is “no.” Although the example shows a strategy module; the same is true for template modules.

TPS Builder will not allow this type of conflict to occur. Following are the most common situations:

Conflict occurs during strategy operation:

Situation	Result
You try to add a strategy to a module, either by creating a new strategy or by moving an existing strategy.	An error message will display and the operation will fail.
One of the parameters is template-defined and it is also propagated down on the module.	<i>Example:</i> HPM1's NETWKNUM is set as propagate-down and AI100's template has NETWKNUM set as template-defining or template-defined. You try to add AI100 to HPM1. It will not be added.

20. Improving Configuration Consistency and Efficiency

20.4. Propagate-Down Fundamentals

Conflict develops when setting a control attribute:

Situation	Result
You have created a module using templates without conflicting parameter attributes, but then try to change the control attributes in a way that produces a conflict.	An error message will display and the attribute causing the conflict will not be set. <i>Example 1:</i> AI100 is a component of HPM1. NETWKNUM is set as propagate-down on HPM1, but AI100 NETWKNUM has no attributes set. You then go to AI100's template and try to set NETWKNUM as template-defining. The template-defining attribute will not be set. <i>Example 2:</i> AI100 is a component of HPM1. NETWKNUM on HPM1 has no attributes set, but NETWKNUM is template-defined on AI100. You then try to set HPM1's NETWKNUM as propagate-down. The propagate-down attribute will not be set.

20.5 Configuring the Propagate-Down Attribute

Fundamentals



REFERENCE - INTERNAL

In this section, refer to Propagate-Down Fundamentals.

Propagate-down preparation

Before beginning, complete these steps:

Step	Action
1	Important! If you have not yet done so, read the discussion of “Propagate-Down Fundamentals” in this section.
2	Identify the modules for which you want to set propagate-down parameters.
3	Control attributes for strategies (blocks or modules) can be configured only if the strategy is contained in a module. Therefore, place strategies to be configured into a module before continuing.

20. Improving Configuration Consistency and Efficiency

20.5. Configuring the Propagate-Down Attribute

How to configure propagate-down

To configure this attribute for a selected module component:

Step	Action
1	Complete the preparation described previously.
2	Right-click the module for which control attributes are to be configured. (If in a drawing, left-, then right-click the module.)
3	In the resulting menu, choose Control Attributes .
4	In the Control Attributes dialog box, choose the Propagate-Down tab.
5	Click a parameter name in the Parameters list.
6	If the parameter represents an array, enter a valid index number.
7	Click ADD→ . The parameter name will be placed in the Propagate-Down Parameters list.
8	Repeat as needed for additional parameters.
9	If needed, when finished with the Propagate-Down tab, click another tab in the dialog box for further control attribute configuration.
10	Click OK when all control attributes are configured.

20.6 Project-Up Fundamentals

Description

The Project-Up attribute applies to any item within a module (template or strategy).

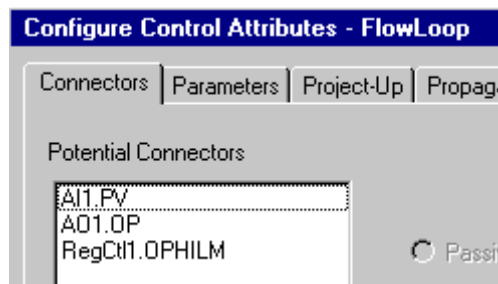
This attribute allows you

- to define connectors for a module containing the item, and
- to configure parameters at the module level, instead of opening the configuration form for every item in the module.

Configuration at the module level becomes possible because a tab containing each parameter set as project-up is added to the module's parameter configuration form.

Module connector use

One important reason to project parameters up is to enable a module to have connectors in a drawing. All parameters you define as project-up become available as connectors for the module. For module connection purposes, build-visible and runtime parameters appear in the module's Control Attributes' Connectors tab:



Streamlining configuration

You may have a module that contains strategies that are significantly configured, with only a few parameters per strategy still to be configured. Rather than opening the configuration form for each of these strategies, you can project them up to the module containing them, so that they are all displayed in its single (module) parameter configuration form. Only build-visible parameters appear on the module parameter form.

20. Improving Configuration Consistency and Efficiency

20.6. Project-Up Fundamentals

Example

For example, here is a Flow Loop module's parameter configuration form. Note that a tab for each block (or module) comprising the module has been established, and parameters defined as project-up are displayed here.

The screenshot shows a 'Parameters' dialog box with a title bar containing a close button. Inside the dialog, there are three tabs: 'AI1', 'Reg1', and 'AO1'. The 'Reg1' tab is currently selected. Below the tabs, there are two parameter rows. The first row is labeled 'Sensor Type' and 'SENSRTYP', with a dropdown menu showing '1.5V'. The second row is labeled 'Pm Module Type' and 'PNTMODTY', with a dropdown menu showing 'HLAI'.

20.7 Configuring the Project-Up Attribute

Preparation

Before beginning, complete these steps:

Step	Action
1	Important! If you have not yet done so, read the discussion of Project-Up Fundamentals in this section.
2	Plan which parameters are to be set as project up.

Procedure

To configure this attribute for a selected item within a module:

Step	Action
1	Complete the preparation described previously.
2	In any module, right-click the item for which control attributes are to be configured. (If in a drawing, left-, then right-click the item.)
3	In the resulting menu, choose Control Attributes .
4	In the Control Attributes dialog box, choose the Project-Up tab.
5	Click a parameter name in the Parameters list.
6	If the parameter represents a buildable array, enter a valid index number. (A buildable parameter is a parameter for which the user specifies a value. Parameters that cannot be built are those for which the user does not specify values, for example, PVFL.)
7	Type an alias for the parameter. You must enter an alias so that connector labels for the module are unique. One approach is to use the strategy name (or an abbreviation for it), followed by the parameter name, such as "AI1PV." Alias names are only for connectors on drawing symbols. The true parameter name is used in the configuration form. Do NOT use spaces in the name. You may use an underbar to separate parts of the name.
8	Click ADD→ . The parameter name will be placed in the Project-up Parameters list.
9	Repeat as needed for additional parameters.
10	If needed, when finished with the Project-Up tab, click another tab in the dialog box for further control attribute configuration.
11	Click OK when all control attributes are configured.

20.8 Using Custom Help with Libraries and Templates

Description

A WinHelp Help topic can be attached to any TPS Builder library or template. In this way, you can add custom information, such as site-specific instructions or personal usage notes, regarding the use of libraries and templates.

Custom Help for templates can be viewed from a library, project, or drawing.

Add user-specific Help to a library or template

Follow these steps to attach your own WinHelp topics to libraries or templates in TPS Builder:

Step	Action
1	Using any Help authoring program (for example, Microsoft Help Workshop or RoboHELP), create one or more WinHelp files containing topics specific to the TPS Builder library(ies) and/or template(s) of interest. Maintain the Help source files in a location of your choice.
2	Place the compiled Help file(s) in the HWIAC/databases/TPS Builder/data directory (or equivalent location, if you did not use the default installation). Include the .hlp file and related .cnt (contents) file if one exists. Important! Place the files in the specified directory so that they will be included in any migration or export/import activities.
3	Identify the Help file name, plus numerical topic ID for each Help topic to be used.
4	Open the library to which Help is to be added, or the library containing the template to which Help is to be added.
5	Inside the library window, • For library Help: Right-click the library name. On the resulting menu, choose Identification. • For template Help: Right-click the desired template name. On the resulting menu, choose Identification.

20. Improving Configuration Consistency and Efficiency

20.8. Using Custom Help with Libraries and Templates

Step	Action
6	In the Identification dialog box, enter the Help file name and topic ID to be added. This entry is not case-sensitive, and it is not necessary to include the ".hlp" extension. Important! Even though Winhelp appends .hlp to the file name while opening the helpfile, you should enter the file extension. If .hlp is not included, the Export function will not be able to locate the helpfile because it only searches the data directory for the filename entered.
7	Click OK. The specified Help topic is now attached to the selected library or template.

View user-specific library or template Help

After you have added Help to a library or template, you can view it by using one of the following procedures.

Step	Action
1	If you are in • a library – inside the library window, right-click the library or template of interest. • a project – right-click the strategy of interest. • a drawing – right-click the drawing symbol of interest.
2	On the resulting menu, choose Library Help or Template Help (as applicable). The Help topic will open.

Whenever Winhelp fails to locate a helpfile, you will be promoted to locate the file manually. Note that this information will not be recorded for subsequent operations.

20.9 Decreasing Configuration Time

Description

Several methods exist for streamlining your configuration activities. Using these methods can greatly increase the efficiency of your parameter configuration activities.

Planning is required to make the most of these methods. An understanding of your templates and their usage, as well as how modules are being used to create control schemes, is also needed.

Methods

The methods are as follows. Specific procedures are presented in "Related topics" following this discussion.

- Defining certain parameter values on a template, so that a default is provided that will be correct in the majority of cases.
- Defining certain parameter values on a template, so that they cannot be changed for any strategies or specialized templates created from that template.
- Defining certain parameter values at the (template or strategy) module level, so that the values are automatically added to all blocks and modules within that module.
- Exposing parameters at the module level, so that you do not need to select every element of a module.
- Bulk building or bulk editing similar templates or strategies. These TPS Builder functions allow you to use Microsoft Excel and its spreadsheet and macro capabilities to quickly assign values to similar items.
- Duplicating strategies. If a few strategies are quite similar, you can configure one, duplicate it the number of times needed, then modify the duplicated strategies as needed.

In addition, you can hide template-defined parameters on configuration forms so that you see only those parameters you need to configure. To do so, from the Options menu, choose Parameter Configuration > Hide Read-Only.

21. Working with Parameters

21.1 Overview

Description

The Parameters function in TPS Builder refers to the creation and use of custom parameters. A custom parameter is a user-defined parameter that is added to the standard parameter list of a template. Custom parameters may be

- TPS Network-loadable custom data segments for use in the Application Module, or
- Non-loadable parameters, for engineering use. For example, Memory Unit (MU) usage, Processor Unit (PU) usage, or estimated engineering effort can be defined as custom parameters.

21.2 View Parameters

Viewing a list of parameters

You can see the parameters related to a particular strategy using either the parameter configuration form or the Strategy Browser.

To view parameters using the parameter form:

Step	Action
1	Choose the parameter form's appearance from Options > Parameter Configuration choices.
2	Open the project containing the strategy of interest.
3	Right-click the strategy.
4	On the resulting menu, choose Parameters .
5	Examine the parameter list as needed.

21. Working with Parameters

21.2. View Parameters

To view parameters using the Strategy Browser:

Step	Action
1	Open the project containing the strategy of interest.
2	Right-click the strategy.
3	On the resulting menu, choose Parameters.
4	Locate a parameter which is configured with a reference to another parameter.
5	Click the  button to access the Strategy Browser.
6	In the Strategy Browser, locate and select the strategy of interest.
7	Examine the associated parameter list as needed.

Custom parameters

To view all custom parameters that have been defined in TPS Builder:

Step	Action
1	From the Parameter menu, choose Parameters.
2	In the Parameter dialog box, ensure that the List Custom Parameters Only option is selected.
3	Examine the parameter list as needed.

All TPS Builder parameters

You can see a list of all parameters in existence in TPS Builder by choosing Parameter > Parameters, and clearing the List Custom parameters Only option. This list is of marginal use, however, because it does not associate the parameters with specific strategies or templates.

21.3 Create Custom Parameters

To create a custom parameter:

Step	Action
1	If the new parameter is for a custom data segment and requires custom enumeration types, create the enumeration types first.
2	With any project, library, or drawing open, choose Parameters from the Parameter menu.
3	In the Parameters dialog box, click New .
4	If the custom parameter will make use of the parameter list \$REG_CTL, then choose type Reference, and choose the \$REG_CTL parameter group included in TPS Builder.
5	Complete the New Parameter dialog box. Make sure that all applicable entries are made. For example: • If you are creating a parameter of type enumeration, make sure that you choose the enumeration types and parameter initialization value. • If you are creating a parameter indexed by enumeration, make sure you choose the enumeration index.
6	Click OK .
7	If needed, perform other activities in the Parameters dialog box, such as creating other custom parameters, editing custom parameter definitions, or deleting custom parameters.
8	When finished, click Close .

21.4 View Parameter Details

Custom parameters

To view details of a custom parameter:

Step	Action
1	With any project, library, or drawing open, choose Parameters from the Parameter menu.
2	Make sure the List Custom Parameters Only option is selected.
3	In the Parameters list, click the name of the parameter of interest.
4	If the parameter has not yet been added to a group that has been added to a template, click the Edit button. Note that you can make changes to the settings, so if this is not intended, ensure you cancel when finished. If the parameter has been added to a group that has been added to a template, then click the View button. No changes can be made to the parameter definition in this case.
5	View the settings in the Edit Parameters dialog box.
6	When finished, click Cancel .
7	If needed, perform other activities in the Parameters dialog box, such as viewing other custom parameter details, creating new custom parameters, editing custom parameter definitions, or deleting custom parameters.
8	When finished, click Close .

System parameters

To view details of a system parameter:

Step	Action
1	With any project, library, or drawing open, choose Parameters from the Parameter menu.
2	Clear the List Custom Parameters Only option.
3	In the Parameters list, click the name of the parameter of interest. If multiple versions of the parameter you are interested in appear, view details of each one until you find the one of interest.
4	Click the View button.
5	View the settings in the Edit Parameters dialog box. Note that you cannot make changes to the settings, so everything in the resulting dialog box is grayed out.
6	When finished, click OK or Cancel .
7	If needed, perform other activities in the Parameters dialog box, such as viewing other parameter details, creating new custom parameters, editing custom parameter definitions, or deleting custom parameters.
8	When finished, click Close .

21.5 Edit Parameters

Process overview

To edit the definition of a custom parameter, first ensure that the parameter has *not* been assigned to a parameter group that has been added to a template.



ATTENTION

You cannot change a custom parameter that is part of a template.

If you need to edit such a parameter, first remove its association from the template. This can be done only if the template has not been used to create strategies or specialize templates. Use the following procedures

- to identify the parameter group(s) and template(s),
- to remove the associations between these items, then
- to edit the parameter definition.

Determine parameter usage

If you do not know if a parameter has been assigned, follow these steps:

Step	Action
1	From the Parameter menu, choose Parameters .
2	In the Parameters dialog box, click the name of the parameter in question.
3	If the second button in the dialog box is Edit , you can click it and modify the parameter definition. If the button is View (not Edit), then the parameter has been assigned. Continue with the next step.
4	Click the Activate option in the Used in Templates box. Templates to which the parameter group has been assigned will be listed. Make a note of the template(s).

Work with parameter groups

To disassociate the parameter group(s) from the template(s), follow the procedures explained in Working with Parameter Groups, Edit Parameter Groups, in this document.

After these steps are completed, you can edit the parameter definition.

Edit the parameter definition

To edit a custom parameter's definition:

Step	Action
1	Complete the previous procedures to ensure the parameter has not been used in any parameter group assigned to a template.
2	With any project, library, or drawing open, choose Parameters from the Parameter menu.
3	In the Parameters dialog box, make sure the List Custom Parameters Only option is selected.
4	In the Parameters list, click the name of the parameter of interest.
5	Click the Edit button.
6	Make needed changes to the settings in the Edit Parameters dialog box.
7	When finished, click OK .
8	If needed, perform other activities in the Parameters dialog box, such as viewing other custom parameter details, creating new custom parameters, editing custom parameter definitions, or deleting custom parameters.
9	When finished, click the Close button.

**ATTENTION**

If you change the parameter name, this change will not be reflected in the **Parameters** dialog box until after you close it.

21. Working with Parameters

21.6. Delete a Parameter

21.6 Delete a Parameter

Parameter group membership

To delete a custom parameter, it cannot belong to a parameter group. If you are not certain:

Step	Action
1	With any project, library, or drawing open, choose Parameters from the Parameter menu.
2	In Parameters dialog box Parameters list, select the parameter to be deleted.
3	If the Delete button is available, you can click it to delete the parameter. If the Delete button is unavailable, click the Activate option in the Used in Parameter Groups box. Parameter groups to which the parameter has been assigned will be listed.
4	If you still want to delete the parameter, make a note of group(s), and follow the "Edit Parameters" steps to find parameter usage and remove associations.

Deleting the parameter

To delete a custom parameter that is not part of a parameter group:

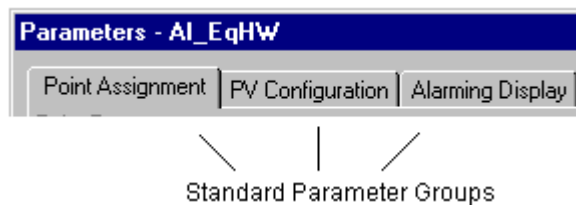
Step	Action
1	Complete the previous procedure to ensure the parameter has not been used in any parameter group.
2	With any project, library, or drawing open, choose Parameters from the Parameter menu.
3	Make sure the List Custom Parameters Only option is selected.
4	In the Parameters list, click the name of the parameter to be deleted.
5	Click the Delete button. In response to the confirmation message, click Yes .

22. Working with Parameter Groups

22.1 Overview

Description

A parameter group is a logical grouping of parameters. The group is visible as a labeled tab in a parameter configuration form. For example:



All parameter configuration forms include Honeywell-assigned (standard) parameter groups for configuring system parameters.

Types of parameter groups

Parameter groups are of the following four types:

- User-assigned parameter groups for custom parameters
 - Used for custom data segments and parameter lists that are loaded to the TPS system.
 - Used for engineering information that is not loaded to the system, for example memory usage.
- Honeywell-assigned parameter groups for system parameters
- Honeywell-provided \$REG_CTL parameter group, for use with AM custom parameters
- Parameter groups created as a result of setting the project-up control attribute for system parameters

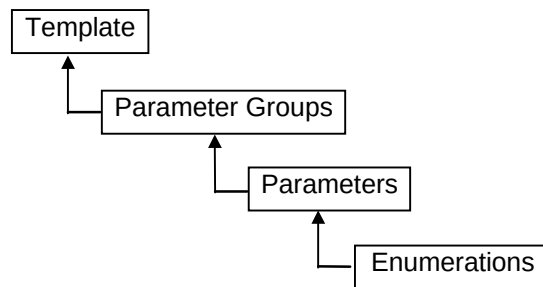
You need be concerned only with the user-assigned parameter group, as the others are either already in place, or created automatically in conjunction with the project-up function.

22. Working with Parameter Groups

22.1. Overview

Relationship of elements

The relationship of a template's parameter-related elements can be represented as follows:



Therefore, in order to assign a parameter group to a template, custom parameters must be defined. In order to define custom parameters, custom enumerations may need to be defined.

Planning

Planning is essential for effective use of parameter groups. Keep these facts in mind when planning for parameter groups:

- Parameter groups must be created and assigned to a template *before* the template is used for creating strategies or specializing templates. If a template has been used, you cannot add a parameter group to it.
- While you can create two different parameter groups that include the same parameter(s), you cannot assign both groups to the same template.
- You cannot add a parameter group to a template module that contains template blocks or other modules. The module must be "empty."

What do you want to do?

You can modify parameter configuration forms by :

- creating a parameter group;
- editing (changing the content of) a parameter group;
- deleting a parameter group;
- adding a parameter group to a template; or
- removing a parameter group from a template.

These procedures are discussed in this section.

22.2 View Parameter Groups

View strategy parameter groups

To view the parameter groups of a particular system strategy, access the strategy's parameter configuration form, with Options > Parameter Configuration set to Tabbed dialog box.

- For strategy blocks, double-click the strategy name in a project or strategy drawing symbol in a drawing.
- For strategy modules, right-click the module name in a project, or the drawing symbol in a drawing and choose Parameters. A module has parameters only if a) it is a TPS system entity defined in TPS Builder as a module, such as a device control point; b) it is a module in which parameters have been set as project-up; or c) it is a module created from a template for which one or more parameter groups have been defined.

Each tab in the configuration form is a parameter group. Some parameter groups depend on particular configurations. For example, for a regulatory control strategy, only the Point Assignment parameter group appears unless the strategy's control algorithm is changed from Null to a specific algorithm. The parameter groups then correspond to the configuration requirements of the algorithm.

To view all parameter groups possible for a particular strategy, view the parameter groups of the template from which the strategy was created, using the following procedure.

View template parameter groups

To view the names of all parameter groups for a specific template:

Step	Action
1	Open the library containing the template of interest.
2	Right-click the template, and on the resulting menu, choose Parameter Groups.
3	In the resulting template Parameter Groups dialog box, view the list of assigned parameter groups. This list includes all system and custom parameter groups associated with the template.
4	To see (view only) the details of a particular parameter group, double-click the parameter group in the Parameter Groups dialog box. (Note: In TPS Builder R110, the group's parameters are not listed.)

View custom parameter groups

To view a list of all custom parameter groups that have been defined in TPS Builder:

Step	Action
1	Ensure that at least one library, project, or drawing is in the TPS Builder workspace (open or minimized).
2	From the Parameter menu, choose Parameter Groups .
3	In the resulting Parameter Groups dialog box, view the list of custom parameter group names.

22.3 Create Custom Parameter Group

To create a custom parameter group:

Step	Action
1	If the new parameter group is for a custom data segment or parameter list and requires custom parameters with custom enumeration types, create the parameters and enumeration types first.
2	With any project, library, or drawing open, choose Parameter Groups from the Parameter menu.
3	In the Parameter Groups dialog box, click the New button.
4	Complete the New Parameter Group dialog box.
5	Click OK .
6	If needed, perform other activities in the Parameter Groups dialog box, such as creating other custom parameter groups, editing parameter group definitions, or deleting parameter groups.
7	When finished, click the Close button.

22.4 View Parameter Group Details

Custom parameter groups

To view details regarding custom parameter groups:

Step	Action
1	From the Parameter menu, choose Parameter Groups .
2	In the Parameter Groups list, select the parameter group of interest.
3	To identify the templates to which this parameter group has been assigned, choose Activate in the Used in Templates section. If the parameter group has been assigned to a template, the template name(s) will appear in the resulting list. Otherwise, the list will be blank.
4	To view the parameter group's contents and configuration, click the Edit or View button: • If the parameter group has not been assigned to a template, the button will read Edit. You can read and modify the definition of the group. • If the group has been assigned to a template, the button will read View. You can read the definition of the group (which appears grayed out).

System parameter groups

You cannot view complete details of system parameter groups from the Parameter Groups dialog box. If you are interested in the details of a particular system parameter group, you can locate the templates to which it is assigned, and view the parameter configuration form containing the parameter group. Each parameter group is a tab in the configuration form.

Depending on the selections made in the configuration form, the template may or may not show all parameter groups and/or details of the parameter group you are interested in. You may need to modify the configuration in order to activate the parameter group of interest. (For example, change a template's control algorithm from Null to an actual algorithm. If you take an action such as this solely to view parameter group details, make sure that you cancel your actions when finished.)

22.5 Edit Parameter Groups

Process overview

To edit the definition of a custom parameter group, first ensure that the group has *not* been assigned to a template.



ATTENTION

You cannot change a parameter group that is part of a template.

If you need to edit such a parameter group, first remove its association from the template. This can be done only if the template has not been used to create strategies or specialize templates. Use the following procedures

- to identify the template(s) to which the parameter group has been assigned,
- to remove the associations between these items, then
- to edit the parameter group definition.

Determine parameter group usage

If you do not know if a parameter group has been assigned, follow these steps:

Step	Action
1	From the Parameter menu, choose Parameter Groups.
2	In the Parameter Groups dialog box, click the name of the parameter group in question.
3	If the second button in the dialog box is Edit, then click it to modify the parameter group definition. If the button is View (not Edit), then the parameter group has been used. Continue with the next step.
4	Click the Activate option in the Used in Templates box. Templates to which the parameter group has been assigned will be listed. Make a note of the template(s). Close the Parameter Groups dialog box.

22. Working with Parameter Groups

22.5. Edit Parameter Groups

Verify no template usage

Before removing the association between the parameter group and template, you must verify that the template has not been used to create strategies or specialize templates.



ATTENTION

If the template has been used, you cannot edit the parameter group.

To check for template usage, follow these steps:

Step	Action
1	Open the library containing the template identified in the previous procedure.
2	Right-click the template and choose Identification on the resulting menu.
3	In the Identification dialog box, click Dependencies.
4	In the Template Dependencies dialog box, ensure that no strategies or specialized templates are based on the template.
5	Click Close, then OK, to close the Dependencies and Identification dialog boxes, respectively.

Remove association with template

For templates that have not been used, remove the association between the parameter group and template as follows:

Step	Action
1	Open the library containing the template identified in the previous procedure.
2	Right click the template and choose Parameter Groups from the resulting menu.
3	In the Parameter Group dialog box, Assigned Parameter Groups list, click the group identified in the previous procedure.
4	Click the Remove button. The parameter group remains intact, but its association with the selected template is removed.
5	If needed, repeat for other templates identified in the previous procedure.

Edit parameter group

To edit the parameter group definition:

Step	Action
1	With any project, library, or drawing open, choose Parameter Groups from the Parameter menu.
2	In the Parameter Group dialog box, make sure the List Custom Parameter Groups Only option is selected.
3	In the Parameter Groups list, click the name of the group of interest.
4	Click the Edit button.
5	Make needed changes to the settings in the Edit Parameter dialog box.
6	When finished, click OK. Note that if you change the parameter group name, this change will not be reflected in the Parameter Groups dialog box until after you close it.
7	If needed, perform other activities in the Parameter Groups dialog box, such as viewing other custom parameter group details, creating, editing, or deleting custom parameter groups.
8	When finished, click the Close button.

22. Working with Parameter Groups

22.6. Delete a Parameter Group

22.6 Delete a Parameter Group

Template assignment

To delete a custom parameter group, it cannot be assigned to any template. If you are not certain:

Step	Action
1	With any project, library, or drawing open, choose Parameter Groups from the Parameter menu.
2	In the Parameter Groups dialog box, click the name of the parameter group in question.
3	If the Delete button is available, you can click it to delete the parameter group. If the Delete button is unavailable, click the Activate option in the Used in Templates box. Templates to which the parameter group has been assigned will be listed.
4	If you still want to delete the parameter group, make a note of the templates, and follow the "Edit Parameter Groups" steps to find parameter group usage and remove associations.

Delete the parameter group

To delete a parameter group that is not assigned to a template:

Step	Action
1	Complete the previous procedure to ensure the parameter group has not been assigned to any template.
2	With any project, library, or drawing open, choose Parameter Groups from the Parameter menu.
3	In the Parameter Groups list, click the name of the parameter to be deleted.
4	Click the Delete button. In response to the confirmation message, click Yes.

22.7 Add Parameter Groups to a Template

Parameter group uses

Add a parameter group to a template to create:

- a custom data segment and/or parameter list for loading to the TPS system.
- a convenient location for entering and viewing engineering information, such as processing units or memory usage.

Before you begin

Before you add a parameter group to a template:

Step	Action
1	Create the group. Refer to Create Custom Parameter Group in this section.
2	Ensure that all custom parameters and related enumerations included in the parameter group have been defined. You cannot modify custom parameters or enumerations once the parameter group they have been assigned to is assigned to a template.
3	If you are going to assign the parameter group to a template module, ensure that the module is empty (that is, it contains no template blocks or modules).

How to add a parameter group to a template

To add a parameter group to a template, follow these steps:

Step	Action
1	Open the library containing the template to which a parameter group is to be added.
2	Right-click the template to be changed.
3	On the resulting menu, choose Parameter Groups.
4	In the left column, click the parameter group you want to add, and then click the Add ---> button. The selected parameter group is added to the list of assigned parameter groups.
5	Click OK to implement the change and close the dialog box.

22. Working with Parameter Groups

22.8. Remove Parameter Groups from a Template

Errors

If the group(s) you are adding contains a parameter that is a duplicate of one already in the template (in another group), then you will receive an error message, and the groups you have selected will not be added.

22.8 Remove Parameter Groups from a Template

When a parameter group can be removed

Custom parameter groups that have been added to a template can be removed, as long as the template has not been used to create a strategy or specialize a template (or, if it is a template module, no template blocks or modules have been added to it).

How to remove a parameter group from a template

To remove a parameter group, follow these steps:

Step	Action
1	Open the library containing the template to be changed.
2	Right-click the template to which the parameter group is to be assigned.
3	On the resulting menu, choose Parameter Groups.
4	In the right column, click the parameter group you want to remove.
5	Click the Remove button.
6	Click OK to implement the change and close the dialog box.

23. Working with Enumeration Types

23.1 Overview

Description

The Enumeration Types function in TPS Builder refers to the creation and use of custom enumerations. An enumeration type is a data type consisting of two or more states that are stored in a parameter.

Custom enumeration example

For example, if you define a custom parameter related to estimated engineering effort, you might want to define a level of difficulty as Easy, Moderate, and Complex. This could be accomplished by defining corresponding custom enumerations.

Self-defined enumerations

Note that this function does not refer to self-defined enumerations, which are configured in the STATE1 and STATE2 parameters of applicable data points.

23.2 Create a New Enumeration Type

To create a custom enumeration type:

Step	Action
1	With any project, library, or drawing open, choose Enumeration Types from the Parameter menu.
2	In the Enumeration Types dialog box, click the New button.
3	Complete the New Enumeration Type dialog box.
4	For the enumeration states defined, evaluate the order in which they appear in the list of current enumeration states. When the enumeration is used for a custom parameter, this order is used for 1) the low and high range for parameter initialization, and 1) the order of states in the selection lists on the New/Edit Parameter dialog box and configuration form. Regarding item a, the first enumeration in the list in the dialog box becomes the low range, and the last enumeration in the list becomes the high range.

23. Working with Enumeration Types

23.3. View Enumeration Types

Step	Action
5	If you need to rearrange the order of enumeration states, refer to the New Enumeration Type dialog box topic in the online Help.
6	When the enumeration type and states are defined as needed, click OK.
7	If needed, perform other activities in the Enumeration Types dialog box, such as creating other custom enumeration types, editing or deleting enumeration types.
8	When finished, click the Close button.

23.3 View Enumeration Types

To view all custom enumeration types those have been defined in TPS Builder:

Step	Action
1	From the Parameter menu, choose Enumeration Types.
2	In the Enumeration Types dialog box, ensure that the List Custom Enum. Types Only option is selected.
3	View the resulting list of custom enumeration types.

23.4 View Enumeration Type Details

Custom enumeration types

To view details of a custom enumeration type:

Step	Action
1	With any project, library, or drawing open, choose Enumeration Types from the Parameter menu.
2	Make sure the List Custom Enum. Types Only option is selected.
3	In the Enumerations list, click the name of the enumeration of interest.
4	If the enumeration has not yet been added to a parameter, click the Edit button. Note that you can make changes to the settings, so if this is not intended, ensure you cancel when finished. If the enumeration has been added to a parameter, then click the View button. No changes can be made to the parameter definition in this case.
5	View the information in the New/Edit Enumeration Type dialog box.
6	When finished, click Cancel.
7	If needed, perform other activities in the Enumeration Types dialog box, such as viewing other enumeration details, creating new enumeration types, editing custom enumeration type definitions, or deleting custom enumeration types.
8	When finished, click the Close button.

23. Working with Enumeration Types

23.4. View Enumeration Type Details

System enumeration types

To view details of a system enumeration type:

Step	Action
1	With any project, library, or drawing open, choose Enumeration Types from the Parameter menu.
2	Clear the List Custom Enum. Types Only option.
3	In the Enumerations list, click the name of the enumeration of interest. If multiple versions of the enumeration you are interested in appear, then view details of each one until you find the one of interest.
4	Click the View button.
5	View the settings in the Edit Enumeration Types dialog box. Note that you cannot make changes to the settings, so everything in the resulting dialog box is grayed out.
6	When finished, click OK or Cancel.
7	If needed, perform other activities in the Enumeration Types dialog box, such as viewing other enumeration details, creating new enumeration types, editing custom enumeration type definitions, or deleting custom enumeration types.
8	When finished, click the Close button.

23.5 Edit Enumeration Types

Process overview

To edit the definition of a custom enumeration type, first ensure that it has *not* been used by a parameter.



ATTENTION

You cannot change a custom enumeration type once it is used in a parameter definition.

If you need to edit such an enumeration, first remove its association from the parameter. This can be done only if the parameter has not been used to create strategies or specialize templates. Use the following procedures

- to determine whether the enumeration has been used;
- to identify the parameter group(s) and template(s) to which the parameter using the enumerations has been assigned;
- to remove the associations between these items, then;
- to edit the enumeration definition.

Determine enumeration usage

If you do not know if a custom enumeration has been used, follow these steps:

Step	Action
1	From the Parameter menu, choose Enumeration Types.
2	In the Enumerations Types dialog box, click the enumeration in question.
3	If the second button in the dialog box is Edit, then click it to modify the enumeration definition. If the button is View (not Edit), then the enumeration has been used. Continue with the next step.
4	Click the Activate option in the Used in Parameters box. Parameters which have used the enumeration type will be listed. Make a note of the parameter(s). Close the Enumeration Types dialog box.

23. Working with Enumeration Types

23.5. Edit Enumeration Types

Remove association

If you need to edit an enumeration type that has been used, you need to remove the association between the enumeration type and the parameter(s) that have used it. To do so:

Step	Action
1	From the Parameter menu, choose Parameters.
2	In the Parameters dialog box, make sure the List Custom Parameters Only option is selected.
3	In the list of custom parameters, select the parameter identified in the previous procedure.
4	If the second button in the dialog box is Edit, click it and go to step 5. If the second button is View, then you must remove the association between this parameter and parameter group(s) to which it has been assigned. Refer to the procedures in Edit Parameter Groups.
5	In the Edit Parameters dialog box, either • change the parameter type to something other than enumeration, or • change the enumeration type to one other than the one you need to edit.
6	Click OK, then close the Parameters dialog box.

Edit the enumeration type

To edit the enumeration type definition:

Step	Action
1	Complete the previous procedures to ensure the enumerations have not been used.
2	With any project, library, or drawing open, choose Enumeration Types from the Parameter menu.
3	In the Enumerations list, click the name of the enumeration to be modified.
4	Click the Edit button.
5	Make needed changes to the settings in the Edit Enumeration Type dialog box.

23. Working with Enumeration Types

23.5. Edit Enumeration Types

Step	Action
6	When finished, click OK.
7	If needed, perform other activities in the Enumerations dialog box, such as viewing other custom parameter details, creating new custom parameters, editing custom parameter definitions, or deleting custom parameters.
8	When finished, click the Close button.



ATTENTION

If you change the enumeration type name, this change will not be reflected in the Enumerations dialog box until after you close it.

23.6 Delete Enumeration Type

Enumeration usage

To delete a custom enumeration type, first ensure that it has *not* been used by a parameter. You cannot delete a custom enumeration type if it is used in a parameter definition. If you are not certain:

Step	Action
1	With any project, library, or drawing open, choose Enumeration Types from the Parameter menu.
2	In Enumeration Types dialog box, select the enumeration to be deleted.
3	If the Delete button is available, you can click it to delete the enumeration. If the Delete button is unavailable, click the Activate option in the Used in Parameters box. Parameters to which the enumeration has been assigned will be listed.
4	If you still want to delete the parameter, make a note of parameter(s), and follow the steps in Edit Enumeration Types to find parameter usage and remove associations.

Delete the enumeration

To delete a custom enumeration that has not been used by a parameter:

Step	Action
1	With any project, library, or drawing open, choose Enumeration Types from the Parameter menu.
2	Make sure the List Custom Parameters Only option is selected.
3	In the Enumerations list, select the enumeration to be deleted.
4	Click the Delete button. In response to the confirmation message, click Yes.

23.7 Create Custom Data Segments

Description

Application Module (AM) Custom, Regulatory, or Switch points can include custom data segments (CDSs), which are defined in Control Language (CL) files. For a review of CDSs, you may want to see the Control Language Application Module Overview.

Custom data segments can be created and used in TPS Builder by following the procedures explained next.

Create a CDS

A CDS is represented as a parameter group in TPS Builder. To create a custom data segment in TPS Builder, follow these steps:

Step	Action	Reference
1	Create the custom parameters to be included in the custom data segment, including custom enumerations if required. Ensure that all parameters are defined as TPS Network parameters.	Create Custom Parameters
2	Create a new parameter group that includes all of the custom parameters. Ensure that the Load to System option is selected, and a shortname/file name is entered (you can leave that blank and use the parameter group name as the name if you follow TPS system naming conventions).	Create Custom Parameter Group
3	Once you are ready to transfer configuration data to the TPS Network, you can create a CL file for the CDS, as explained next.	

Create CL file

Follow these steps to create a CL file for the CDS, from a TPS Builder parameter group:

Step	Action
1	From the Parameter menu, choose Parameter Groups.
2	In the Parameter Groups dialog box, select the parameter group that is to become a CDS CL file.
3	Click the Build CDS button. In the resulting dialog box, choose a location for the CL file. It is recommended that all CL files be kept in <i>Projectx\program</i> , where <i>Projectx</i> is the name of your project. (The complete directory path depends on your TPS Builder installation.) Do not change the file name displayed. This name becomes the name of the CL file. It is not necessary to add the .cl extension (TPS Builder adds it)
4	Click Save. The CL file will be created, including appropriate header and end entries.

Edit a CDS

You can modify a CDS by modifying the custom parameters and/or parameter groups you have used to create the CDS, then following the previous steps to create a CL file. This will overwrite the original CL file.

You can also use the CL Editor to modify the file directly. If you take this approach, the file will no longer match the custom parameters and parameter groups in TPS Builder, unless you backbuild the CL file. To backbuild a CDS for an existing parameter group, you must first delete the parameter group. (You cannot overwrite an existing parameter group.)

Delete a CDS

To delete a CDS, delete the custom parameters and parameter group in TPS Builder. Refer to:

- Delete a Parameter
- Delete a Parameter Group

To delete a CL file, locate it in My Computer or Explorer, then use standard Windows NT/XP procedures to delete it.

Add a CDS to a template

To add a CDS to a template, assign the CDS parameter group to the template.

23.8 Parameter Lists

Description

A parameter list is a list of the names and types that the parameters of a TPS system data point can have. At compile time, the compiler can refer to a parameter list for a description of the referenced data points. For a review of parameter lists, you may want to see the Control Language Application Module Overview.

Parameter lists can be created and used in TPS Builder by following the procedures explained next.

You can also backbuild parameter lists. Refer to Backbuilding CL Files in the Importing TPS Entity Data from TPS Files (Backbuilding) section of this document.

Create a parameter list

A parameter list is represented as a parameter group in TPS Builder. To create a parameter list in TPS Builder, follow these steps:

Step	Action	Reference
1	Create the custom parameters to be included in the parameter list, including custom enumerations if required. Ensure that all parameters are defined as TPS Network parameters.	Create Custom Parameters
2	Create a new parameter group that includes all of the custom parameters. Ensure that the Load to System option is selected, and a shortname/file name is entered (you can leave that blank and use the parameter group name as the name if you follow TPS system naming conventions).	Create Custom Parameter Group
3	Once you are ready to transfer configuration data to the TPS Network, you can create a CL file for the CDS, as explained next.	

23. Working with Enumeration Types

23.8. Parameter Lists

Create a CL file for the parameter list

Follow these steps to create the CL file from a TPS Builder parameter group:

Step	Action
1	From the Parameter menu, choose Parameter Groups.
2	In the Parameter Groups dialog box, select the parameter group that is to become a parameter list.
3	Click the Build Parameter List button.
4	In the resulting dialog box, choose a location for the CL file. It is recommended that all CL files be kept in the <i>Projectx</i> /program directory, where <i>Projectx</i> is the specific project directory. (The complete directory path depends on your TPS Builder installation.) Do not change the file name displayed. This name becomes the name of the CL file. It is not necessary to add the .cl extension (TPS Builder adds it).
5	Click Save. The CL file will be created, including appropriate header and end entries.

Edit a parameter list

You can modify a parameter list by modifying the custom parameters and/or parameter groups you have used to create the parameter list, then following the previous steps. This will overwrite the original CL file.

You can also access the CL Editor and modify the file directly. If you take this approach, the file will no longer match the custom parameters and parameter groups in TPS Builder. To backbuild a parameter list for an existing parameter group, you must first delete the parameter group. (You cannot overwrite an existing parameter group.)

Delete a parameter list

To delete a parameter list, delete the custom parameters and parameter group in TPS Builder. Refer to:

- Delete a Parameter
- Delete a Parameter Group

To delete an associated CL file, locate it in My Computer or Explorer, then use standard Windows NT/XP procedures to delete it.

24. Determining Object Relationships

24.1 Overview

Description

The Relationship Browser enables you to evaluate the associations between several objects within TPS Builder. You can determine relationships between and view details of the following objects:

- Projects
- Libraries
- Template
- Strategies
- GUS Pictures
- OLE Documents
- Views
- Attached Documents

Examples

The following table offers explains of the types information you can obtain from the relationship browser.

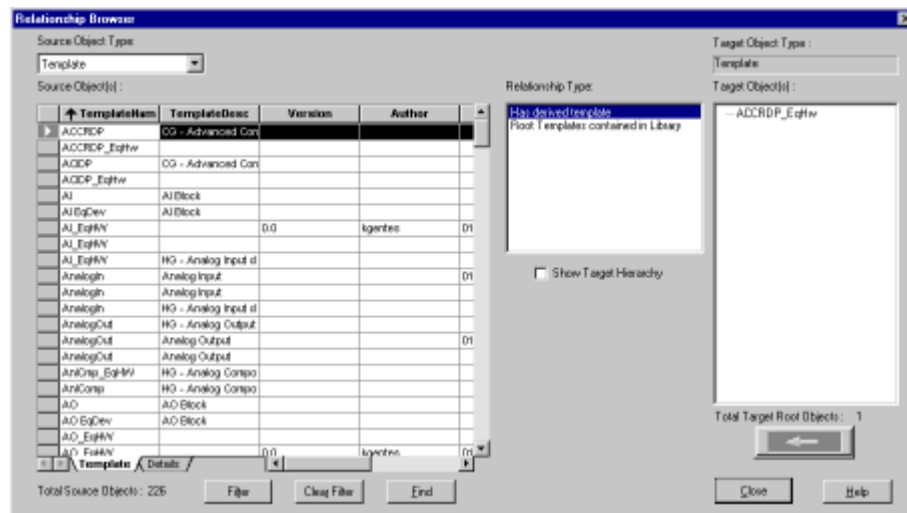
Object	You can see...
Project	A list of all projects, and for a specific project, determines the composition views to which it belongs. For each view related to the project, you can determine the strategies contained in that view. Details about the project are also available (such as creation and modification dates; release related to the project).
Composition View	A list of all views (system and user-created), and for a specific view, identify all templates that have been configured with that view membership.
Strategy	A list of all strategies, and for a specific strategy, identify • the project it is associated with, • the view(s) it appears in, and • the template from which it was created.

24. Determining Object Relationships

24.1. Overview

Display

The Relationship Browser is shown in the following figure.



Activities

- Use the Relationship Browser
- Filter Relationship Browser Contents
- Clear Relationship Browser Filter
- Sort Relationship Browser Contents
- Find a Relationship Browser Object

24.2 Use the Relationship Browser

Description

You may open the Relationship Browser and locate needed information of various levels of detail. You can choose a type of object to evaluate, such as projects, templates, strategies, and so forth, and find quite a bit of information in the browser with one simple selection.

For more detail, you can choose a specific object, and work your way through various levels of information, as explained in the following procedures.

Basic procedure

To use the Relationship Browser, follow these steps:

Step	Action
1	With any project, library, or drawing open, choose Relationship Browser from the Tools menu.
2	In the browser, choose the type of object you want to evaluate (Project, Template, Strategy, and so forth) from the Source Object Type list. The browser presents all objects of the type chosen in the Source Object(s) list.
3	Evaluate the resulting list of Source Objects.
4	Navigate in the Source Objects table as explained in the Relationship Browser dialog box topic.
5	Continue as needed by evaluating a) relationships of a specific source object, b) details of a specific relationship, or c) details about a specific target object. Each of these procedures is a continuation of the previous one, and all are presented below.

24. Determining Object Relationships

24.2. Use the Relationship Browser

Evaluate relationships of a specific object

To evaluate the relationship of one object to others, follow these steps:

Step	Action
1	In Source Object(s), select a specific object by positioning the cursor over the leftmost square of the object's row, so that the cursor is black arrow:  . Click the mouse button. The row will be selected (highlighted) and information will appear in the Relationship Type field.
2	Evaluate the Relationship Types. The Relationship Types depend on the object selected. For example, a strategy has at least a relationship with a template ("Is created from") and a project ("Is contained in the xxx view of"). It may belong to more than one view, and may have attached documents or other relationships.

Evaluate details of a specific relationship

For more detail about a specific relationship:

Step	Action
1	In the Relationship Types list click the relationship about which you want more information. The applicable details will appear in the Target Objects list. For example, for a project, all strategies in the project's root level will appear if you select a particular view in the Relationship Types list.
2	If an object might have a further level of detail, then click the Show Target Hierarchy option to access the additional detail. For example, if a project includes modules (as most projects do), then you can view the contents of all modules if the Show Target Hierarchy option is selected.
3	To view a complete hierarchy, locate a module or other item that includes a hierarchy and click the + sign next to the module name.

Evaluate a specific target object

Once you have details in the Target Object(s) list, you may want additional details about a specific object in that list. To begin a new evaluation based on a selected target object:

Step	Action
1	In the Target Object(s) list, click the item about which you want more information. You can select multiple items by using Shift+click (a contiguous group of items) or Ctrl+click (noncontiguous items). The arrow button below the list will become available (highlighted), 
2	Click the arrow button to change the Source Object to the item(s) you highlighted in the Target Object(s) list.
3	To find relationships for a new source object, repeat the evaluation of relationships and/or the evaluation of details. Note that you must select a new source object to obtain its relationship information (merely transferring it from the Target Objects list does <i>not</i> change the browser contents).
4	Repeat as needed.

24.3 Filter Relationship Browser Contents

Description

The Relationship Browser content (Source Object list) can be filtered, by column, based on various user-set conditions. These conditions may be based on some value appearing in a cell or the value supplied by the user.

Basic facts about filtering and filtering methods are described in the following paragraphs.

Multiple filters

You can perform multiple filters, one at a time, to meet your requirements. For example, you could filter the strategy list first by author, then by specific type of strategy name, and even further to a specific date, if required.

Filter persistence

The filter is active only as long as you remain in a particular Source Object(s) list. For example, if you filter the Strategy list, change the Source Object Type to Template, then return to the Strategy list, the filtering will be gone.

Filter on current cell value

This filter option is the simplest. It limits the Source Object(s) list to objects matching exactly the cell you choose. For example, you may limit the list to only templates authored by "Smith."

To filter in this manner,

Step	Action
1	From the Tools menu, choose Relationship Browser. In the browser, choose the desired Source Object Type.
2	In the Source Object(s) list (either tab), click in an individual cell containing the object matching your filter criterion.
3	Click the Filter button to open the Filter dialog box.
4	In the Filter dialog box, the Current Selection option is selected by default. Leave this selection unchanged.
5	Click OK. The Source Object list will be limited to objects matching the one you chose in the previous steps.
6	The filtered list is in whatever order the column was in before filtering. You may want to sort it in ascending or descending order. Refer to "Sort Relationship Browser Contents."

Filter using wildcard characters

A wildcard is a character that represents one or more characters. Including wildcards in your filter allows you to filter to a general level of similarity, such as filtering for all strategies with names beginning with "AI," without regard to what follows that prefix.

To filter using wildcards:

Step	Action
1	From the Tools menu, choose Relationship Browser. In the browser, choose the desired Source Object Type.
2	Click in an individual cell in the Source Object(s) list (either tab) you want to filter.
3	Click the Filter button to open the Filter dialog box.
4	In the Filter dialog box, choose the Wild Cards option.

24. Determining Object Relationships

24.3. Filter Relationship Browser Contents

Step	Action
5	In the Wild Cards field, type the desired term, and use an asterisk (*) and/or question mark (?) as follows: • Use the asterisk to match one or more characters. For example, AI* for all strategies beginning with "AI." Your results would include AI100 as well as AIFlow. • Use the question mark to match one single character or a specific number of characters. For example, AI?? would find AI10, AIAA, but not AI1 or AI1000. Note: Filtering is not case sensitive. Typing "eq" will include all capitalization versions of that term.
6	If desired, set multiple wildcard criteria by separating each criterion with a semicolon (;), without having a space before or after the semicolon. Use no spaces before or after the semicolon. For example: • D*;AI* will list objects in the selected column that begin with either D or AI, such as D100, DvCtl, AI100. • *Eq;*HW will list objects in the column ending with either Eq or HW, such as CB_HW and LogGate EQ.
7	Click OK. The Source Object list will be limited to objects matching those chosen by the previous steps.
8	The filtered list is in the order that the column was before filtering. You may want to sort it in ascending or descending order. Refer to Sort Relationship Browser Contents in this section.

Filter using SQL syntax

Description

This filter option allows you to specify criteria for the filter using a SQL (Structured Query Language) statement. The SQL statement is based on the column heading names in the Source Object(s) list (for example, TemplateName, Author).

Use standard SQL syntax, including the following symbols and terms: =, >, <, and, or, not, like.

If you are not familiar with SQL syntax, study the following examples, and obtain an appropriate reference if necessary.

Examples

Examples of SQL syntax filter statements are as follows. Assume that the Source Object type is Strategy for these examples:

- `nodetyp = "apm" and ntwknum = "8"`

This statement limits the Source Object(s) list to APM strategies with a network number of 8.

- `strategyname like "d*"`

This statement limits the Source Object(s) list to strategies beginning with D.

Procedure

To filter using a SQL statement:

Step	Action
1	From the Tools menu, choose Relationship Browser. In the browser, choose the desired Source Object Type. Choose the desired Source Object(s) list tab.
2	Click the Filter button to open the Filter dialog box.
3	In the Filter dialog box, choose the SQL Syntax option.
4	In the SQL Syntax field, type a SQL statement representing the needed filter.
5	Click OK. The Source Object list will be limited to objects matching those chosen by the previous steps.
6	The filtered list is in whatever order the column was in before filtering. You may want to sort it in ascending or descending order. Refer to "Refer to Sort Relationship Browser Contents" in this section.

24. Determining Object Relationships

24.3. Filter Relationship Browser Contents

Filter on date

This filter option allows you to filter the Source Object(s) list by date. The radio button for this filter condition is disabled if selection is made as the entire Column (not a cell). To filter in this manner, follow these steps:

Step	Action
1	From the Tools menu, choose Relationship Browser. In the browser, choose the desired Source Object Type. Choose the main Source Object(s) tab (the Details tab contains no date information).
2	Click the Filter button to open the Filter dialog box.
3	In the Filter dialog box, choose the Date option, then choose either DateCreated or DateLastModified from the pulldown menu.
4	In the From field, enter a specific starting date and time, in the format set for your Windows environment. (This format will be presented in the field, for example, mm/dd/yy hh:mm:ss.) For example, 08/28/06 08:00:00.
5	In the To field, enter a specific ending date and time, in the same format. For example, 08/28/06 16:30:00. Note that the examples in steps 4 and 5 show how to specify a single shift.
6	Click OK. The Source Object list will be limited to objects matching those chosen by the previous steps.
7	The filtered list is in whatever order the column was in before filtering. You may want to sort it in ascending or descending order.



TIP

To see all objects for a specific day (24-hour period), enter the date you want in the From field, and the next day in the To field. You do not need to enter time (hours, etc.).

For example, for 08/28/06 only, select 08/28/06 and 08/29/06 as the From and To dates.

24.4 Clear Relationship Browser Filter

If you have filtered a specific Source Object(s) list in the Relationship Browser, any of the following actions will clear the filter:

- Click the Clear Filter button.
- Choose another Source Object Type.
- Close the Relationship Browser.

24.5 Sort Relationship Browser Contents

Description

The Source Object(s) list can be sorted in ascending or descending order by column. For example, in the Strategy list, you can sort by strategy name from A to Z (ascending), or from Z to A (descending).

Order of blank cells

Blank cells are placed ahead of cells with data for ascending order, or after cells with data in descending order.

Sorting symbols

Sorting is shown by the sorting symbol in each column's heading:

Symbol	Meaning
None	Before any sorting occurs, no symbol indicates default sorting.
◊	After you have sorted a column, this symbol indicates release of sorting preference for a column.
↑	An "up" triangle indicates user-request for ascending sort order for the column (lowest to highest, 0 - 9 and A - Z).
↓	A "down" triangle indicates user-request for descending sort order for the column (highest to lowest, 9 - 0 and Z - A).

Multiple sorts not allowed

The Source Object(s) list can be sorted by only one column at a time. (You cannot perform multiple sorts.)

24. Determining Object Relationships

24.6. Find a Relationship Browser Object

Procedure

To sort the Source Object(s) list, follow these steps:

Step	Action
1	In the Relationship Browser, choose the desired Source Object Type.
2	In the Source Object(s) list, position the cursor in the desired column heading so that the cursor becomes a down arrow, ↓.
3	Right-click once. Depending on the previous sort action on the column (whether or not some type of sorting is still in effect), a sorting symbol will appear as previously defined.
4	To change to another sort order, right-click in the column heading again. The sequence is ↑ (ascending), ↓ (descending), no symbol or ⬮ (default).

The sorting order remains until you select another Source Object Type, or close the Relationship Browser.

Clear sort order

To clear a particular column's sorting order, use any of these methods:

- Right-click in the sorted column's heading until the default sort symbol appears, ⬮.
- Sort a different column.
- Choose another Source Object Type.
- Close the Relationship Browser.

24.6 Find a Relationship Browser Object

Description

You can search for a particular entry in any single column in the Relationship Browser. You cannot search across columns.

Procedure

To find an entry in the Relationship Browser, follow these steps:

Step	Action
1	From the Tools menu, choose Relationship Browser. In the browser, choose the desired Source Object Type.
2	If desired, filter or sort the Source Object(s) list contents before performing the Find.
3	In the Source Object(s) list, position the cursor in the column in which you want to find something.
4	Click in the first cell of the column in which you want to search (or, to search above or below a particular point, click in the cell where you want to begin the search).
5	Click the Find button.
6	In the Find dialog box, type all or part of the needed term or phrase. For example, you might type "mc" to find author names McAfee, McConnell, and McDonald.
7	If desired, choose the Match Case option to limit the search to the exact capitalization used.
8	Choose a search direction (Up or Down; Down is the default).
9	Click the OK button. The first instance of the term or phrase, in the column selected, will be highlighted.
10	Click the Find Next button to find the next instance. Repeat as needed.

When all instances found

When no more instances are found, the system will beep. The Find function searches up or down for one pass, from the point it started. It does not "wrap" around in the column.

Search another column

To search another column, simply click in the column, and repeat the procedure. The Find dialog box remains open until you click the Cancel button, or the **X** in the upper right corner of the dialog box.

24. Determining Object Relationships

24.6. Find a Relationship Browser Object

Honeywell

Honeywell Process Solutions
1860 W. Rose Garden Lane
Phoenix, AZ 85027 USA